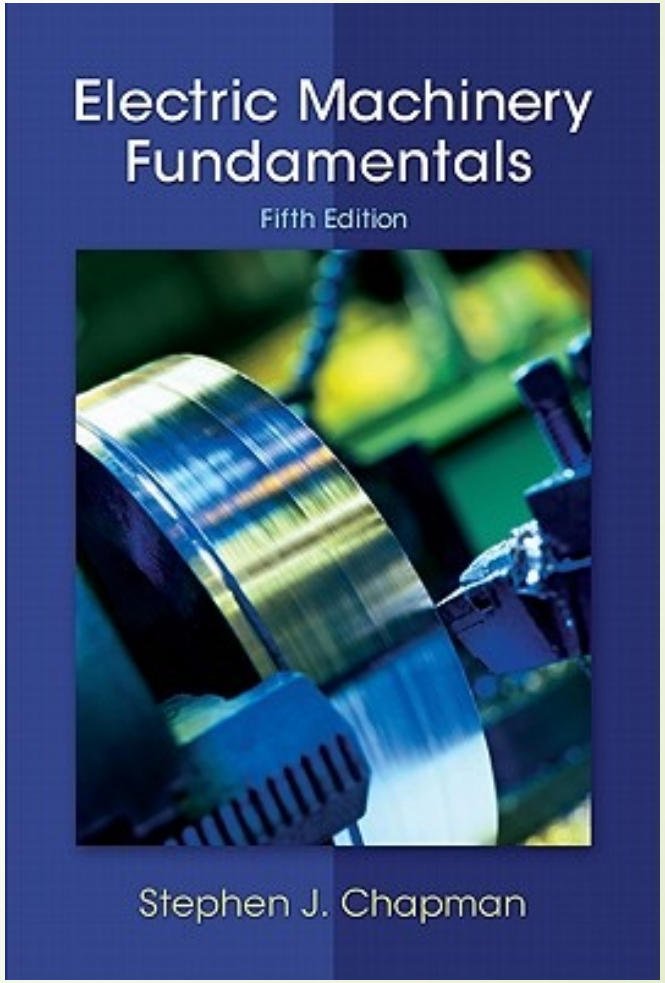
# Chapter 1 Introduction to Machinery Principles

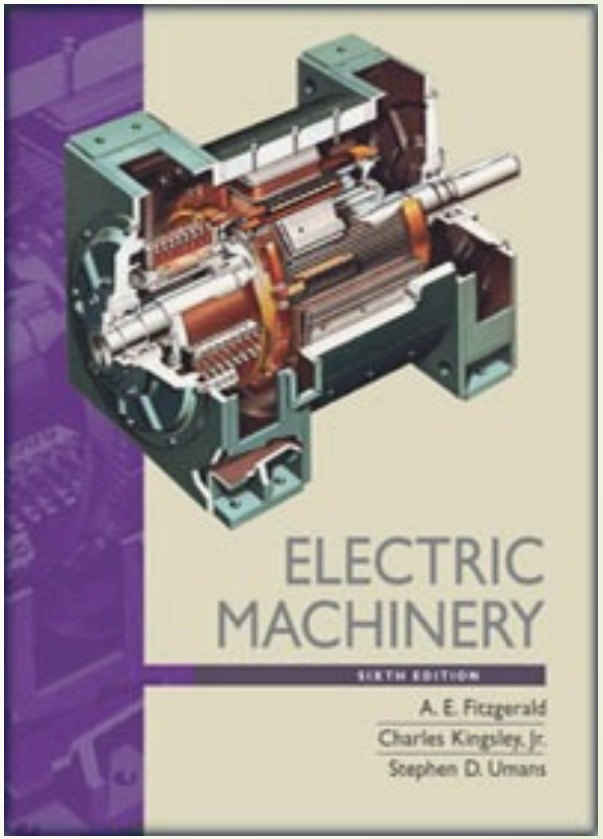
# Text book and supplementary materials of this course

- Stephen J. Chapman
- 5th edition (Feb. 18, 2011)



# Reference book

- A. E. Fitzgerald, *Electric Machinery*, McGraw-Hill , 6th edition (July 25, 2002)



# Electric Machinery Principles





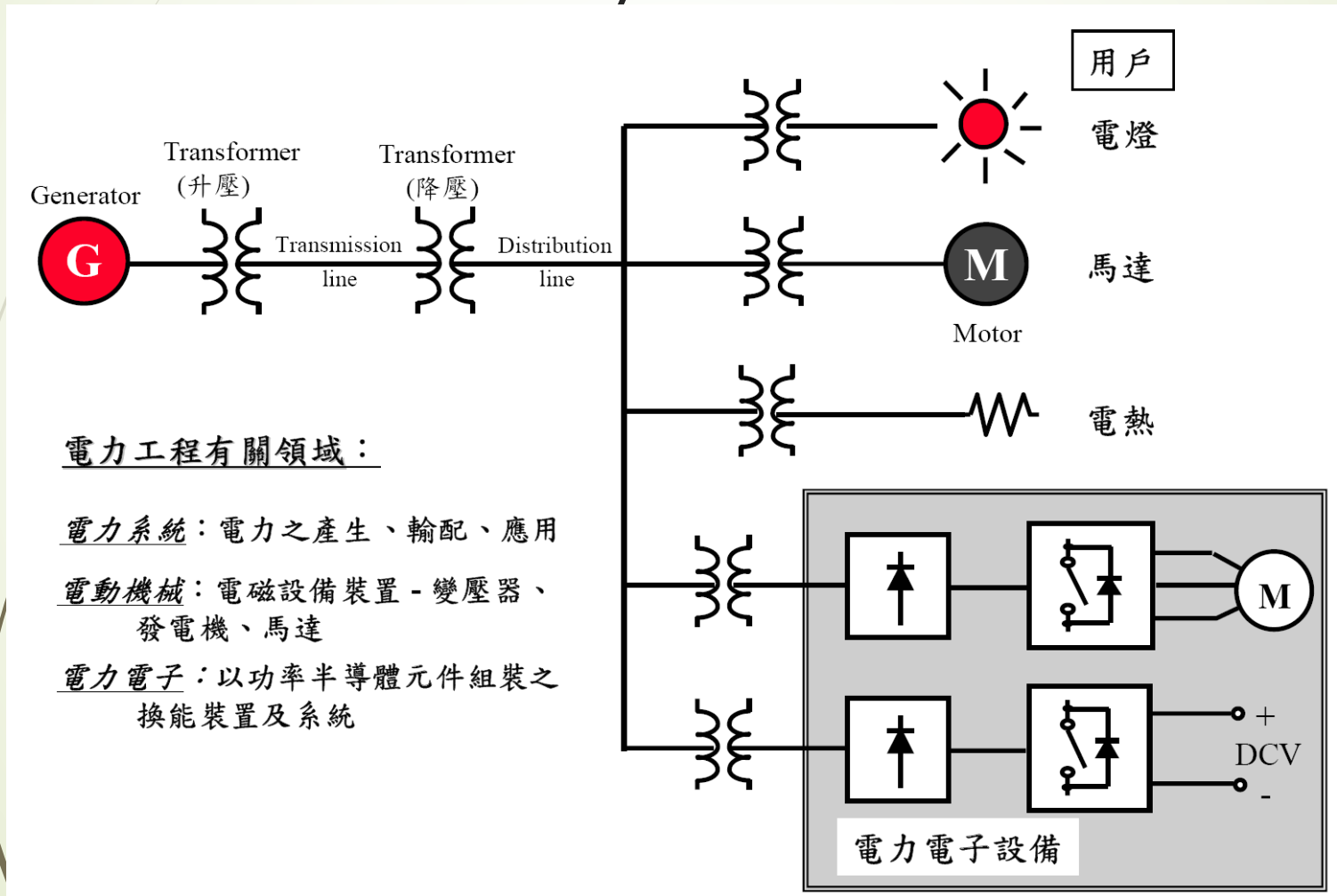




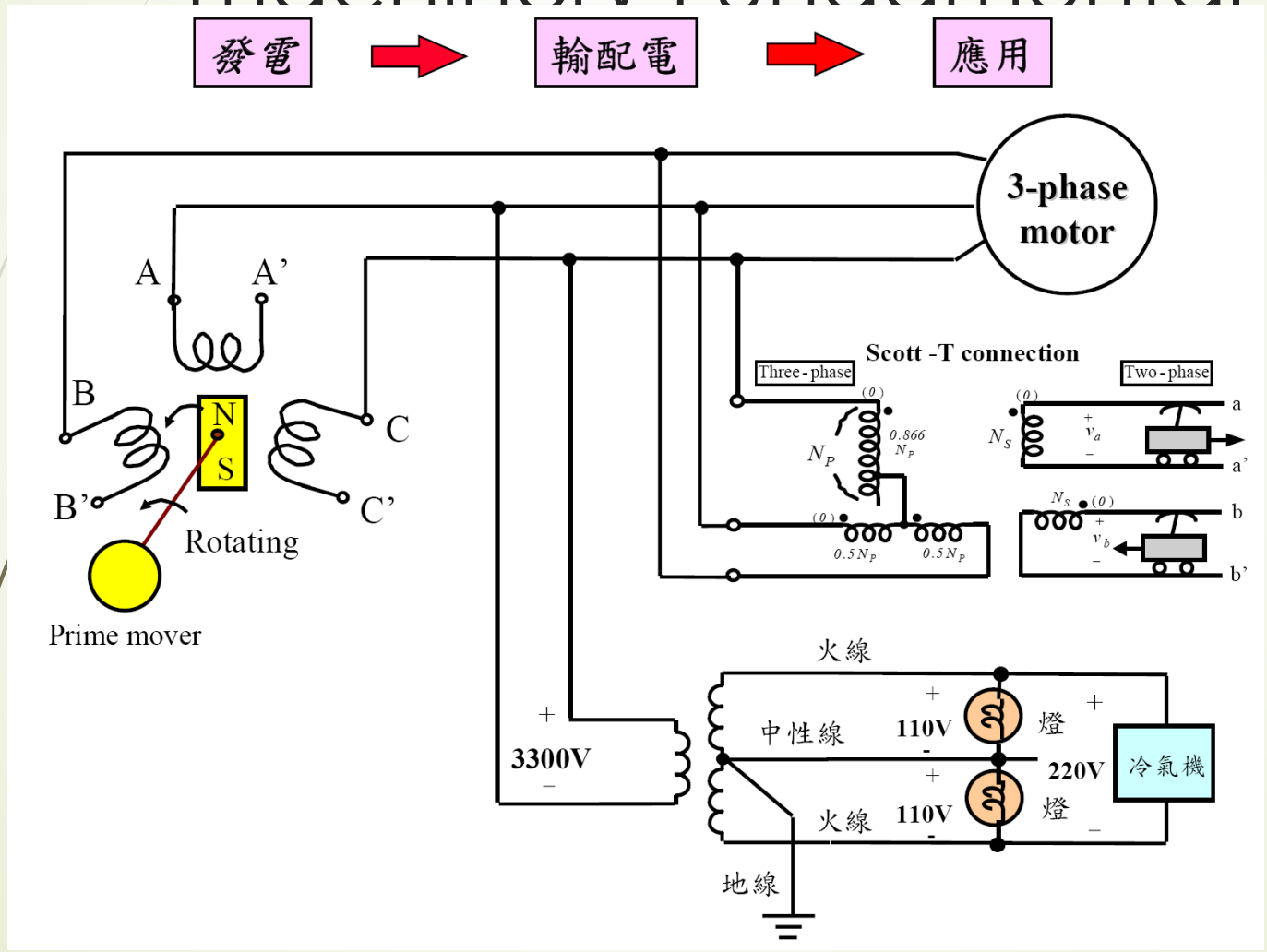




# Introduction to Electric machinery Fundamental



# Introduction to Electric machinery Fundamental

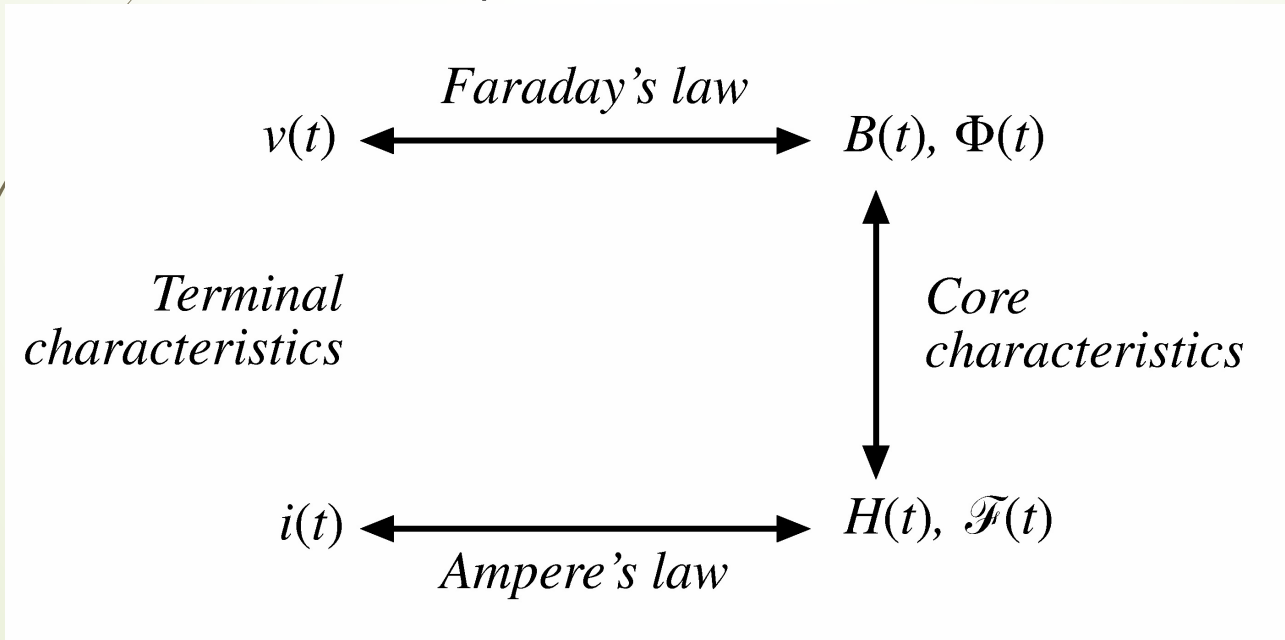


# What to learn in this course

- ***Energy Conversion*** schemes are the key ideas introduced in this course
- Which types of energy conversion are concerned?
- Electric energy to electric energy
  - **Transformer**
- Electric energy to mechanical energy
  - **Motor**
- Mechanical energy to electric energy
  - **Generator**
- Magnetic energy is essential !

# Course Outlines - Overview of relative electromagnetic theories (3wks)

- Magnetic field: Ampere's law
- Magnetic flux: magnetic material, hysteresis characteristics
- Transformer: Faraday's law, Len's law



# Course Outlines - Overview of relative electromagnetic theories (conti)

- Magnetic circuit
- Motor/generator: Induced voltage, induced force

# Course Outlines - Transformer (3wks)

- Ideal/non-ideal transformer
- Equivalent transformer circuit
- Voltage regulation, efficiency

# Course Outlines - Basic electric machine (motor/generator) theories (3wks)

- AC machine : induction machine, synchronous machine
- DC machine : separated excited, shunt excited, series excited, compound excited
- How the motor rotates ?
  - Torque/speed
- How the generator to build output voltage ?
  - Voltage/current

# Course Outline - induction (asynchronous) machine (3wks)

- Induction motor (IM) – the most widely used ac motor in the world
  - Structure and operation theories of IM
  - Equivalent circuit of IM
  - Torque/speed characteristics
  - Basic motor control
- Induction generator (seldom used)
  - Output voltage control
  - Voltage/current characteristics

# Course Outline - synchronous machine (3wks)

- Synchronous generator (SG) – the most widely used generator in the world
  - Structure and operation theories of SG
  - Equivalent circuit of SG
  - Voltage/current characteristics
  - Parallel operation
- Synchronous motor
  - Operation principles
  - Starting of synchronous motor
  - Torque/speed characteristics

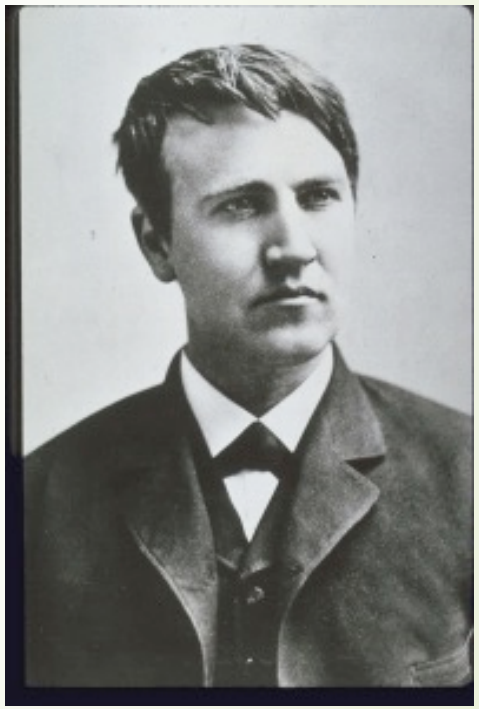
# History of Electric Machinery

DC generator, driven by steam engines



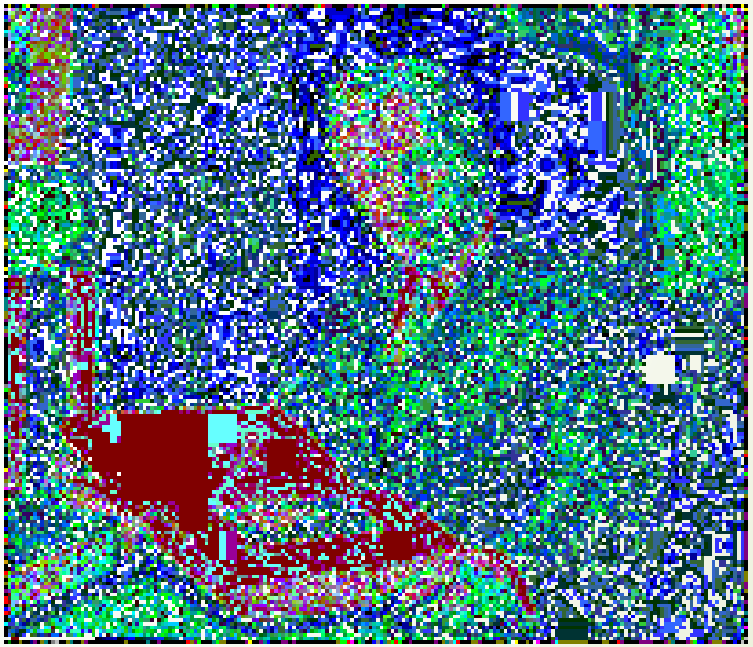
Waterwheel-driven DC generator installed in Appleton, Wisconsin

1882 Thomas A. Edison opens Pearl St. Station, NYC

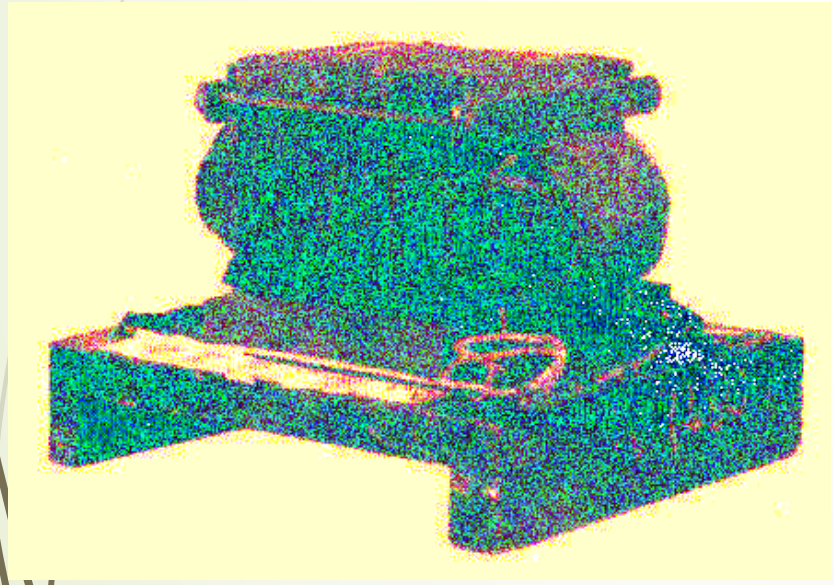


# History of Electric Machinery

**1884** **Frank J. Sprague produces DC motor for Edison systems**



# History of Electric Machinery



**1885** **William Stanley develops commercially practical transformer**



# History of Electric Machinery




**1888**

**Nikola Tesla presents paper on two-phase ac induction and synchronous motors**



# Today's development

- DC Machine
  - Motor
  - Generator
- Transformer
  - Single phase
  - Three phases
- AC Machine
  - Synchronous machine – motor, generator
  - Asynchronous machine (induction machine) – motor, generator

# Today's development and future trends

- Micro-step stepping motor
- Permanent magnet synchronous motor (PMSM)
  - Brushless dc motor (BLDCM)
- Linear motor
- Reluctance motor
  - Synchronous reluctance
  - Switched reluctance
- Ultrasonic motor
- Bionic robotics
- MEMS motor

# Course relation

- 碩/博班入學與高考科目
- It is the fundamental course of the electrical engineering
- Future courses
  - Power electronics
  - Motor control
  - Electric motor drive
  - Power systems
  - Renewable energy
  - Electrical vehicle

# Chapter 1. Introduction to machinery principles

1. Rotation motion, Newton's law and power relationships
2. The magnetic field
3. Faraday's law
4. Produce an induced force on a wire
5. Produce an induced voltage on a conductor
6. Linear dc machine examples
7. Real, reactive and apparatus power in AC circuits

# Rotation motion, Newton's law and power relationships

- Clockwise (CW) and Counterclockwise (CCW)
  - CCW is assumed as the positive direction, CW is assumed as the negative direction.
- Linear and rotation motion
  - Position and angular
    - (meter) (degree or radian)
  - Speed and angular speed

$$v = \frac{dr}{dt} \qquad \omega = \frac{d\theta}{dt}$$

| | |
|---|---|
| $\omega_m$ | angular velocity expressed in radians per second |
| $f_m$ | angular velocity expressed in revolutions per second |
| $n_m$ | angular velocity expressed in revolutions per minute |

# Rotation motion, Newton's law and power relationships

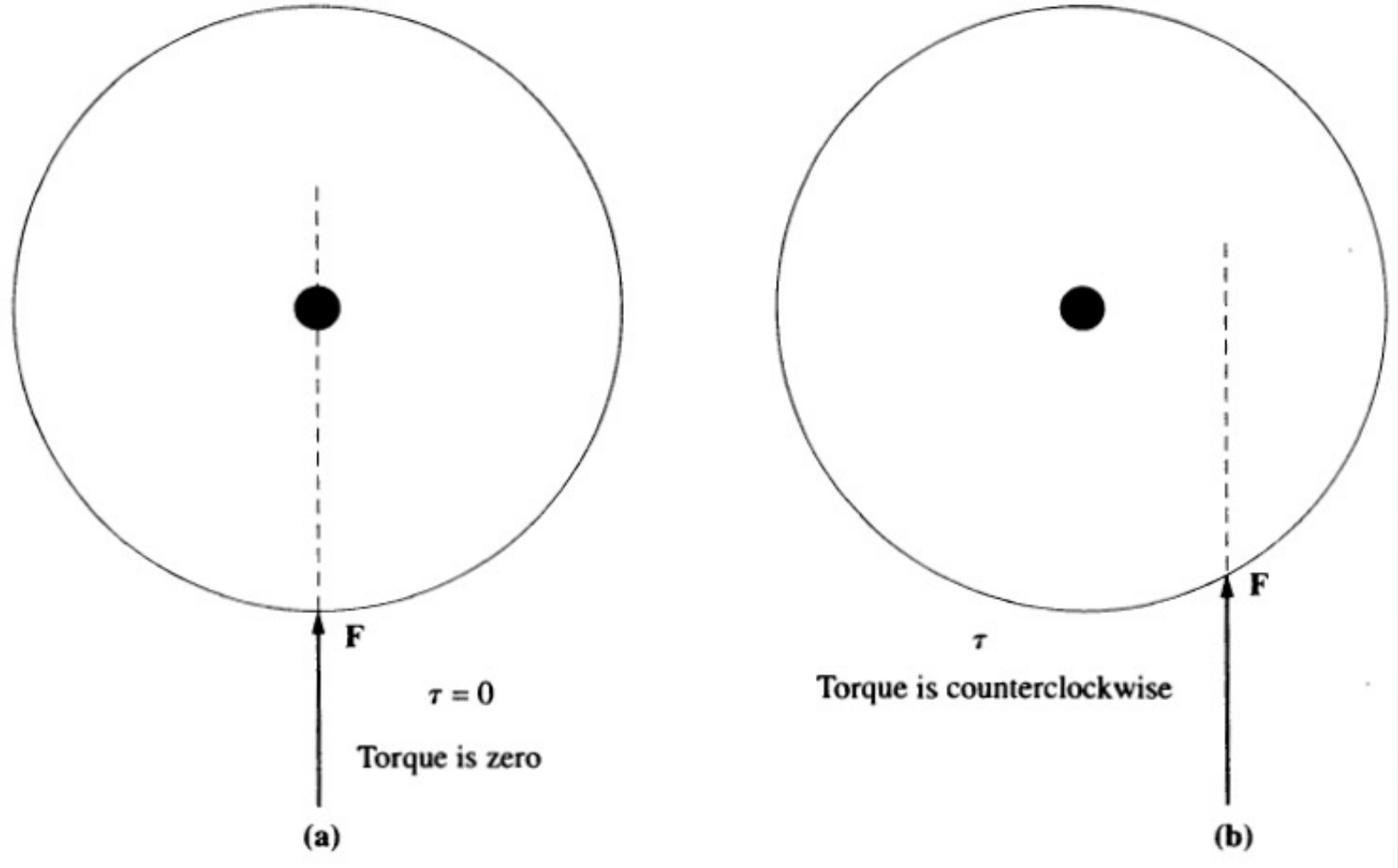
- relatic $n_m = 60 f_m$

$$f_m = \frac{\omega_m}{2\pi}$$

- Acceleration and angular acceleration

$$a = \frac{dv}{dt} \qquad \alpha = \frac{d\omega}{dt}$$

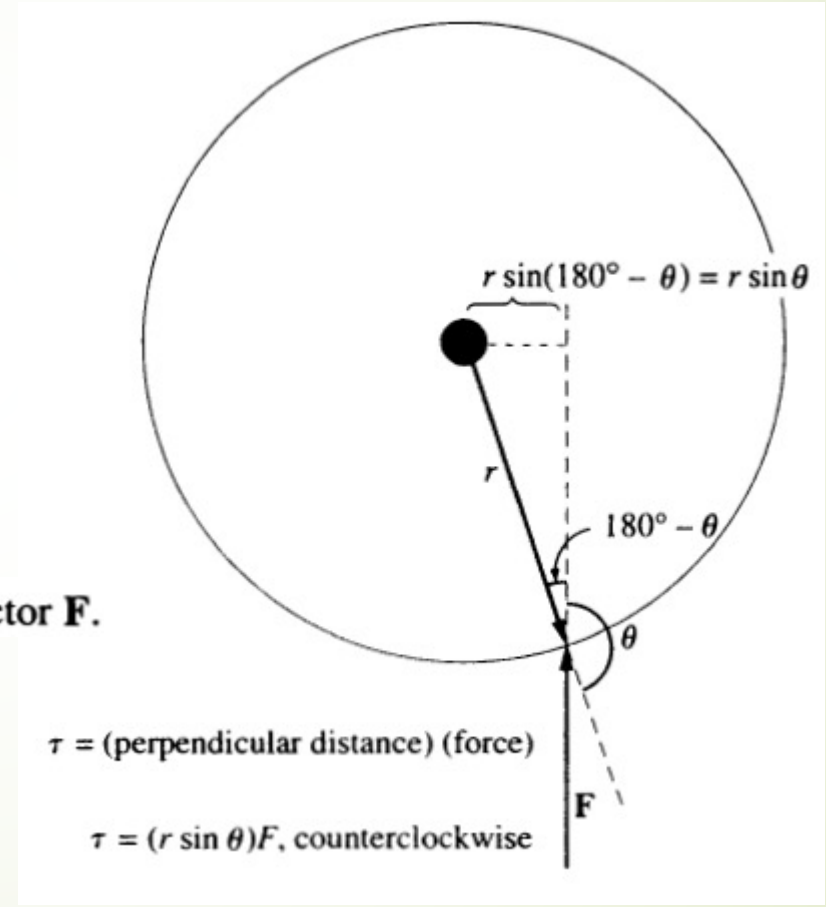
# Torque



# Torque

$$\tau = (\text{force applied})(\text{perpendicular distance})$$
$$= (F)\,(r \sin \theta)$$
$$= rF \sin \theta$$

where $\theta$ is the angle between the vector **r** and the vector **F**.



# Newton's law of rotation

1. F $F = ma$

$F$ = net force applied to an object
$m$ = mass of the object
$a$ = resulting acceleration

2. Tor $\tau = J\alpha$

# Torque and Work

For linear motion, work is defined as the application of a *force* through a *distance*. In equation form,

$$W = \int F\, dr \tag{1–9}$$

where it is assumed that the force is collinear with the direction of motion. For the special case of a constant force applied collinearly with the direction of motion, this equation becomes just

$$W = Fr \tag{1–10}$$

The units of work are joules in SI and foot-pounds in the English system.

For rotational motion, work is the application of a *torque* through an *angle*. Here the equation for work is

$$W = \int \tau\, d\theta \tag{1–11}$$

and if the torque is constant,

$$W = \tau\theta \tag{1–12}$$

# Power (rate of doing work)

Power is the rate of doing work, or the increase in work per unit time. The equation for power is

$$P = \frac{dW}{dt} \tag{1–13}$$

It is usually measured in joules per second (watts), but also can be measured in foot-pounds per second or in horsepower.

By this definition, and assuming that force is constant and collinear with the direction of motion, power is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(Fr) = F\left(\frac{dr}{dt}\right) = Fv \tag{1–14}$$

Similarly, assuming constant torque, power in rotational motion is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(\tau\theta) = \tau\left(\frac{d\theta}{dt}\right) = \tau\omega$$

$$P = \tau\omega \tag{1–15}$$

Equation (1–15) is very important in the study of electric machinery, because it can describe the mechanical power on the shaft of a motor or generator.

# Relationships of linear Motion and Revolution

| Linear Motion | | Revolution | |
|---|---|---|---|
| $r$ | *distance* | $\theta$ | *angular* |
| $v$ | $v = \frac{dr}{dt}$ | $w$ | $w = \frac{d\theta}{dt}$ |
| $a$ | $a = \frac{dv}{dt}$ | $\alpha$ | $\alpha = \frac{dw}{dt}$ |
| $f$ | *force* | $\tau$ | *torque* |
| $W$ | $W = fr$ | $W$ | $W = \tau\theta$ |
| $P$ | $P = \frac{dW}{dt} = Fv$ | $P$ | $P = \frac{d}{dt}(\tau\theta) = \tau w$ |

# Conversion between watts and horsepower

1. Watts and horsepower

$$P\,(\text{watts}) = \frac{\tau\,(\text{lb-ft})\; n\;(\text{r/min})}{7.04}$$

$$P\,(\text{horsepower}) = \frac{\tau\,(\text{lb-ft})\; n\;(\text{r/min})}{5252}$$

2. Conversion between two units

- 5252 / 7.04 = 746.02
- 1hp = 746W = 0.746kW
-

# The magnetic field

1. A current-carrying wire produces a magnetic field in the area around it.
2. A time-changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of *transformer action*.)
3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of *motor action*.)
4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of *generator action*.)

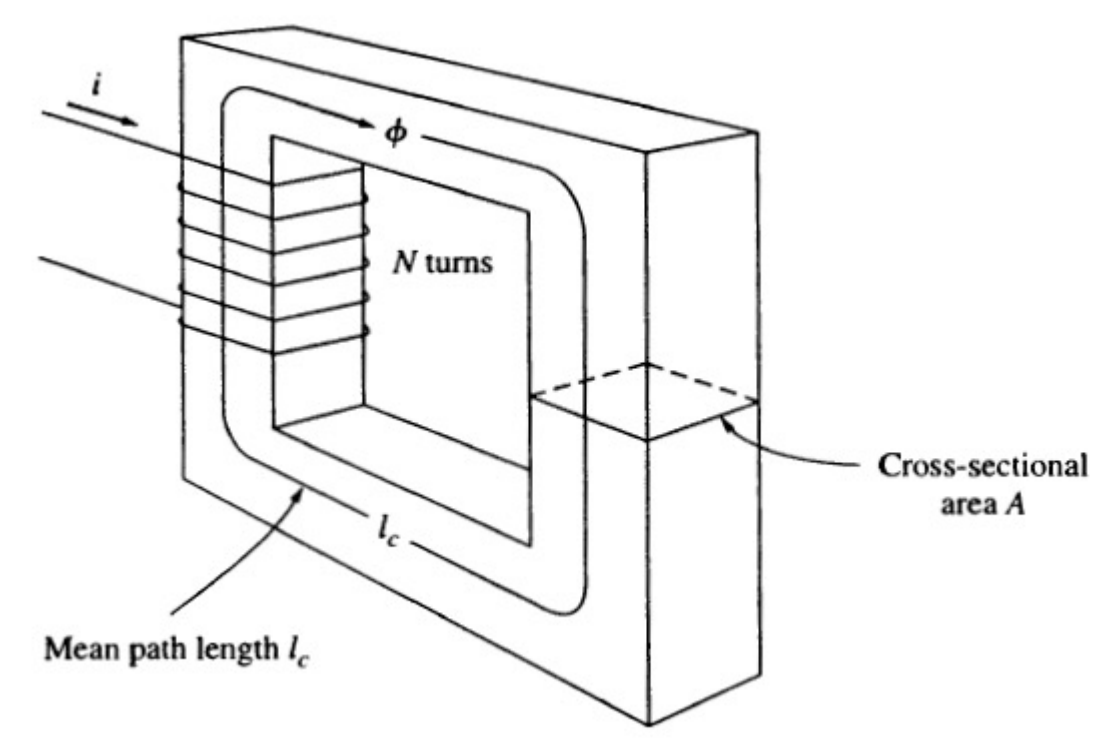
# Producing a magnetic field Ampere's law

1. The magnetic field is produced by Ampere's law
2. The core is a ferromagnetic material

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{net}}$$

$$Hl_c = Ni$$

$$H = \frac{Ni}{l_c}$$



# From the magnetic field to magnetic flux density

1. When the magnetic field is applied on a ferromagnetic material, the magnetic flux density **B** will be produced

$$\mathbf{B} = \mu\mathbf{H}$$

**H** = magnetic field intensity
$\mu$ = magnetic *permeability* of material
**B** = resulting magnetic flux density produced

The permeability of free space is called $\mu_0$, and its value is

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \qquad (1\text{–}22)$$

The permeability of any other material compared to the permeability of free space is called its *relative permeability:*

$$\mu_r = \frac{\mu}{\mu_0} \qquad (1\text{–}23)$$

# Magnetic flux density and magnetic flux

1. Magnetic flux density

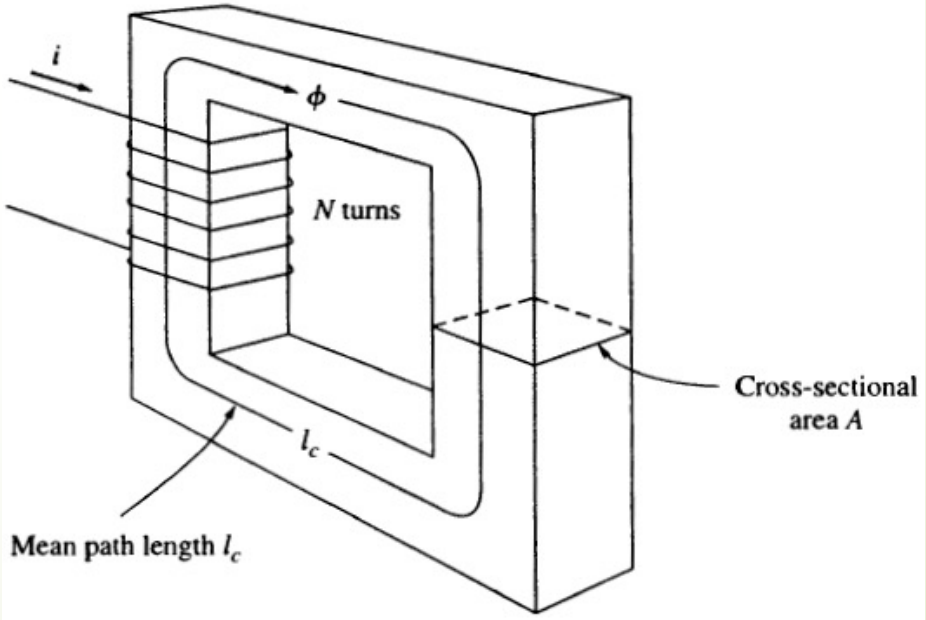
$$B = \mu H = \frac{\mu N i}{l_c}$$

2. Magnetic flux

$$\phi = \int_A \mathbf{B} \cdot d\mathbf{A}$$

$$\phi = BA$$

$$\phi = BA = \frac{\mu N i A}{l_c}$$

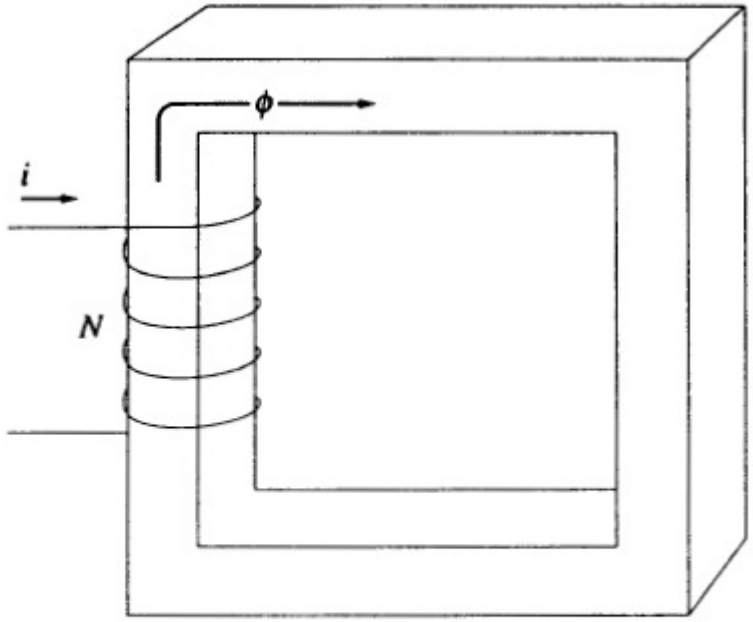


# Magnetic Circuit – magnetomotive force

$$\mathcal{F} = Ni$$

$$\mathcal{F} = \phi \mathcal{R}$$

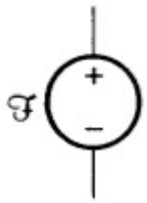





$\mathcal{F}$ = magnetomotive force of circuit

$\phi$ = flux of circuit

$\mathcal{R}$ = *reluctance* of circuit



# Magnetic circuit

1. Magnetic circuit

$$\phi = BA = \frac{\mu NiA}{l_c}$$

$$= Ni\left(\frac{\mu A}{l_c}\right)$$

$$\phi = \mathcal{F}\left(\frac{\mu A}{l_c}\right)$$

$$\mathcal{R} = \frac{l_c}{\mu A}$$

$$\mathcal{F} = \phi\,\mathcal{R}$$

# Electric circuit and magnetic circuit

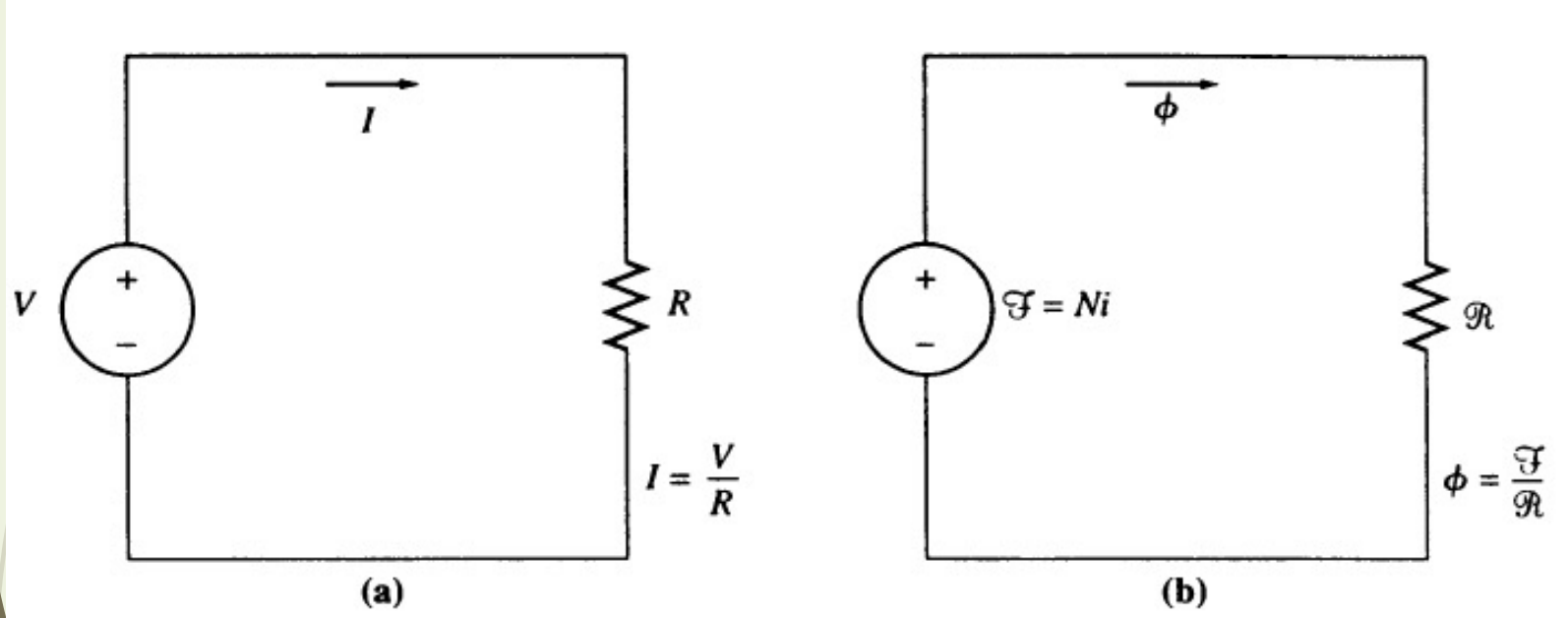


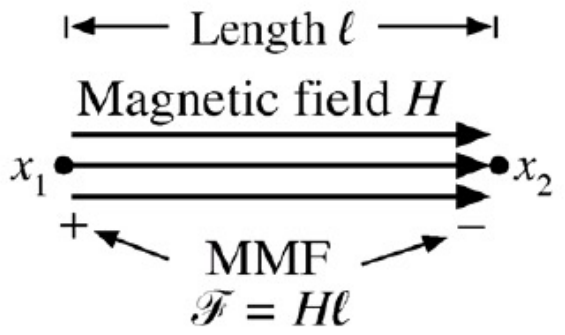

**FIGURE 1–4**
(a) A simple electric circuit. (b) The magnetic circuit analog to a transformer core.

# Magnetic field $H$ and magnetomotive force $\mathcal{F}$

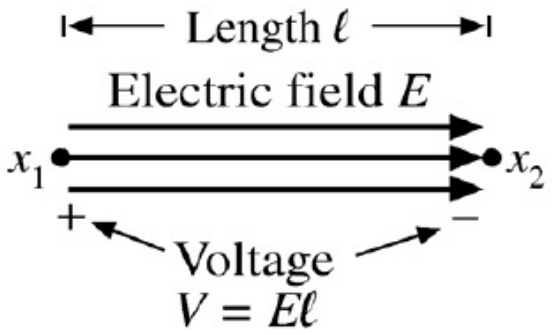
Magnetomotive force (MMF) $\mathcal{F}$ between points $x_1$ and $x_2$ is related to the magnetic field $H$ according to

$$\mathcal{F} = \int_{x_1}^{x_2} \boldsymbol{H} \cdot d\boldsymbol{\ell}$$

*Example: uniform magnetic field of magnitude $H$*




*Analogous to electric field of strength $E$, which induces voltage (EMF) $V$:*

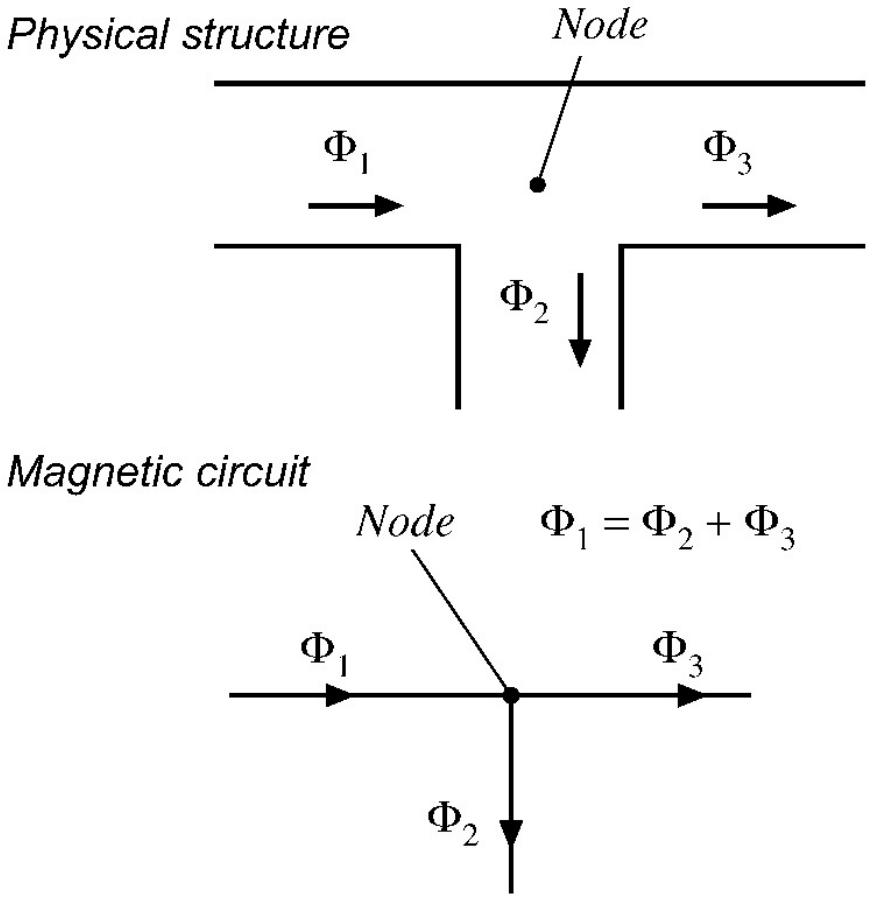


# Magnetic analog of Kirchoff's current law

Divergence of $\boldsymbol{B} = 0$

Flux lines are continuous and cannot end

Total flux entering a node must be zero



# Reluctance in magnetic circuit

1. Series connection

$$\mathcal{R}_{eq} = \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \cdots$$

2. Parallel connection

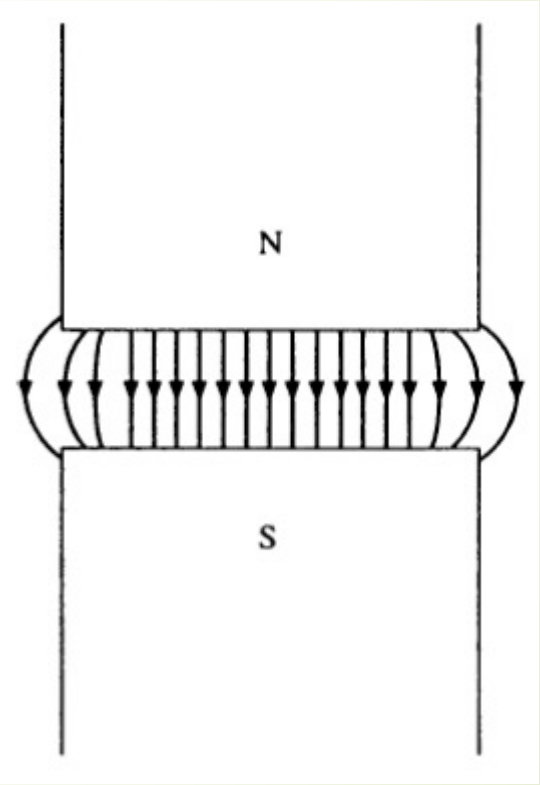
$$\frac{1}{\mathcal{R}_{eq}} = \frac{1}{\mathcal{R}_1} + \frac{1}{\mathcal{R}_2} + \frac{1}{\mathcal{R}_3} + \cdots$$

# The errors in magnetic circuit computation

1. The magnetic circuit concept assumes that all flux is confined within a magnetic core. Unfortunately, this is not quite true. The permeability of a ferromagnetic core is 2000 to 6000 times that of air, but a small fraction of the flux escapes from the core into the surrounding low-permeability air. This flux outside the core is called *leakage flux*, and it plays a very important role in electric machine design.
2. The calculation of reluctance assumes a certain mean path length and cross-sectional area for the core. These assumptions are not really very good, especially at corners.
3. In ferromagnetic materials, the permeability varies with the amount of flux already in the material. This nonlinear effect is described in detail. It adds yet another source of error to magnetic circuit analysis, since the reluctances used in magnetic circuit calculations depend on the permeability of the material.

# The errors in magnetic circuit computation

- 4. Air gap “fringing effect”



# Example 1-1

**Example 1–1.** A ferromagnetic core is shown in Figure 1–7a. Three sides of this core are of uniform width, while the fourth side is somewhat thinner. The depth of the core (into the page) is 10 cm, and the other dimensions are shown in the figure. There is a 200-turn coil wrapped around the left side of the core. Assuming relative permeability $\mu_r$ of 2500, how much flux will be produced by a 1-A input current?

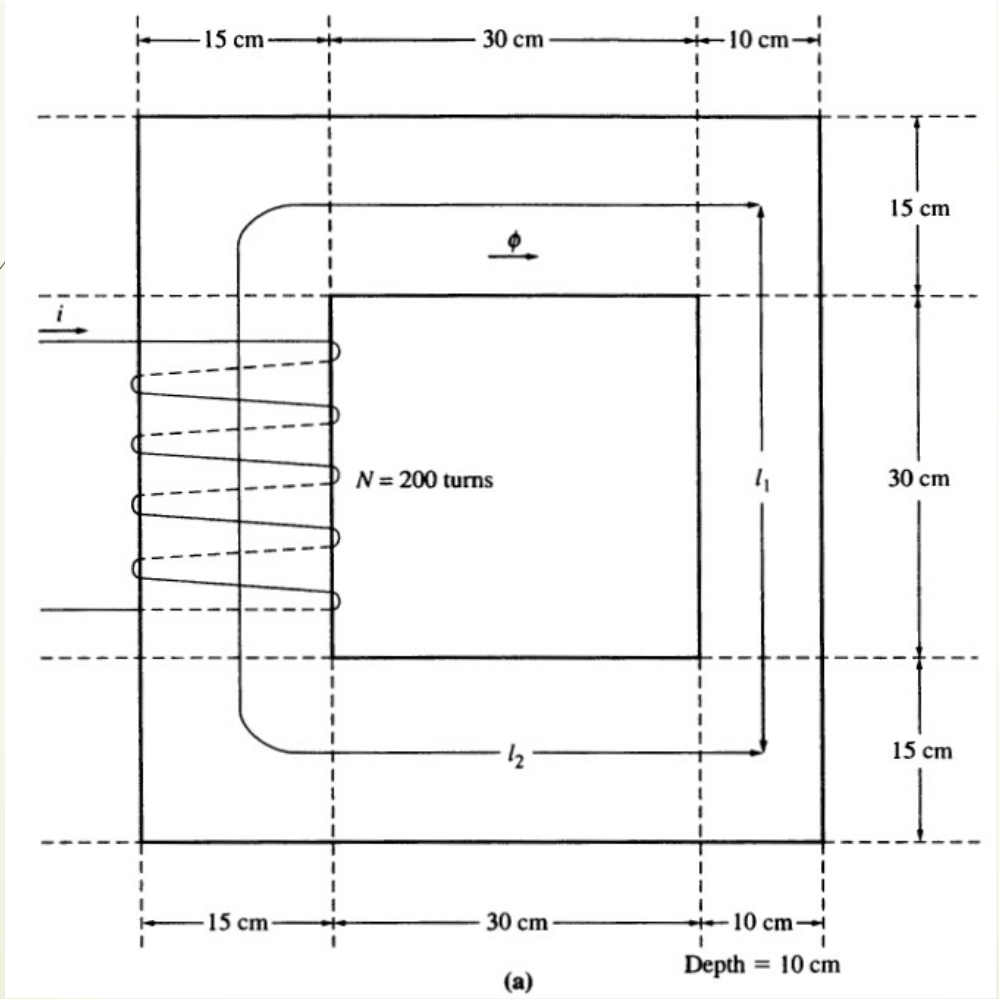



Fig. 1-7a

The mean path length of region 1 is 45 cm, and the cross-sectional area is 10 × 10 cm = 100 cm$^2$. Therefore, the reluctance in the first region is

$$\mathcal{R}_1 = \frac{l_1}{\mu A_1} = \frac{l_1}{\mu_r \mu_0 A_1} \tag{1–32}$$

$$= \frac{0.45 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.01 \text{ m}^2)}$$

$$= 14{,}300 \text{ A} \cdot \text{turns/Wb}$$

The mean path length of region 2 is 130 cm, and the cross-sectional area is 15 × 10 cm = 150 cm$^2$. Therefore, the reluctance in the second region is

$$\mathcal{R}_2 = \frac{l_2}{\mu A_2} = \frac{l_2}{\mu_r \mu_0 A_2} \tag{1–32}$$

$$= \frac{1.3 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.015 \text{ m}^2)}$$

$$= 27{,}600 \text{ A} \cdot \text{turns/Wb}$$

Therefore, the total reluctance in the core is

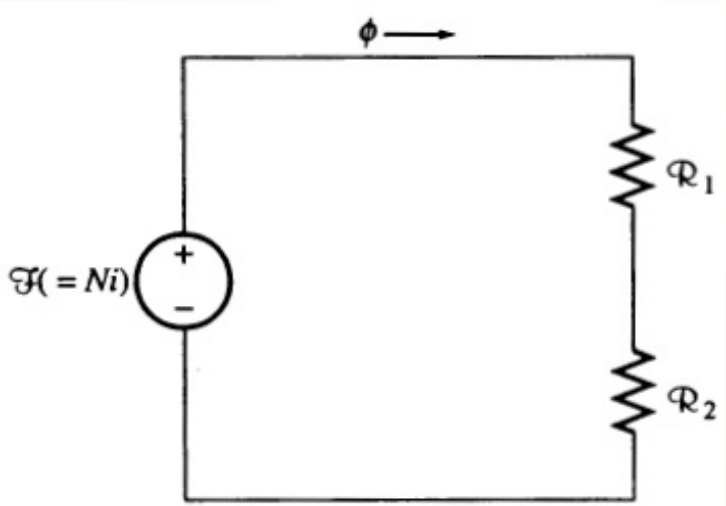
$$\begin{aligned}\mathcal{R}_{eq} &= \mathcal{R}_1 + \mathcal{R}_2 \\ &= 14{,}300 \text{ A} \cdot \text{turns/Wb} + 27{,}600 \text{ A} \cdot \text{turns/Wb} \\ &= 41{,}900 \text{ A} \cdot \text{turns/Wb}\end{aligned}$$

The total magnetomotive force is

$$\mathcal{F} = Ni = (200 \text{ turns})(1.0 \text{ A}) = 200 \text{ A} \cdot \text{turns}$$

The total flux in the core is given by

$$\begin{aligned}\phi &= \frac{\mathcal{F}}{\mathcal{R}} = \frac{200 \text{ A} \cdot \text{turns}}{41{,}900 \text{ A} \cdot \text{turns/ Wb}} \\ &= 0.0048 \text{ Wb}\end{aligned}$$




(b)

# MATLAB Programs

```
% M-file: ex1_1.m
% M-file to calculate the flux in Example 1-1.
l1 = 0.45;                          % Length of region 1
l2 = 1.3;                           % Length of region 2
a1 = 0.01;                          % Area of region 1
a2 = 0.015;                         % Area of region 2
ur = 2500;                          % Relative permeability
u0 = 4*pi*1E-7;                     % Permeability of free space
n = 200;                            % Number of turns on core
i = 1;                              % Current in amps

% Calculate the first reluctance
r1 = l1 / (ur * u0 * a1);
disp (['r1 = ' num2str(r1)]);

% Calculate the second reluctance
r2 = l2 / (ur * u0 * a2);
disp (['r2 = ' num2str(r2)]);
```

```
% Calculate the total reluctance
rtot = r1 + r2;

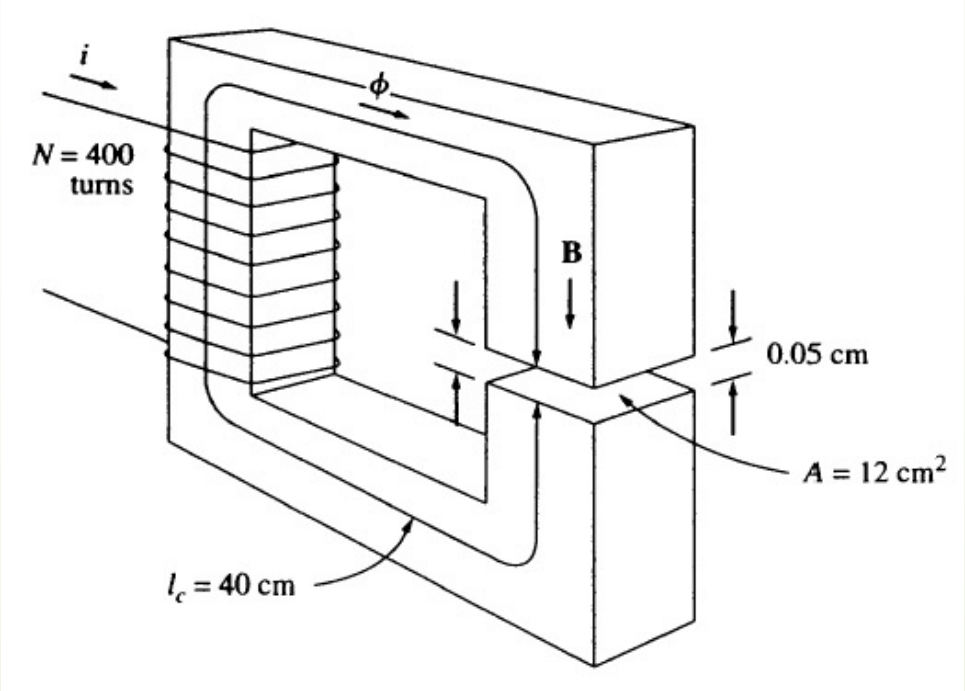
% Calculate the mmf
mmf = n * i;

% Finally, get the flux in the core
flux = mmf / rtot;

% Display result
disp (['Flux = ' num2str(flux)]);
```

# Example 1-2

**Example 1–2.** Figure 1–8a shows a ferromagnetic core whose mean path length is 40 cm. There is a small gap of 0.05 cm in the structure of the otherwise whole core. The cross-sectional area of the core is 12 $cm^2$, the relative permeability of the core is 4000, and the coil of wire on the core has 400 turns. Assume that fringing in the air gap increases the effective cross-sectional area of the air gap by 5 percent. Given this information, find (a) the total reluctance of the flux path (iron plus air gap) and (b) the current required to produce a flux density of 0.5 T in the air gap.



# Solution

***Solution***

The magnetic circuit corresponding to this core is shown in Figure 1–8b.

*(a)* The reluctance of the core is

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_r \mu_0 A_c} \qquad (1\text{–}32)$$

$$= \frac{0.4 \text{ m}}{(4000)(4\pi \times 10^{-7})(0.002 \text{ m}^2)}$$

$A_c$=0.0012 $m^2$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb}$$

$R_c$=66,315 A.turns/Wb

The effective area of the air gap is $1.05 \times 12 \text{ cm}^2 = 12.6 \text{ cm}^2$, so the reluctance of the air gap is

$$\mathcal{R}_a = \frac{l_a}{\mu_0 A_a} \qquad (1\text{–}32)$$

$$= \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7})(0.00126 \text{ m}^2)}$$

$$= 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

# Solution

(a) The ferromagnetic core of Example 1–2. (b) The magnetic circuit corresponding to (a).

Therefore, the total reluctance of the flux path is

$$\mathcal{R}_{eq} = \mathcal{R}_c + \mathcal{R}_a$$
$$= 66{,}300 \text{ A} \cdot \text{turns/Wb} + 316{,}000 \text{ A} \cdot \text{turns/Wb}$$
$$= 382{,}300 \text{ A} \cdot \text{turns/Wb}$$

$R_{eq}$=382,315A.turns/Wb

Note that the air gap contributes most of the reluctance even though it is 800 times shorter than the core.

*(b)* Equation (1–28) states that

$$\mathcal{F} = \phi\mathcal{R} \qquad (1\text{–}28)$$

Since the flux $\phi = BA$ and $\mathcal{F} = Ni$, this equation becomes

$$Ni = BA\mathcal{R}$$

# Solution

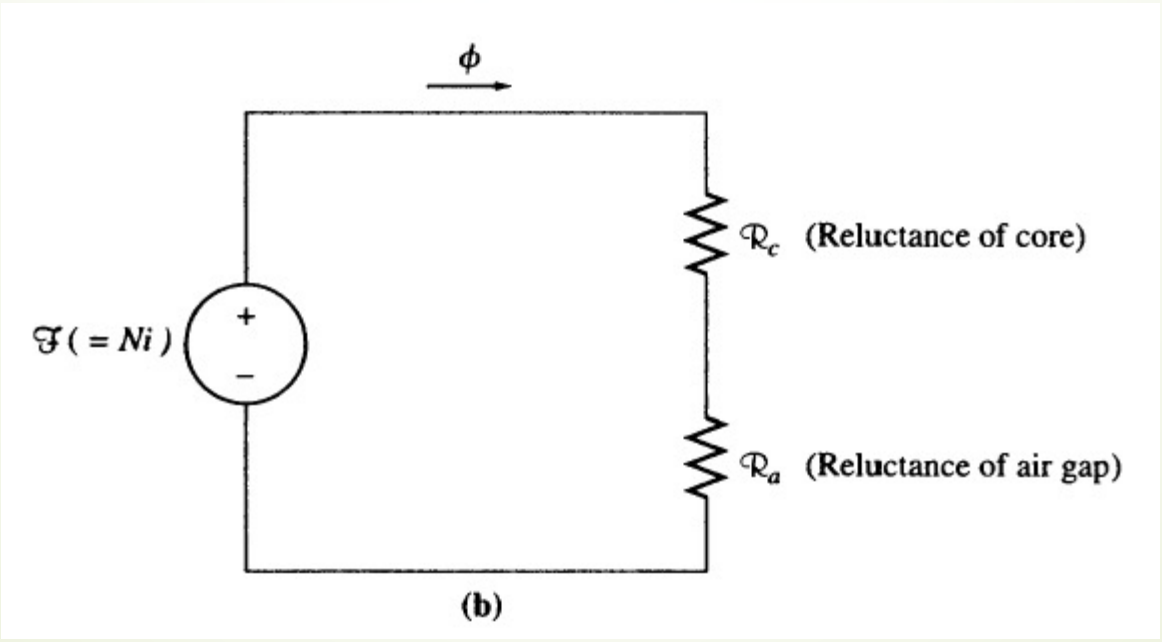
so

$$i = \frac{BA\mathcal{R}}{N}$$

$$= \frac{(0.5\ \text{T})(0.00126\ \text{m}^2)(383{,}200\ \text{A} \cdot \text{turns/ Wb})}{400\ \text{turns}}$$

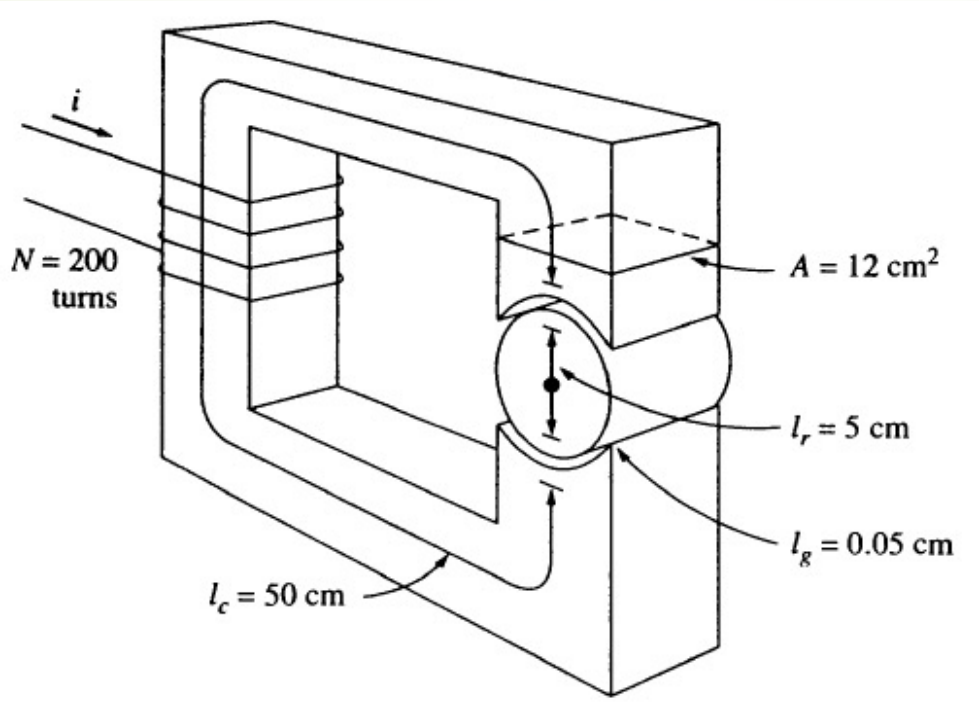
$$= 0.602\ \text{A}$$ i=0.6037A

Notice that, since the *air-gap* flux was required, the effective air-gap area was used in the above equation.



# Example 1-3

**Example 1–3.** Figure 1–9a shows a simplified rotor and stator for a dc motor. The mean path length of the stator is 50 cm, and its cross-sectional area is 12 $cm^2$. The mean path length of the rotor is 5 cm, and its cross-sectional area also may be assumed to be 12 $cm^2$. Each air gap between the rotor and the stator is 0.05 cm wide, and the cross-sectional area of each air gap (including fringing) is 14 $cm^2$. The iron of the core has a relative permeability of 2000, and there are 200 turns of wire on the core. If the current in the wire is adjusted to be 1 A, what will the resulting flux density in the air gaps be?



# Solution 1-13

***Solution***

To determine the flux density in the air gap, it is necessary to first calculate the magnetomotive force applied to the core and the total reluctance of the flux path. With this information, the total flux in the core can be found. Finally, knowing the cross-sectional area of the air gaps enables the flux density to be calculated.

The reluctance of the stator is

$$\mathcal{R}_s = \frac{l_s}{\mu_r \mu_0 A_s}$$

$$= \frac{0.5 \text{ m}}{(2000)(4\pi \times 10^{-7})(0.0012 \text{ m}^2)}$$

$$= 166{,}000 \text{ A} \cdot \text{turns/Wb}$$

The reluctance of the rotor is

$$\mathcal{R}_r = \frac{l_r}{\mu_r \mu_0 A_r}$$

$$= \frac{0.05 \text{ m}}{(2000)(4\pi \times 10^{-7})(0.0012 \text{ m}^2)}$$

$$= 16{,}600 \text{ A} \cdot \text{turns/Wb}$$

# Solution 1-13

The reluctance of the air gaps is

$$\mathcal{R}_a = \frac{l_a}{\mu_r \mu_0 A_a}$$

$$= \frac{0.0005 \text{ m}}{(1)(4\pi \times 10^{-7})(0.0014 \text{ m}^2)}$$

$$= 284{,}000 \text{ A} \cdot \text{turns/Wb}$$

The magnetic circuit corresponding to this machine is shown in Figure 1–9b. The total reluctance of the flux path is thus

$$\mathcal{R}_{eq} = \mathcal{R}_s + \mathcal{R}_{a1} + \mathcal{R}_r + \mathcal{R}_{a2}$$

$$= 166{,}000 + 284{,}000 + 16{,}600 + 284{,}000 \text{ A} \cdot \text{turns/Wb}$$

$$= 751{,}000 \text{ A} \cdot \text{turns/Wb}$$

The net magnetomotive force applied to the core is

$$\mathcal{F} = Ni = (200 \text{ turns})(1.0 \text{ A}) = 200 \text{ A} \cdot \text{turns}$$

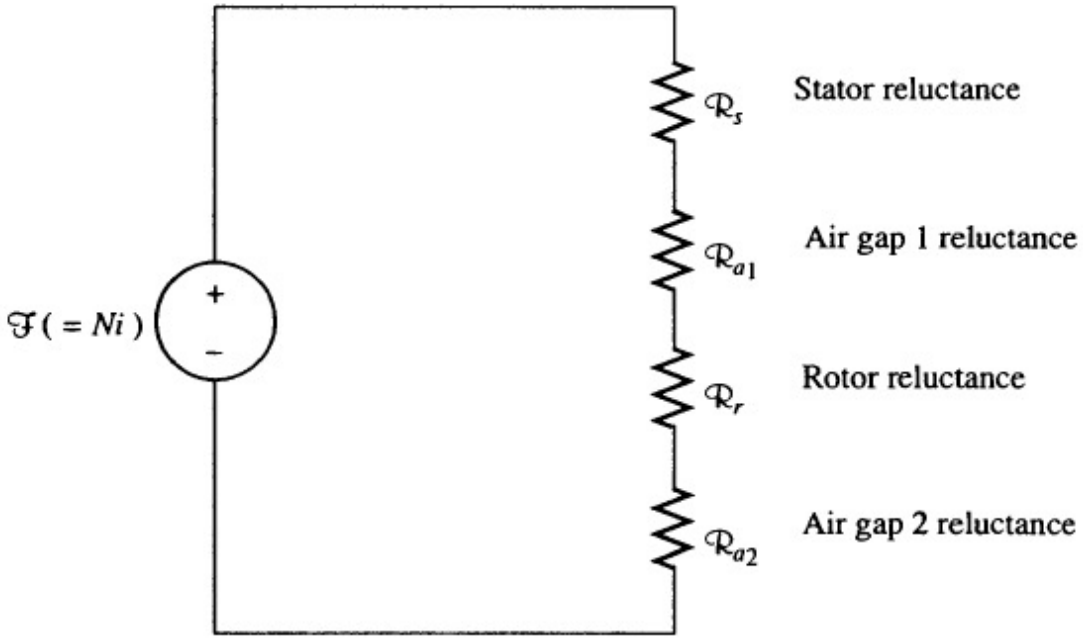
Therefore, the total flux in the core is

# Solution 1-13

$$\phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{200 \text{ A} \cdot \text{turns}}{751{,}000 \text{ A} \cdot \text{turns/Wb}}$$
$$= 0.00266 \text{ Wb}$$

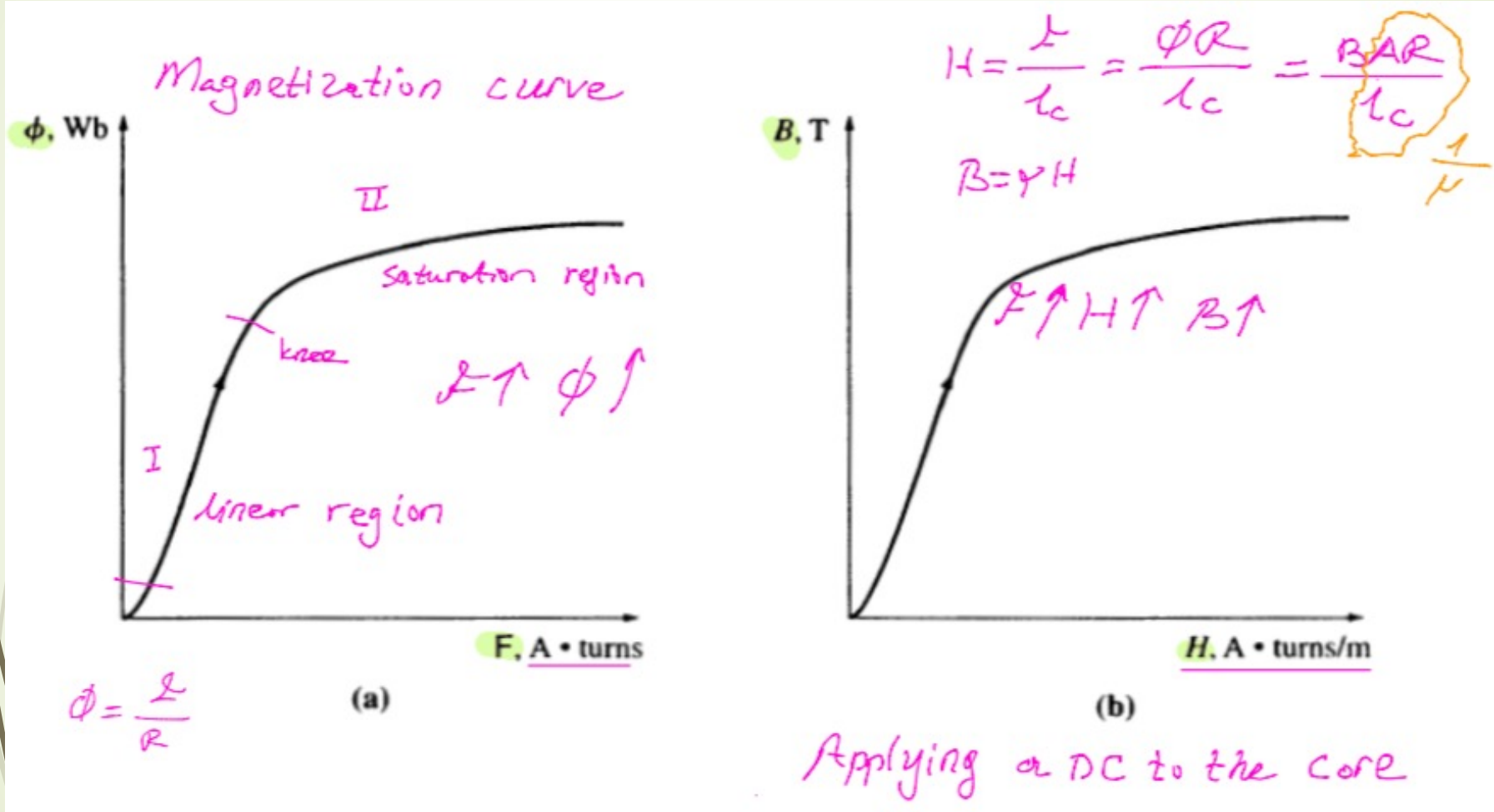
Finally, the magnetic flux density in the motor's air gap is

$$B = \frac{\phi}{A} = \frac{0.000266 \text{ Wb}}{0.0014 \text{ m}^2} = 0.19 \text{ T}$$

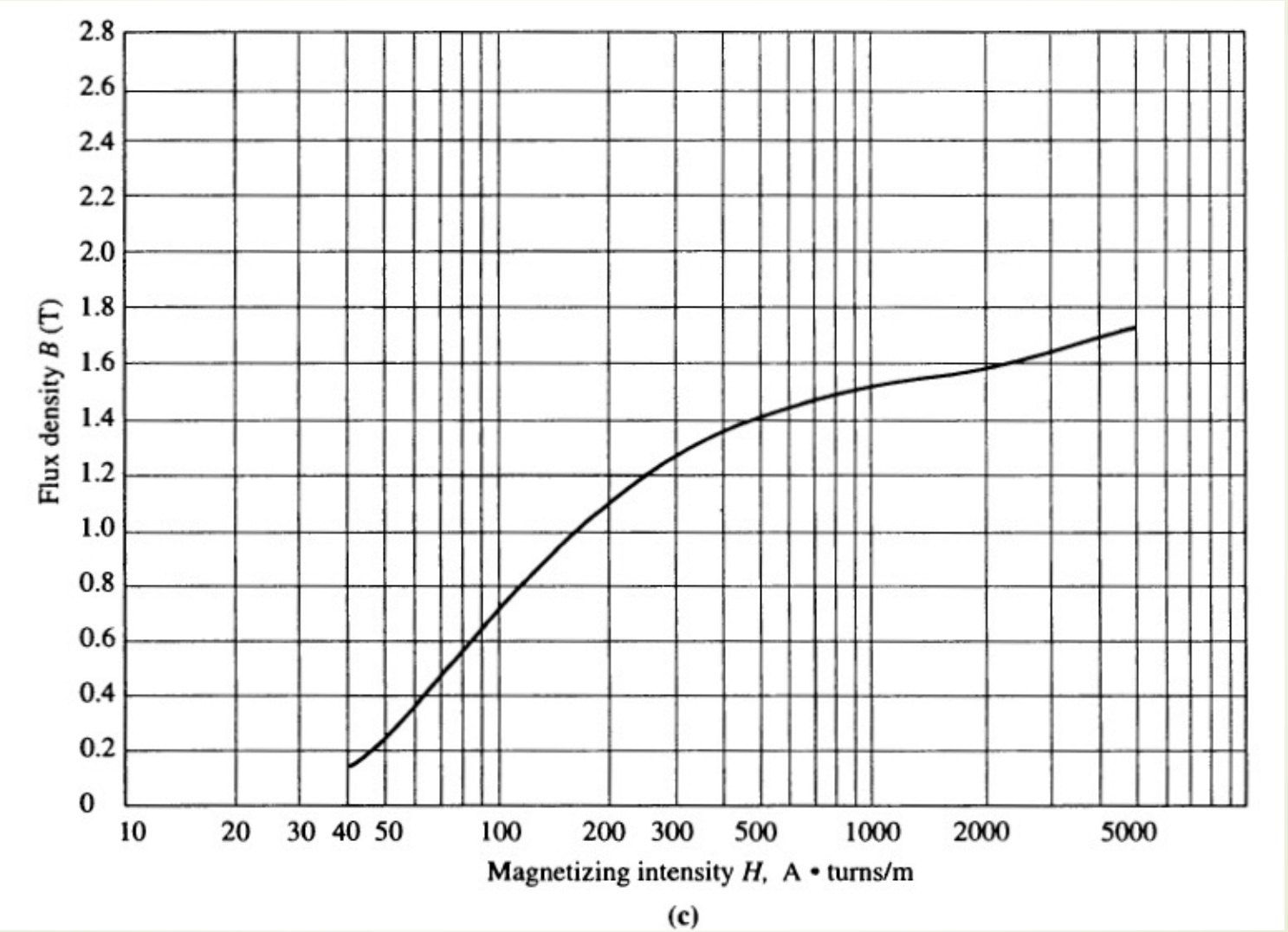



(b)

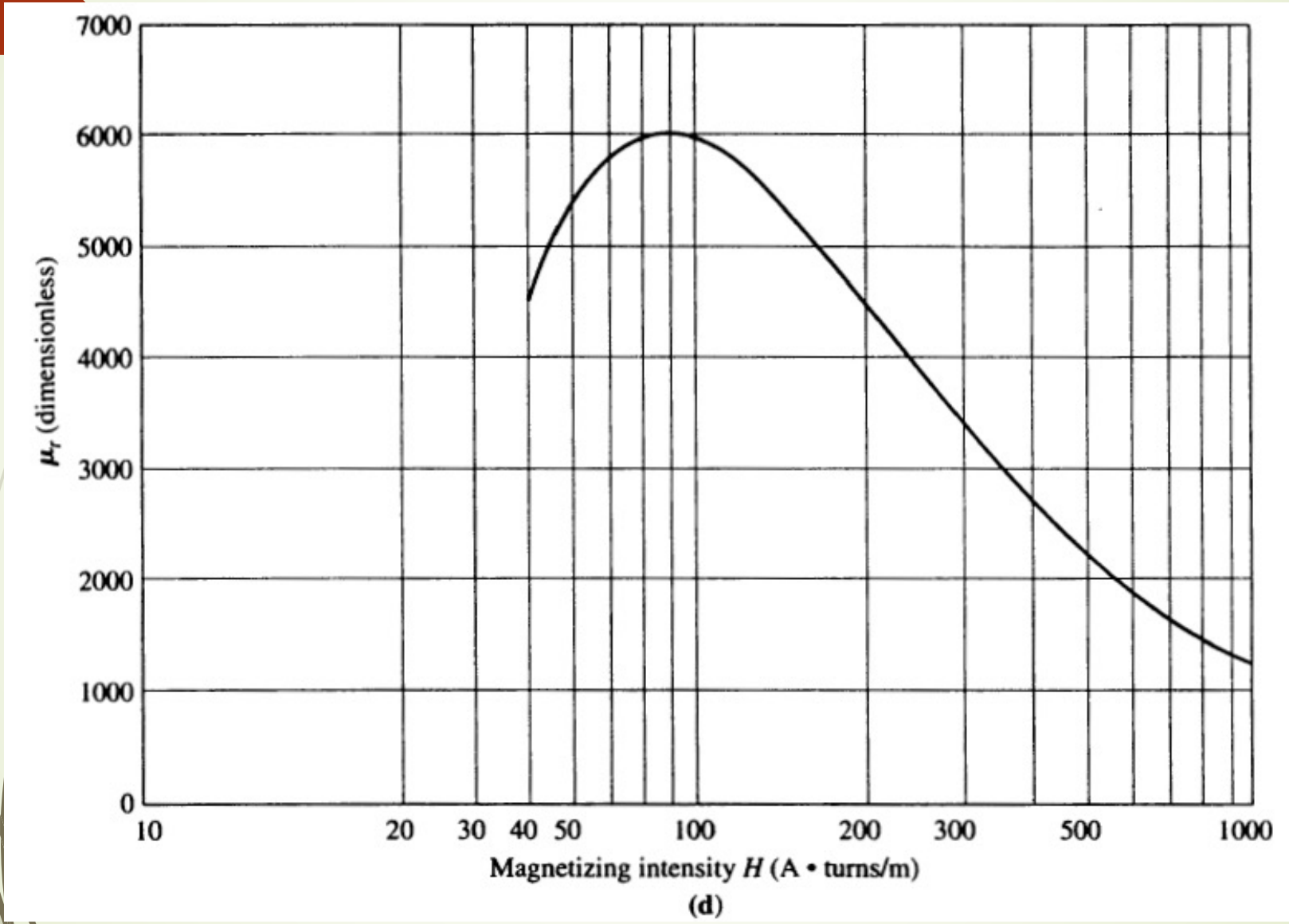
# Magnetic behavior of ferromagnetic material - Saturation



# Magnetic curve for a typical steel



# A plot of relative permeability $\mu_r$



# Example 1-5

**Example 1–5.** A square magnetic core has a mean path length of 55 cm and a cross-sectional area of 150 $cm^2$. A 200-turn coil of wire is wrapped around one leg of the core. The core is made of a material having the magnetization curve shown in Figure 1–10c.

*(a)* How much current is required to produce 0.012 Wb of flux in the core?
*(b)* What is the core's relative permeability at that current level?
*(c)* What is its reluctance?

***Solution***

*(a)* The required flux density in the core is

$$B = \frac{\phi}{A} = \frac{0.012 \text{ Wb}}{0.015 \text{ m}^2} = 0.8 \text{ T}$$

From Figure 1–10c, the required magnetizing intensity is

$$H = 115 \text{ A} \cdot \text{turns/m}$$

From Equation (1–20), the magnetomotive force needed to produce this magnetizing intensity is

$$\mathcal{F} = Ni = Hl_c$$
$$= (115 \text{ A} \cdot \text{turns/m})(0.55 \text{ m}) = 63.25 \text{ A} \cdot \text{turns}$$

so the required current is

$$i = \frac{\mathcal{F}}{N} = \frac{63.25 \text{ A} \cdot \text{turns}}{200 \text{ turns}} = 0.316 \text{ A}$$

# Solution 1-15

*(b)* The core's permeability at this current is

$$\mu = \frac{B}{H} = \frac{0.8\ \text{T}}{115\ \text{A} \cdot \text{turns/m}} = 0.00696\ \text{H/m}$$

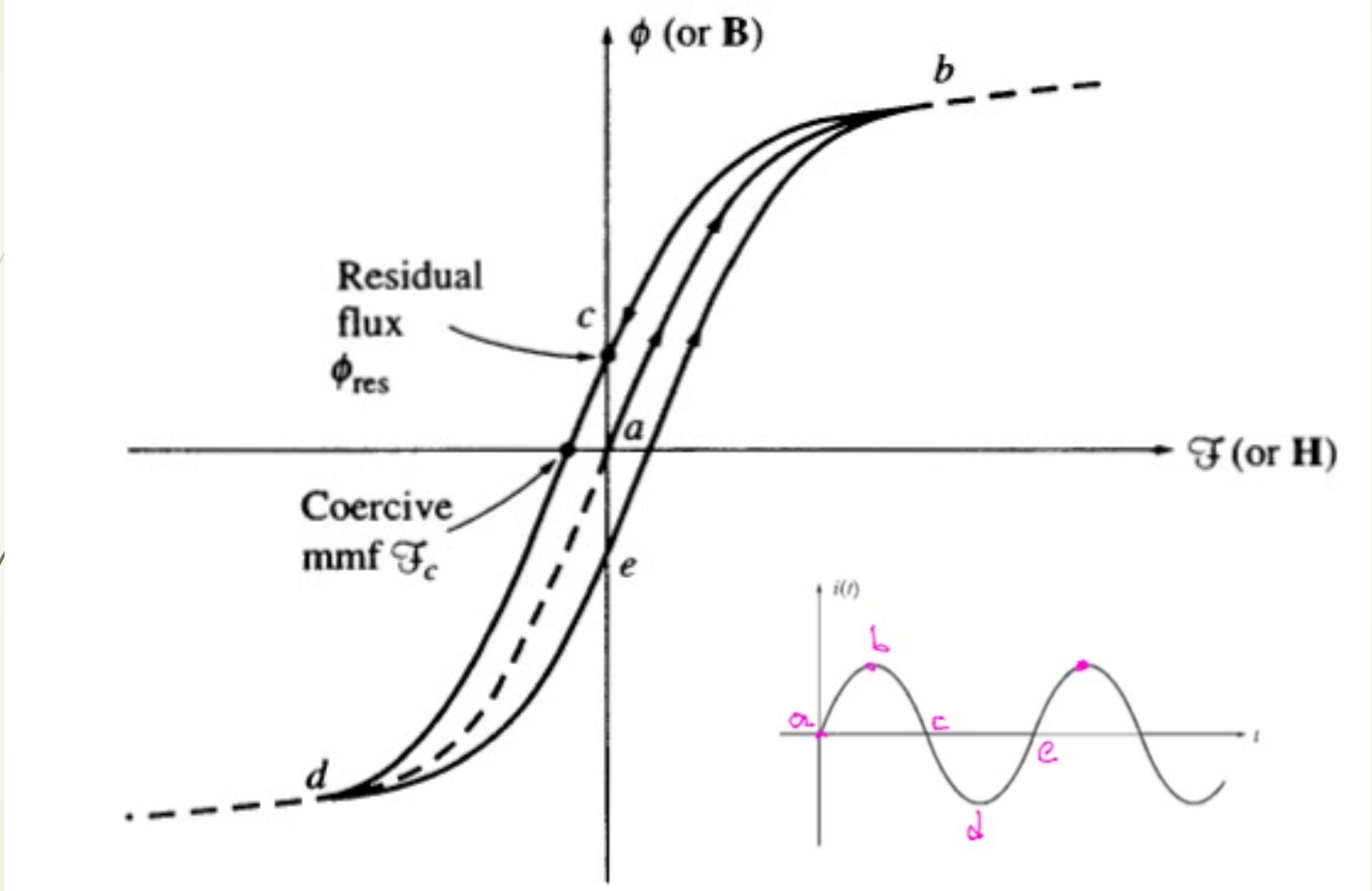
Therefore, the relative permeability is

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.00696\ \text{H/m}}{4\pi \times 10^{-7}\ \text{H/m}} = 5540$$

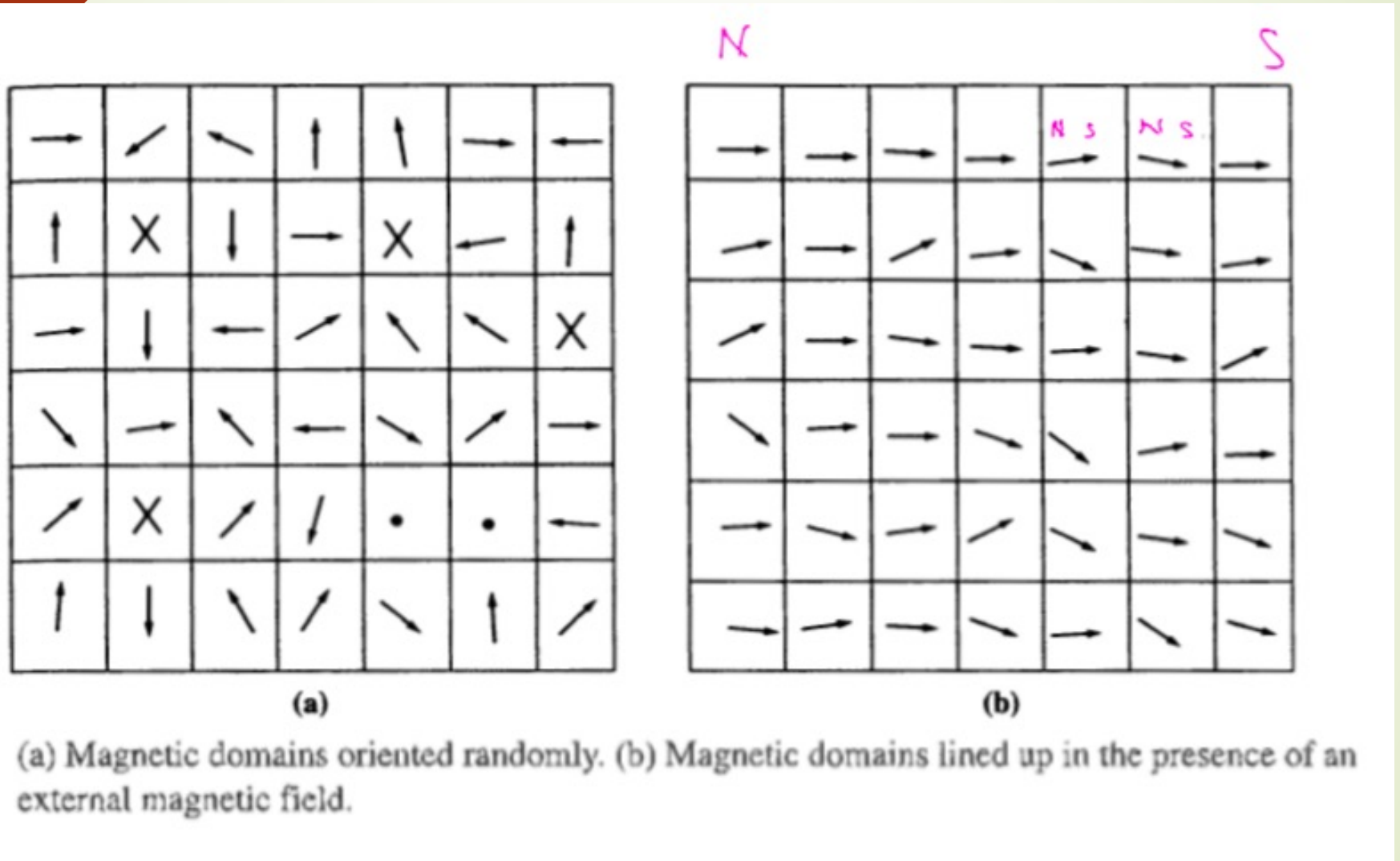
*(c)* The reluctance of the core is

$$\mathcal{R} = \frac{\mathcal{F}}{\phi} = \frac{63.25\ \text{A} \cdot \text{turns}}{0.012\ \text{Wb}} = 5270\ \text{A} \cdot \text{turns/Wb}$$

# Energy loss in ferromagnetic core – hysteresis loss

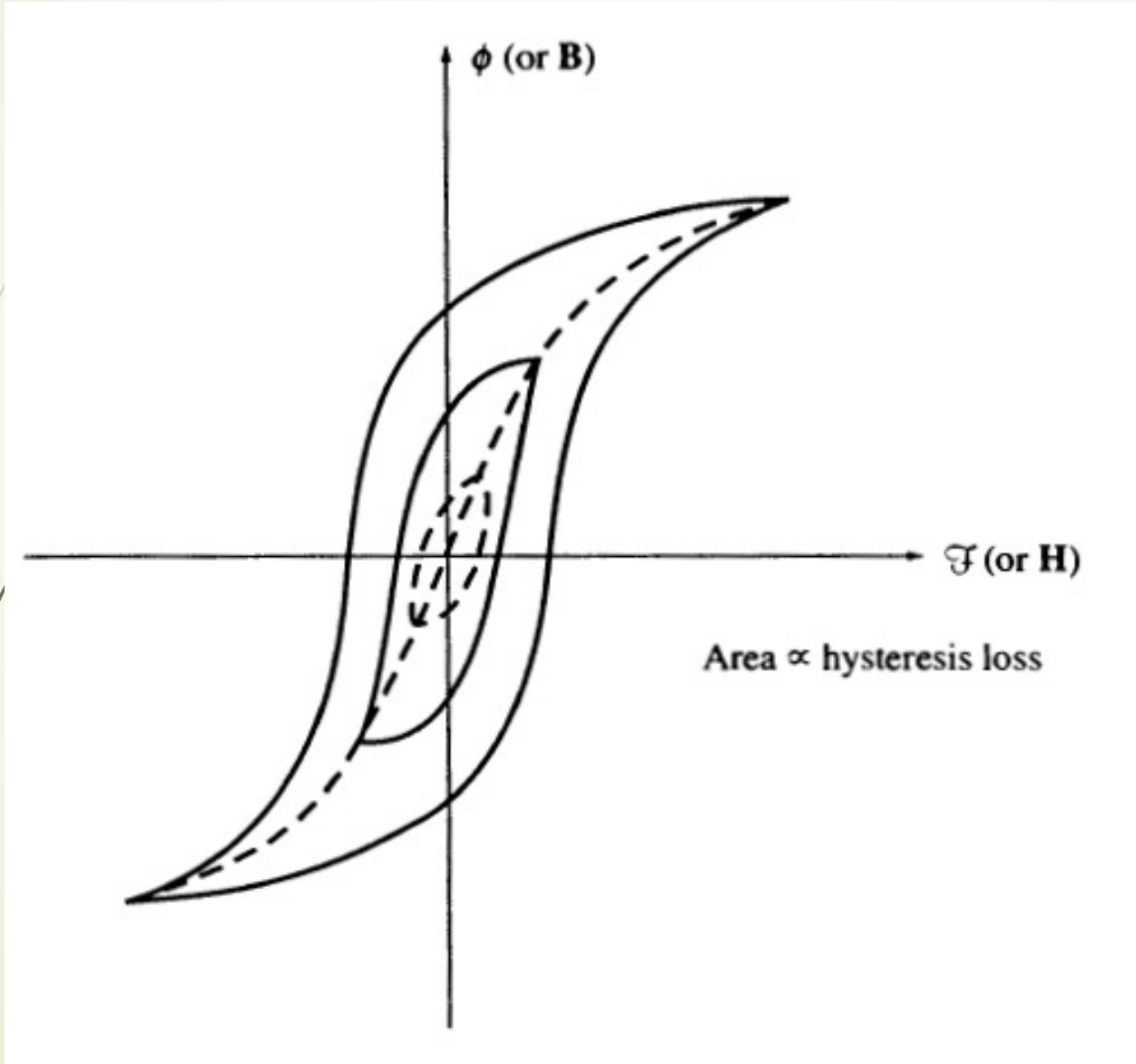


# Hysteresis loop – residual flux

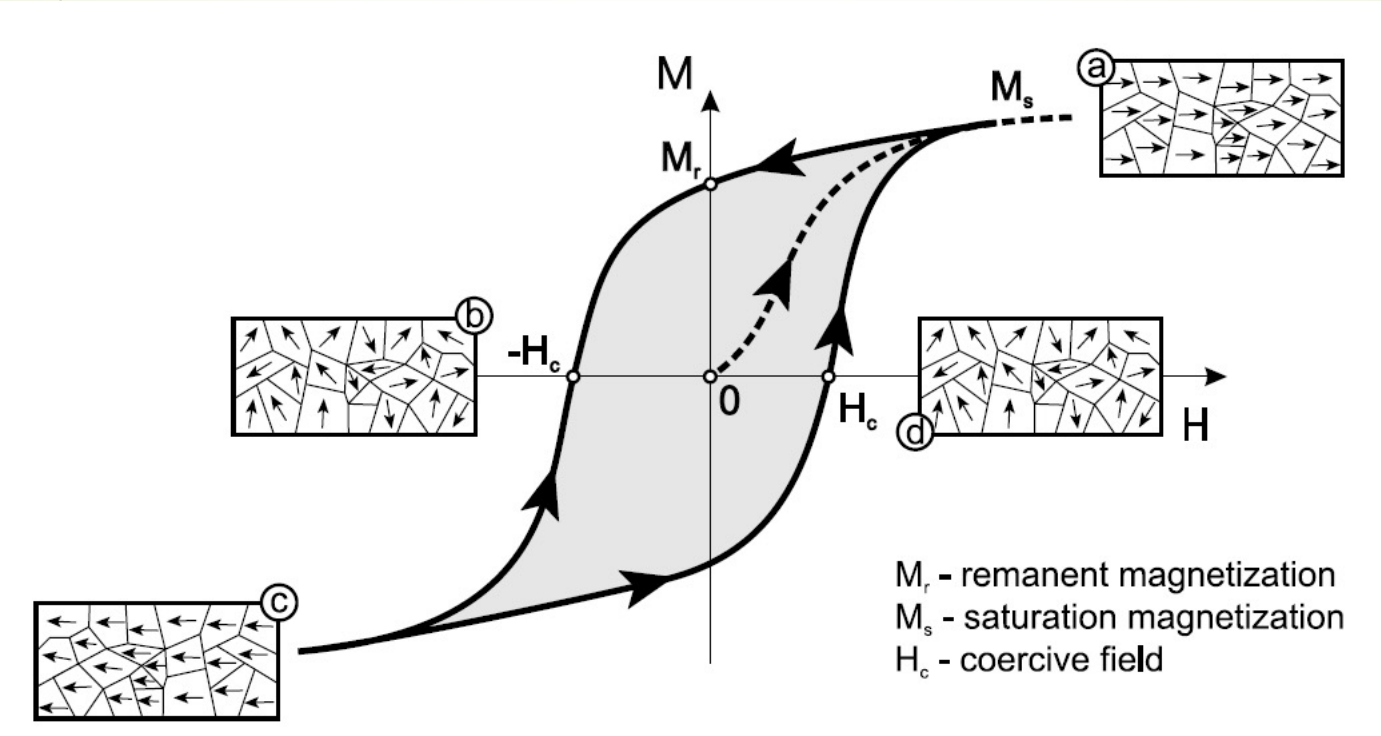



(a) Magnetic domains oriented randomly. (b) Magnetic domains lined up in the presence of an external magnetic field.

# The effect of magnetomotive force on the hysteresis loop



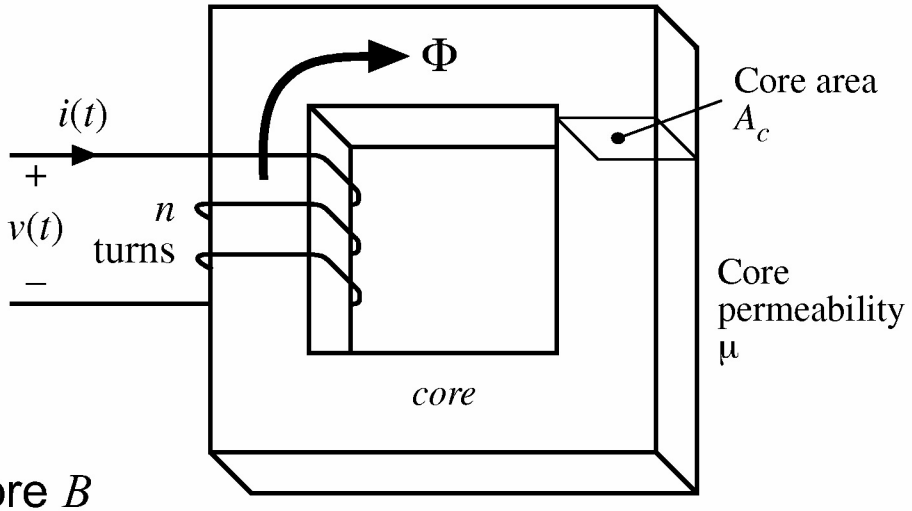
# Magnetization curve



# Hysteresis loss

Energy per cycle $W$ flowing into $n$-turn winding of an inductor, excited by periodic waveforms of frequency $f$:

$$W = \int_{\text{one cycle}} v(t)i(t)\,dt$$




Relate winding voltage and current to core $B$ and $H$ via Faraday's law and Ampere's law:

$$v(t) = nA_c \frac{dB(t)}{dt} \qquad H(t)\ell_m = ni(t)$$

Substitute into integral:

$$W = \int_{\text{one cycle}} \left(nA_c \frac{dB(t)}{dt}\right)\left(\frac{H(t)\ell_m}{n}\right)dt$$

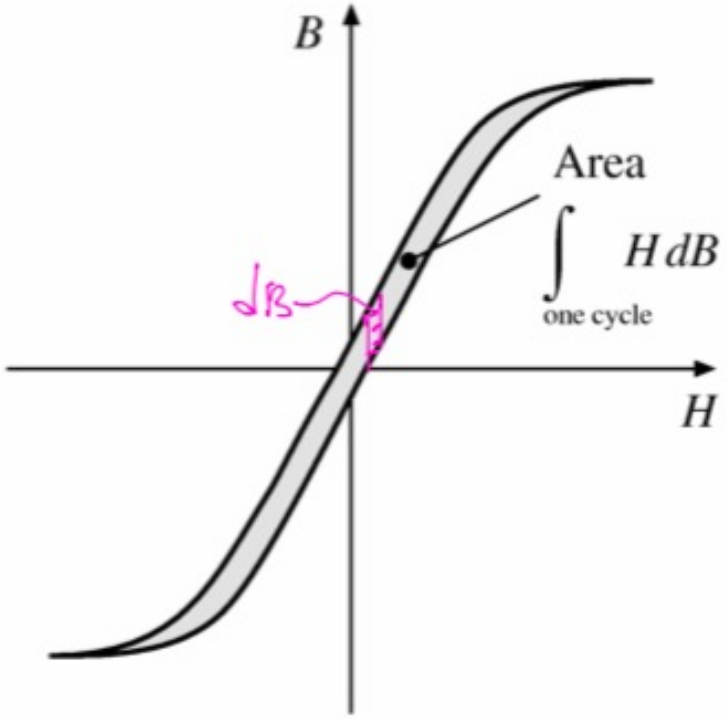
$$= \left(A_c\ell_m\right)\int_{\text{one cycle}} H\,dB$$

# Hysteresis loss

$$W = (A_c \ell_m) \int_{\text{one cycle}} H\,dB$$

Volume of the core

The term $A_c \ell_m$ is the volume of the core, while the integral is the area of the $B$–$H$ loop.




(energy lost per cycle) = (core volume) (area of $B$–$H$ loop)

$P = \frac{W}{t}$

$$P_H = (f)(A_c \ell_m) \int_{\text{one cycle}} H\,dB$$

Hysteresis loss is directly proportional to applied frequency

# Unit conversions

## Units

*Table 12.1. Units for magnetic quantities*

| *quantity* | *MKS* | *unrationalized cgs* | *conversions* |
|---|---|---|---|
| core material equation | $B = \mu_0 \mu_r H$ | $B = \mu_r H$ | |
| $B$ | Tesla | Gauss | $1\text{T} = 10^4\text{G}$ |
| $H$ | Ampere / meter | Oersted | $1\text{A/m} = 4\pi \cdot 10^{-3}$ Oe |
| $\Phi$ | Weber | Maxwell | $1\text{Wb} = 10^8$ Mx<br>$1\text{T} = 1\text{Wb} / \text{m}^2$ |

# Faraday's law – induce voltage from a time-varying magnetic field

1. Induced voltage magnitude and polarity

$$e_{ind} = -\frac{d\phi}{dt}$$

$$e_{ind} = \sum_{i=1}^{N} e_i \qquad \phi = \sum_{i=1}^{N} \phi_i$$

$$e_{ind} = -N\frac{d\phi}{dt} \quad , \quad e_i = \frac{-d\phi_i}{dt}$$

$e_{ind}$ = voltage induced in the coil

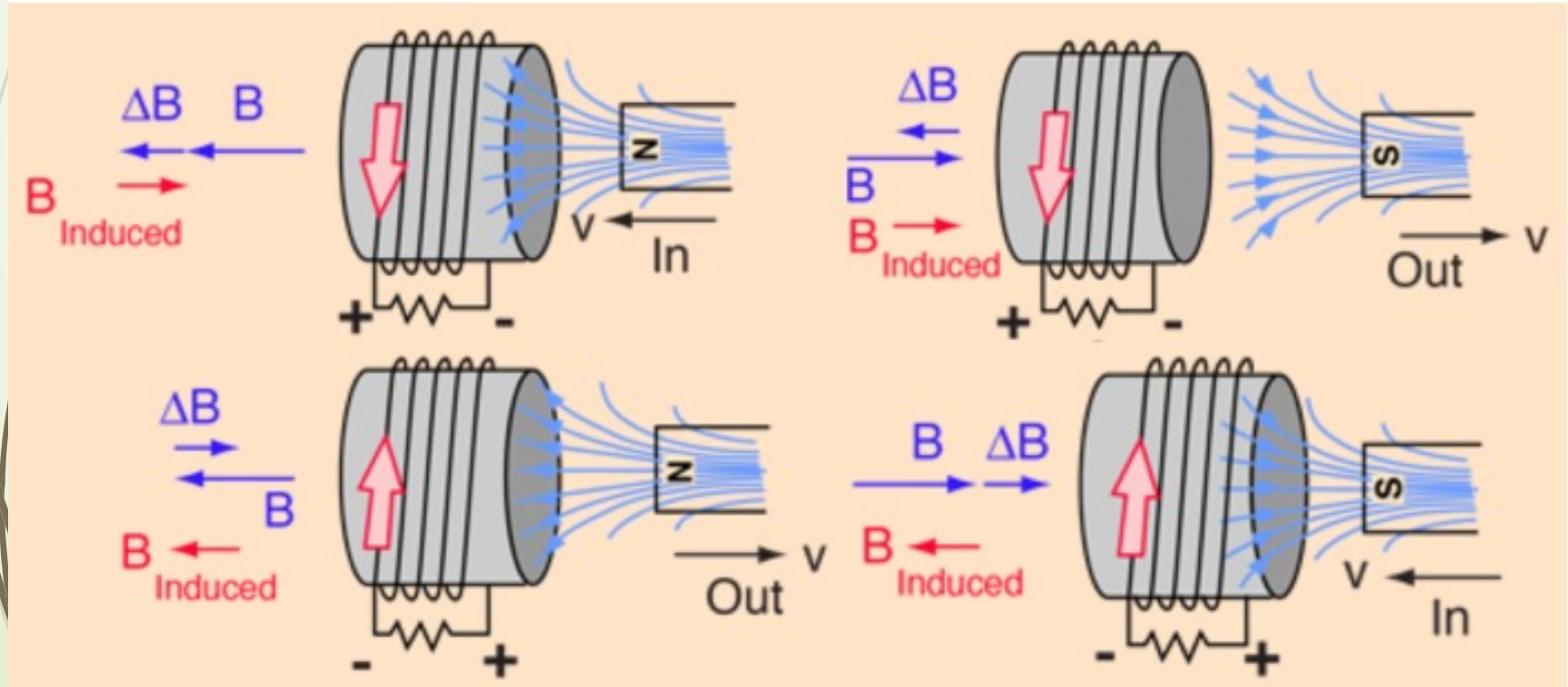
$N$ = number of turns of wire in coil

$\phi$ = flux passing through coil

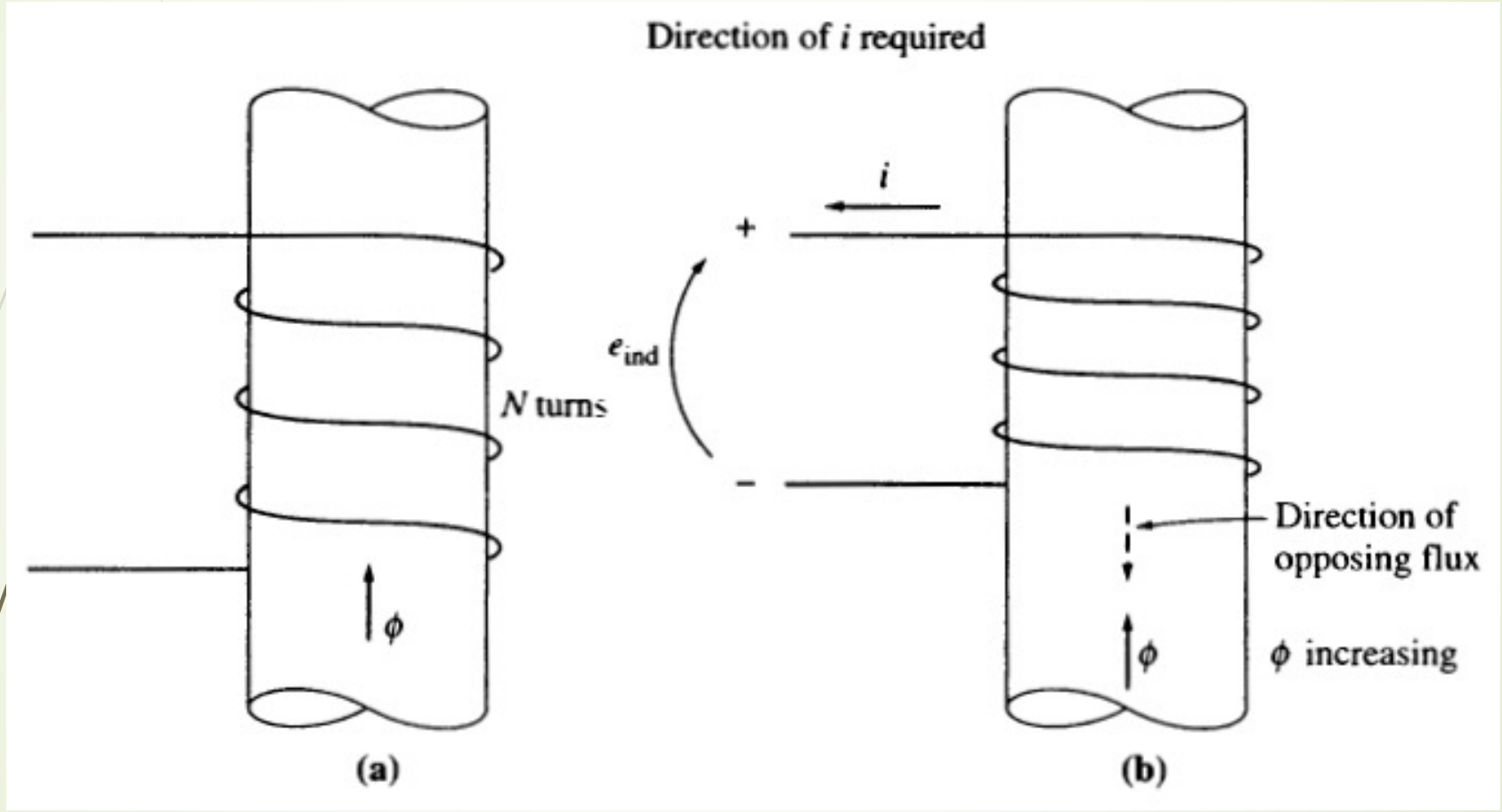
The minus sign in the equations is an expression of *Lenz's law*.

# Lenz's law

When an emf is generated by a change in magnetic flux according to Faraday's Law, the polarity of the induced emf is such that it produces a current whose magnetic field opposes the change which produces it. The induced magnetic field inside any loop of wire always acts to keep the magnetic flux in the loop constant. In the examples below, if the B field is increasing, the induced field acts in opposition to it. If it is decreasing, the induced field acts in the direction of the applied field to try to keep it constant.



# The induced voltage polarity – Lenz's law



# Flux and flux linkage

The term in parentheses in Equation (1–40) is called *the flux linkage* $\lambda$ of the coil, and Faraday's law can be rewritten in terms of flux linkage as

$$e_{\text{ind}} = \frac{d\lambda}{dt} \tag{1–41}$$

where

$$\lambda = \sum_{i=1}^{N} \phi_i \tag{1–42}$$

The units of flux linkage are weber-turns.

# Example 1-6

**Example 1–6.** Figure 1–15 shows a coil of wire wrapped around an iron core. If the flux in the core is given by the equation

$$\phi = 0.05 \sin 377t \quad \text{Wb}$$

If there are 100 turns on the core, what voltage is produced at the terminals of the coil? Of what polarity is the voltage during the time when flux is *increasing* in the reference direction shown in the figure? Assume that all the magnetic flux stays within the core (i.e., assume that the flux leakage is zero).

The core of Example 1–6. Determination of the voltage polarity at the terminals is shown.

# Example 1-6

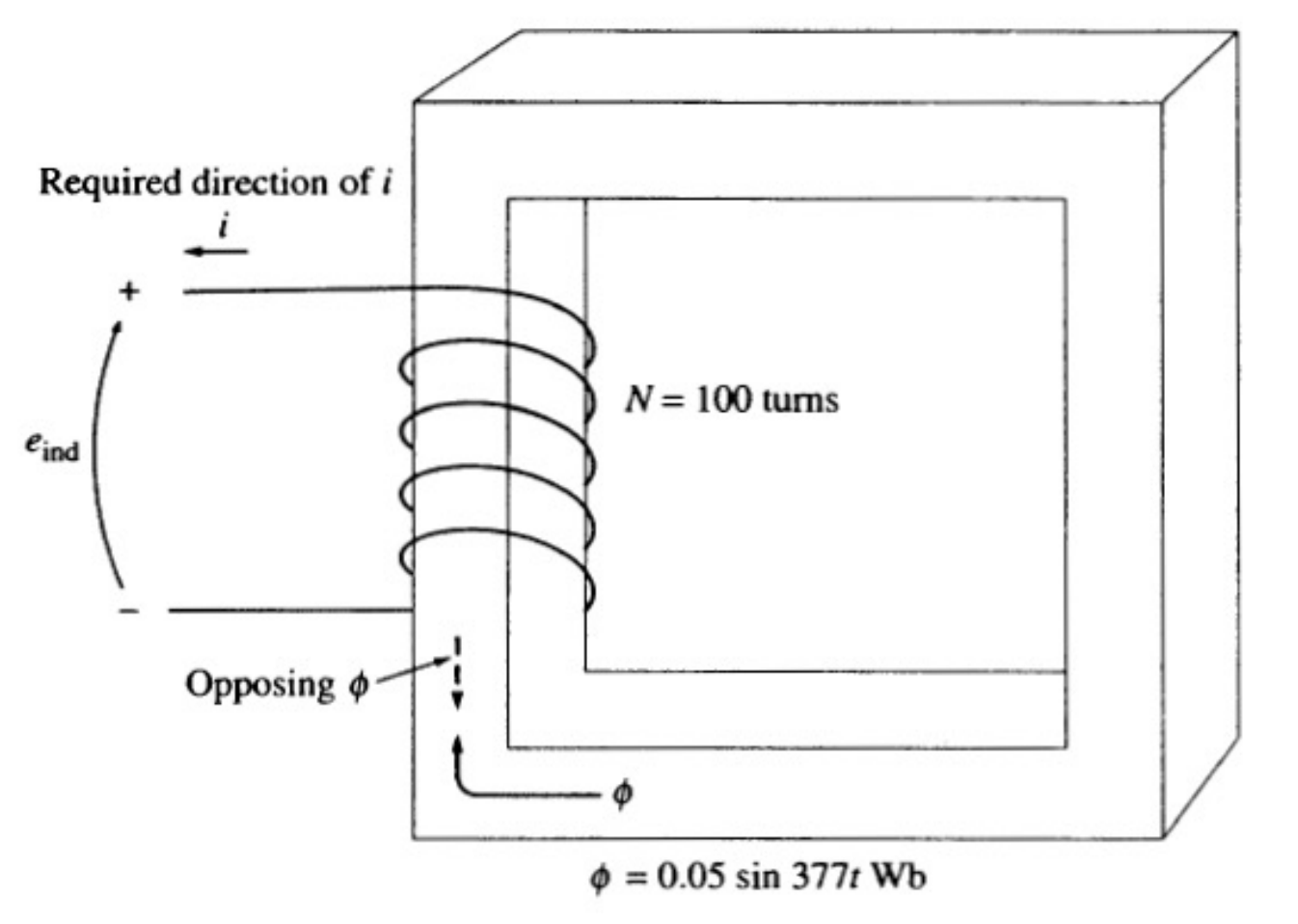



Fig. 1-15

# Example 1-6

***Solution***

By the same reasoning as in the discussion on pages 29–30, the direction of the voltage while the flux is increasing in the reference direction must be positive to negative, as shown in Figure 1–15. The *magnitude* of the voltage is given by

$$e_{\text{ind}} = N\frac{d\phi}{dt}$$

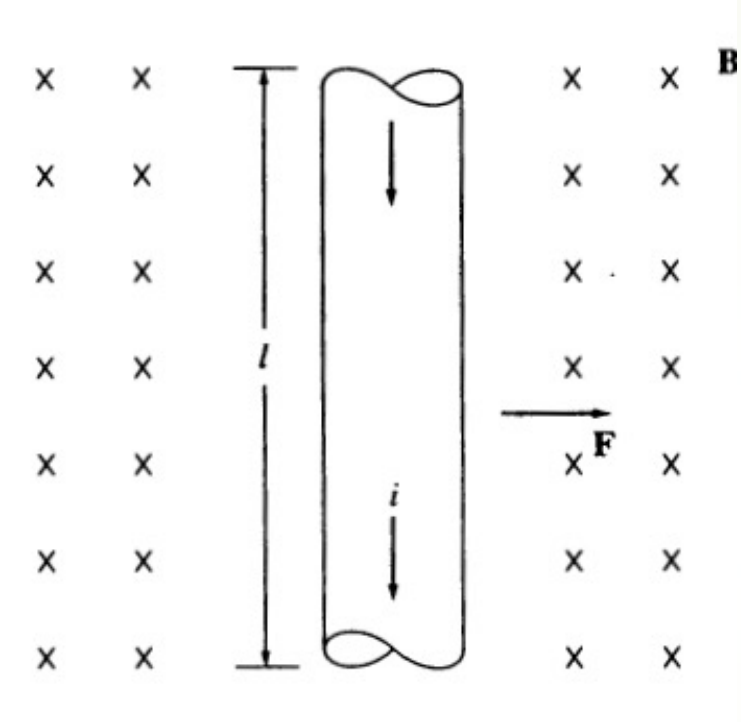
$$= (100 \text{ turns})\frac{d}{dt}(0.05 \sin 377t)$$

$$= 1885 \cos 377t$$

or alternatively,

$$e_{\text{ind}} = 1885 \sin(377t + 90°) \text{ V}$$

# Producing induced force on a wire




Using left hand rule

forefinger → B

middle finger → i

thumb → F

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

$$F = ilB \sin \theta$$

$i$ = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

The induced force on a wire by a current in the presence of a magnetic field is the basis of *motor action.*

# Example 1-7

**Example 1–7.** Figure 1–16 shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T, directed into the page. If the wire is 1.0 m long and carries 0.5 A of current in the direction from the top of the page to the bottom of the page, what are the magnitude and direction of the force induced on the wire?

***Solution***

The direction of the force is given by the right-hand rule as being to the right. The magnitude is given by

$$F = ilB \sin \theta \qquad (1\text{–}44)$$

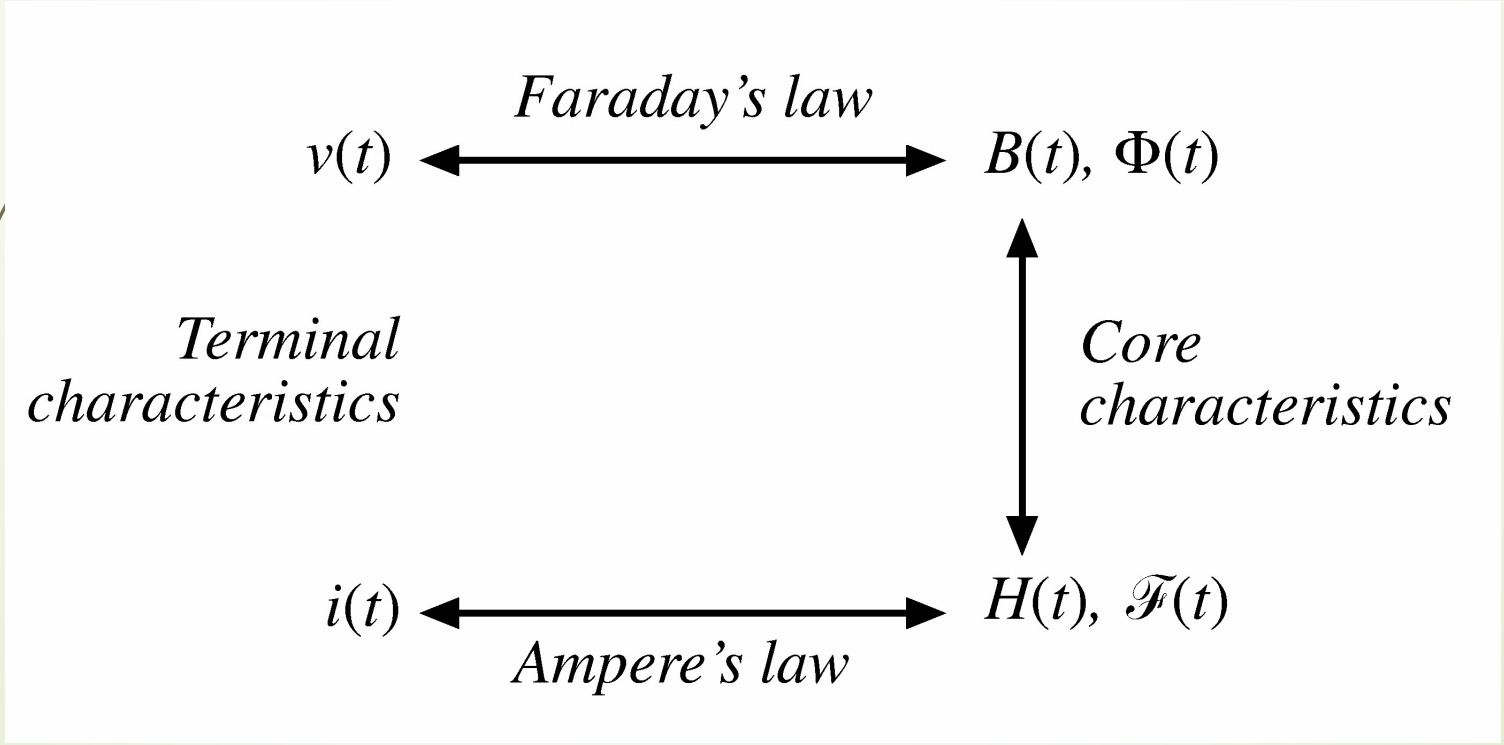
$$= (0.5 \text{ A})(1.0 \text{ m})(0.25 \text{ T}) \sin 90° = 0.125 \text{ N}$$

Therefore,

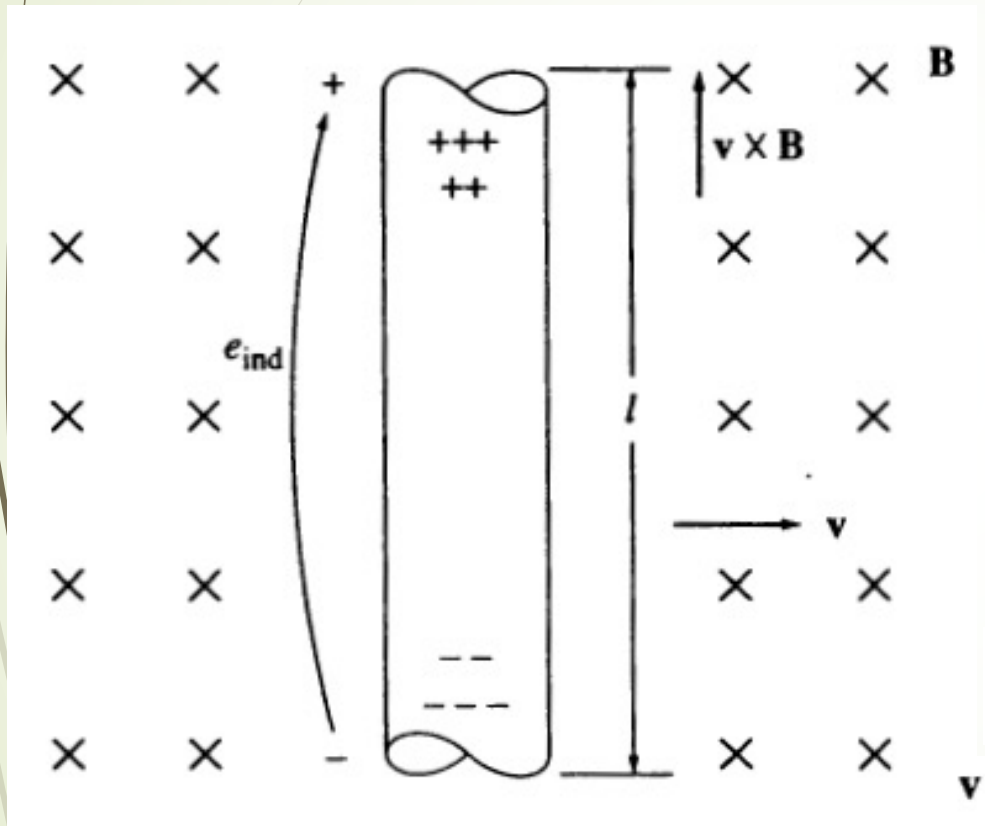
$$\mathbf{F} = 0.125 \text{ N, directed to the right}$$

# Relationship between electric-magnetic variables

- Magnetic field: Ampere's law
- Magnetic flux: magnetic material, hysteresis characteristics
- Transformer: Faraday's law, Len's law



# Induced voltage on a conductor moving in a magnetic field




y

Using right-hand rule

positive polarit is the direction of $\vec{v} \times \vec{B}$




$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

$\mathbf{v}$ = velocity of the wire

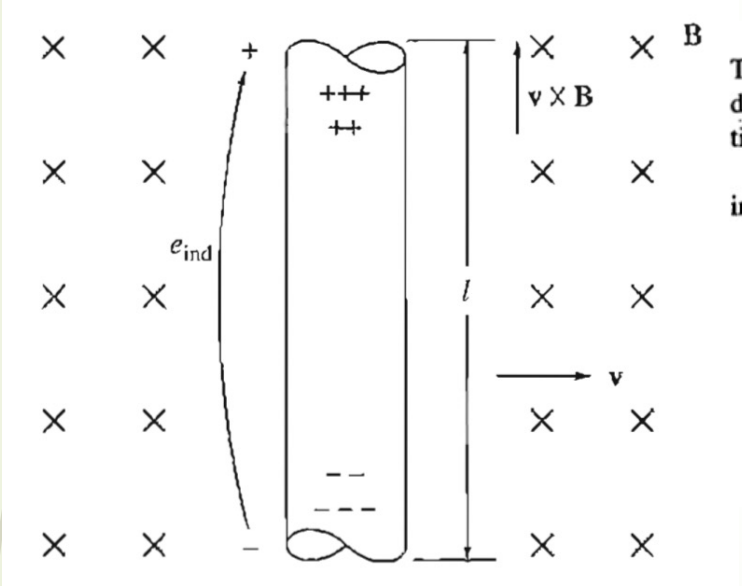
$\mathbf{B}$ = magnetic flux density vector

$\mathbf{l}$ = length of conductor in the magnetic field

The direction of the induced voltage is given by the right-hand rule.

# Example 1-8

**Example 1–8.** Figure 1–17 shows a conductor moving with a velocity of 5.0 m/s to the right in the presence of a magnetic field. The flux density is 0.5 T into the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?




***Solution***

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ in this example is up. Therefore, the voltage on the conductor will be built up positive at the top with respect to the bottom of the wire. The direction of vector **l** is up, so that it makes the smallest angle with respect to the vector $\mathbf{v} \times \mathbf{B}$.

Since **v** is perpendicular to **B** and since $\mathbf{v} \times \mathbf{B}$ is parallel to **l**, the magnitude of the induced voltage reduces to

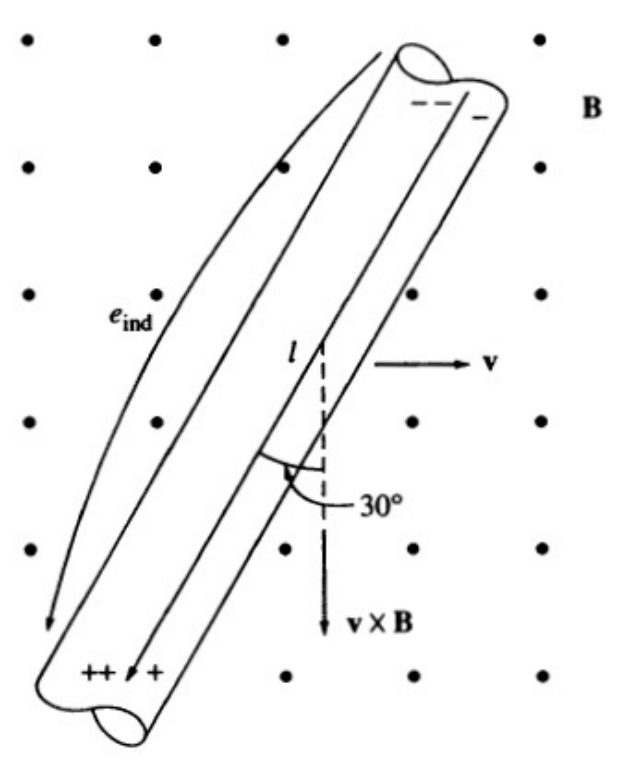
$$\begin{aligned} e_{ind} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45) \\ &= (vB \sin 90°)\, l \cos 0° \\ &= vBl \\ &= (5.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \\ &= 2.5 \text{ V} \end{aligned}$$

Thus the induced voltage is 2.5 V, positive at the top of the wire.

Fig 1.17 A conductor in the presence of a magnetic field

# Example 1-9

**Example 1–9.** Figure 1–18 shows a conductor moving with a velocity of 10 m/s to the right in a magnetic field. The flux density is 0.5 T, out of the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?




***Solution***

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ is down. The wire is not oriented on an up-down line, so choose the direction of l as shown to make the smallest possible angle with the direction of $\mathbf{v} \times \mathbf{B}$. The voltage is positive at the bottom of the wire with respect to the top of the wire. The magnitude of the voltage is

$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45)$$

$$= (vB \sin 90°)\, l \cos 30°$$

$$= (10.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \cos 30°$$

$$= 4.33 \text{ V}$$

Fig 1.18 A conductor in the presence of a magnetic field

# The linear DC machine - a simple example

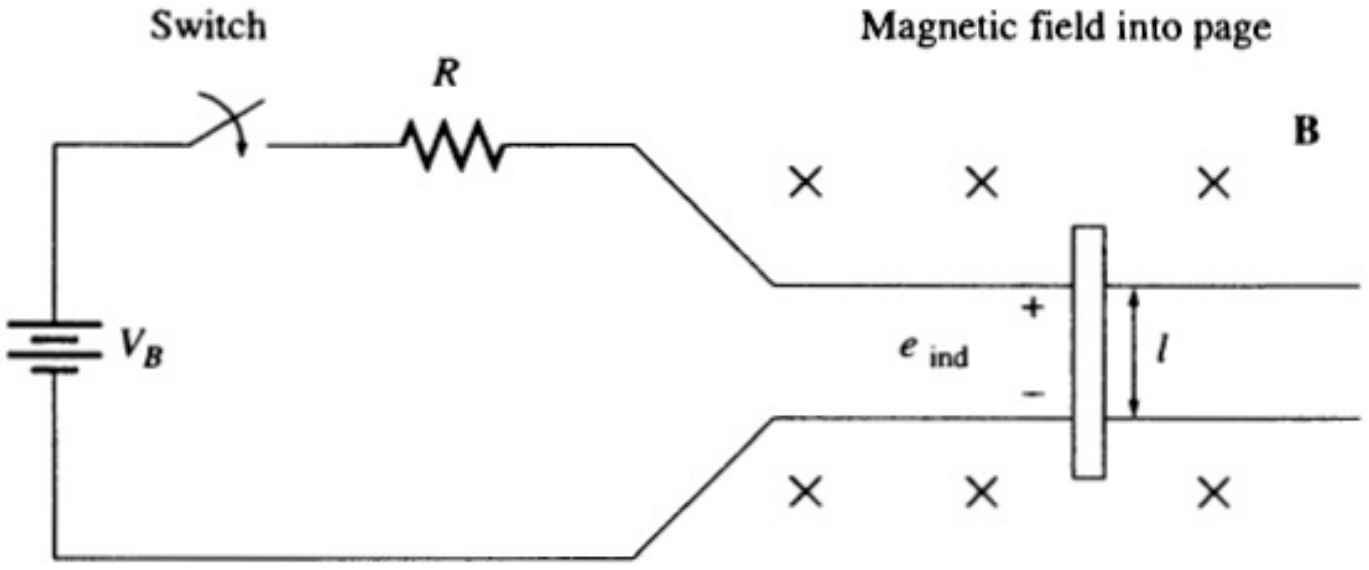



**FIGURE 1–19**
A linear dc machine. The magnetic field points into the page.

- Uniform and constant magnetic field
- Frictionless tracks
- A bar lying across the tracks
- Battery
- Switch

# The linear DC machine

- The simplest and easiest to understand version of a DC machine
- It works on the same principles and exhibits the same behavior as real generators and motors.
- Therefore, it serves as a good starting point in the study of machines.

**Determining the behavior of the machine with four basic equations**

**1.** The equation for the force on a wire in the presence of a magnetic field:

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B}) \qquad (1\text{–}43)$$

where $\mathbf{F}$ = force on wire

$i$ = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

# The linear DC machine

**2.** The equation for the voltage induced on a wire moving in a magnetic field:

$$\boxed{e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}} \qquad (1\text{–}45)$$

where $e_{\text{ind}}$ = voltage induced in wire
$\mathbf{v}$ = velocity of the wire
$\mathbf{B}$ = magnetic flux density vector
$\mathbf{l}$ = length of conductor in the magnetic field

**3.** Kirchhoff's voltage law for this machine. From Figure 1–19 this law gives

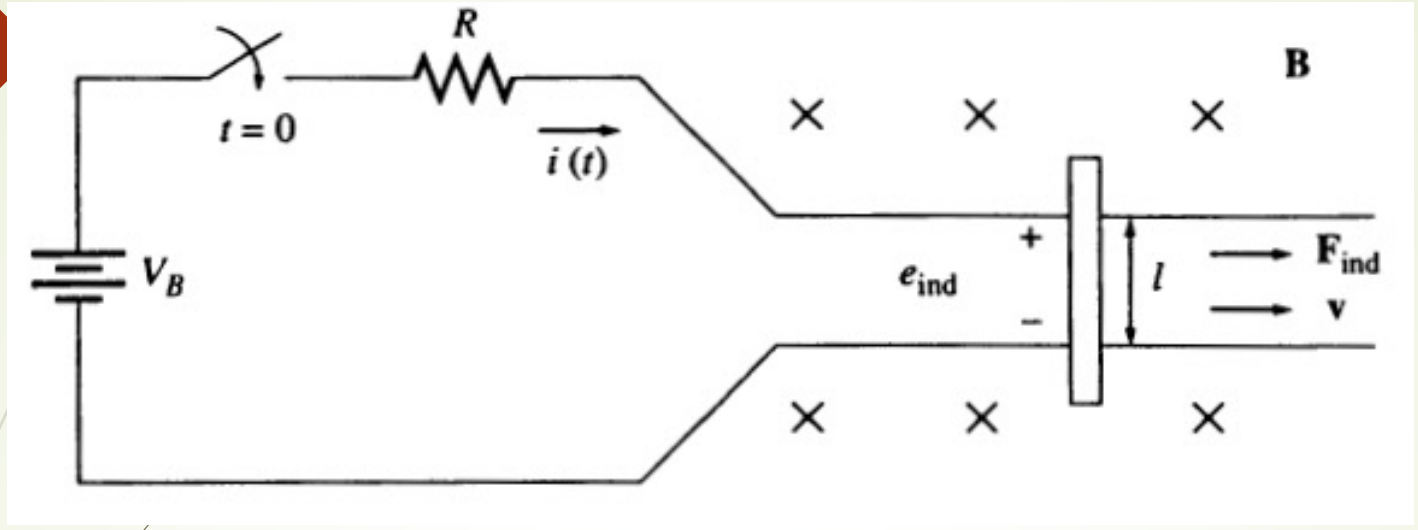
$$V_B - iR - e_{\text{ind}} = 0$$

$$\boxed{V_B = e_{\text{ind}} + iR = 0} \qquad (1\text{–}46)$$

**4.** Newton's law for the bar across the tracks:

$$\boxed{F_{\text{net}} = ma} \qquad (1\text{–}7)$$

# Starting a linear DC machine




1. Current => $i = \dfrac{V_B - e_{ind}}{R}$

2. Induced force => $F_{ind} = ilB$ to the right

3. Induced voltage => $e_{ind} = vBl$ positive upward

$$i\downarrow = \frac{V_B - e_{ind}\uparrow}{R}$$

# Starting a linear DC machine

- Eventually the bar will reach a constant steady-state speed where the net force on the bar is zero.

- This occurs when $e_{ind}$ rises until it equals the voltage $V_B$.
- At that time, the bar will be moving at a speed given by

$$V_B = e_{\text{ind}} = v_{ss} Bl$$

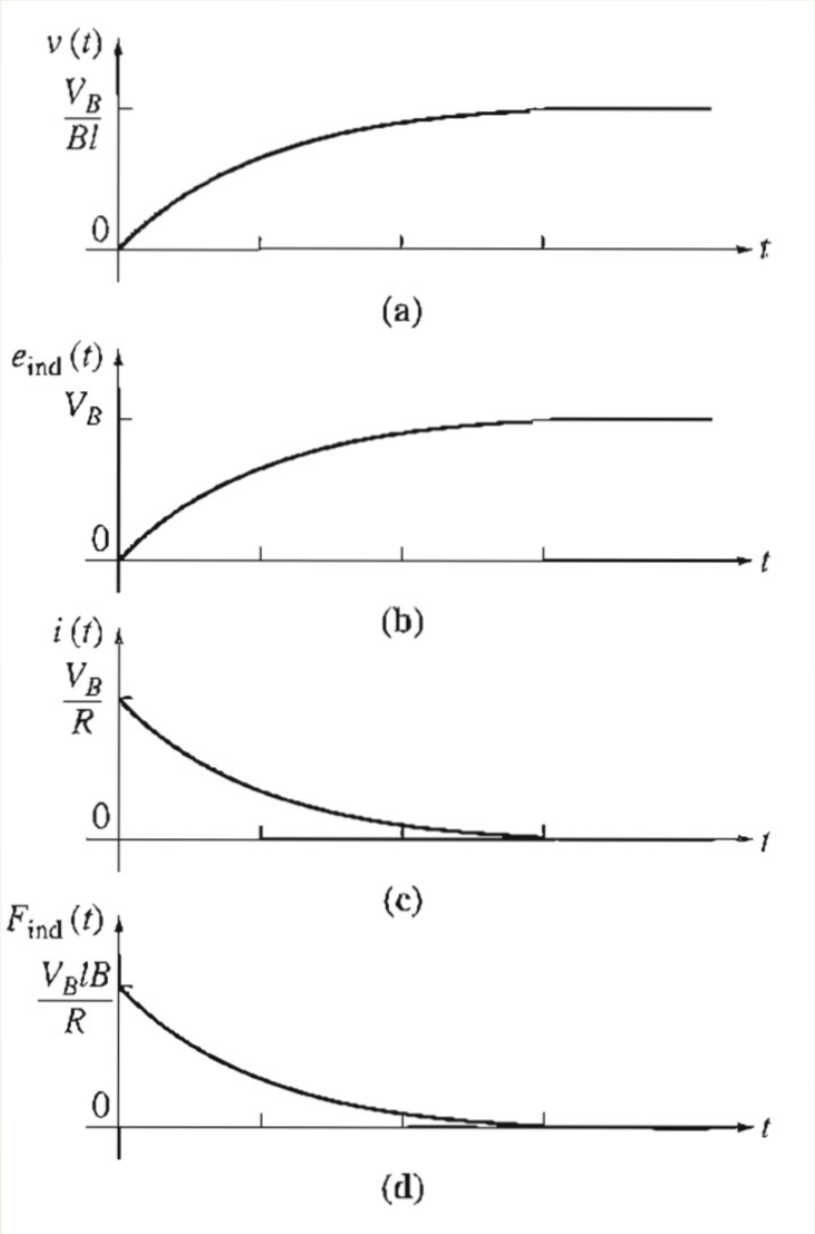
$$v_{ss} = \frac{V_B}{Bl}$$

- The bar will continue to slide along at this no-load speed forever unless some external force disturbs it.

# Summarize of starting a dc machine

1. Closing the switch produces a current flow $i = V_B / R$.
2. The current flow produces a force on the bar given by $F = ilB$.
3. The bar accelerates to the right, producing an induced voltage $e_{ind}$ as it speeds up.
4. This induced voltage reduces the current flow $i = (V_B - e_{ind}\uparrow)/R$.
5. The induced force is thus decreased ($F = i\downarrow lB$) until eventually $F = 0$. At that point, $e_{ind} = V_B$, $i = 0$, and the bar moves at a constant no-load speed $v_{ss} = V_B / Bl$.

# DC linear machine operates at no-load condition




Velocity $v$(t) vs time

Velocity $e_{ind}$(t) vs time

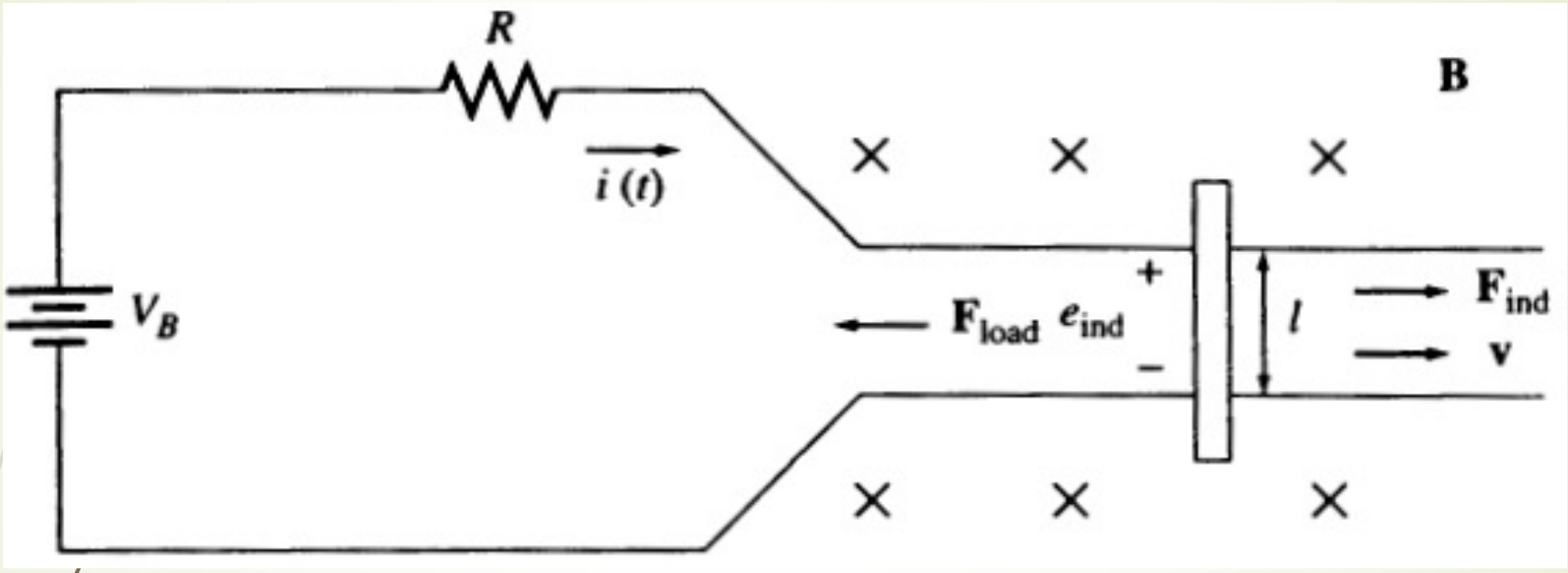
Current $i$(t) vs time

Induced force $F_{ind}$(t) vs time

Fig. 1-21

# Linear dc machine as a motor

- Applying the load ($F_{load}$)




$$F_{net} = F_{load} - F_{ind}$$

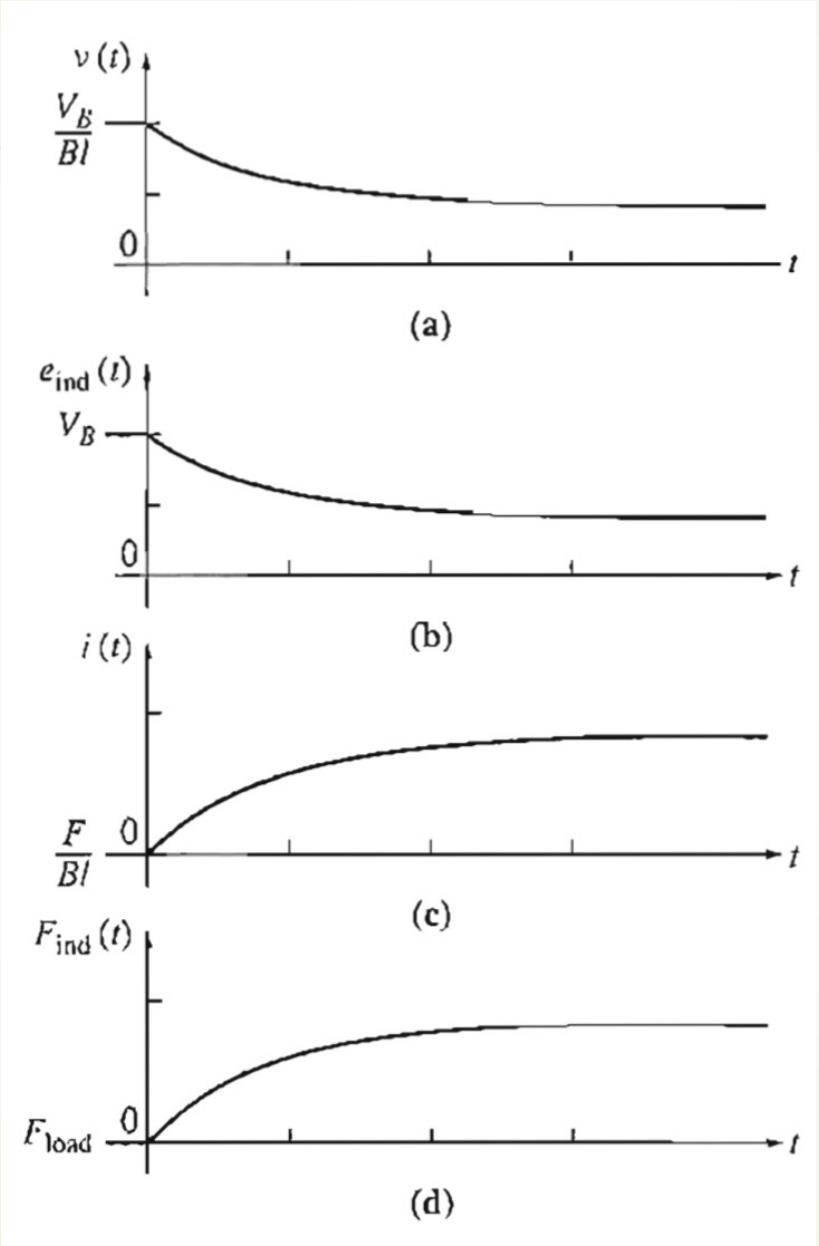
- The conversion of power between mechanical and electrical

$$P_{conv} = e_{ind} i = F_{ind} v$$

# Summarize of a dc motor operation

1. A force $\mathbf{F}_{load}$ is applied opposite to the direction of motion, which causes a net force $\mathbf{F}_{net}$ opposite to the direction of motion.
2. The resulting acceleration $a = F_{net}/m$ is negative, so the bar slows down ($v\downarrow$).
3. The voltage $e_{ind} = v\downarrow Bl$ falls, and so $i = (V_B - e_{ind}\downarrow)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a lower speed $v$.
5. An amount of electric power equal to $e_{ind}i$ is now being converted to mechanical power equal to $F_{ind}v$, and the machine is acting as a motor.

# Graphical plots for motor action




Speed $v$(t) vs time

Induced voltage $e_{ind}$(t) vs tim

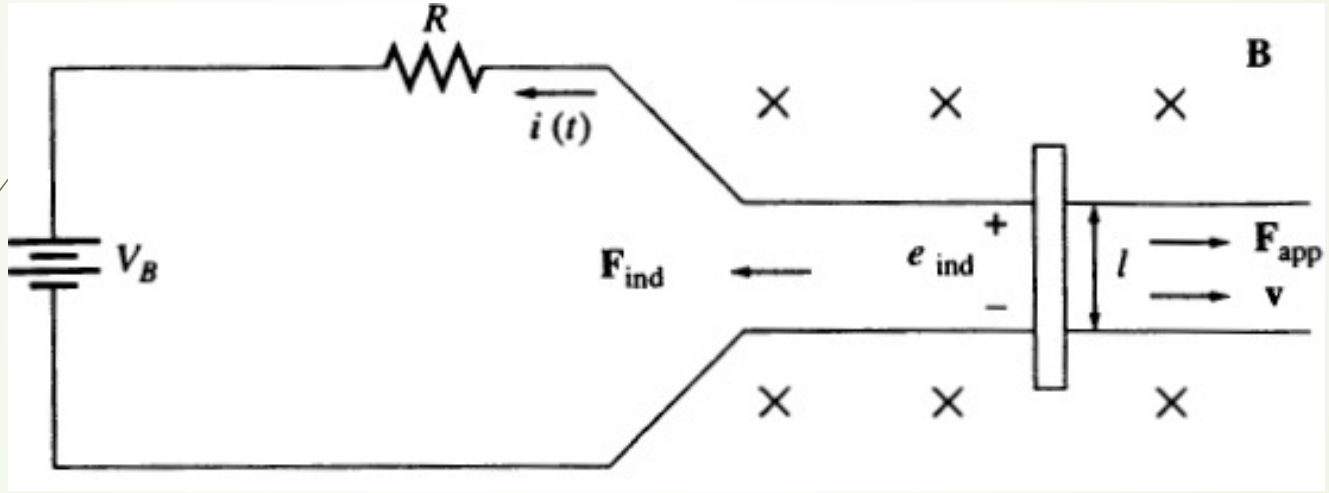
Current $i$(t) vs time

Induced force $F_{ind}$(t) vs time

Fig. 1-23

# Linear dc machine as a generator

- Applying an external force ($F_{app}$) on the moving direction




- The conversion of mechanical power into electrical

$$P_{conv} = Fv = ie_{ind}$$

Fnet & Fapp in the direction of motion.

# Linear dc machine as a generator

- With applying a force to the bar in the direction of motion, the bar positively accelerates and the velocity $v$ of the bar increases.
- As the velocity increases, $e_{\text{ind}} = v\uparrow$ will increase and will be larger than the battery voltage $V_B$.
- When $e_{ind} > V_B$ , the current reverses direction and is now given by the equation

$$i = \frac{e_{\text{ind}} - V_B}{R}$$

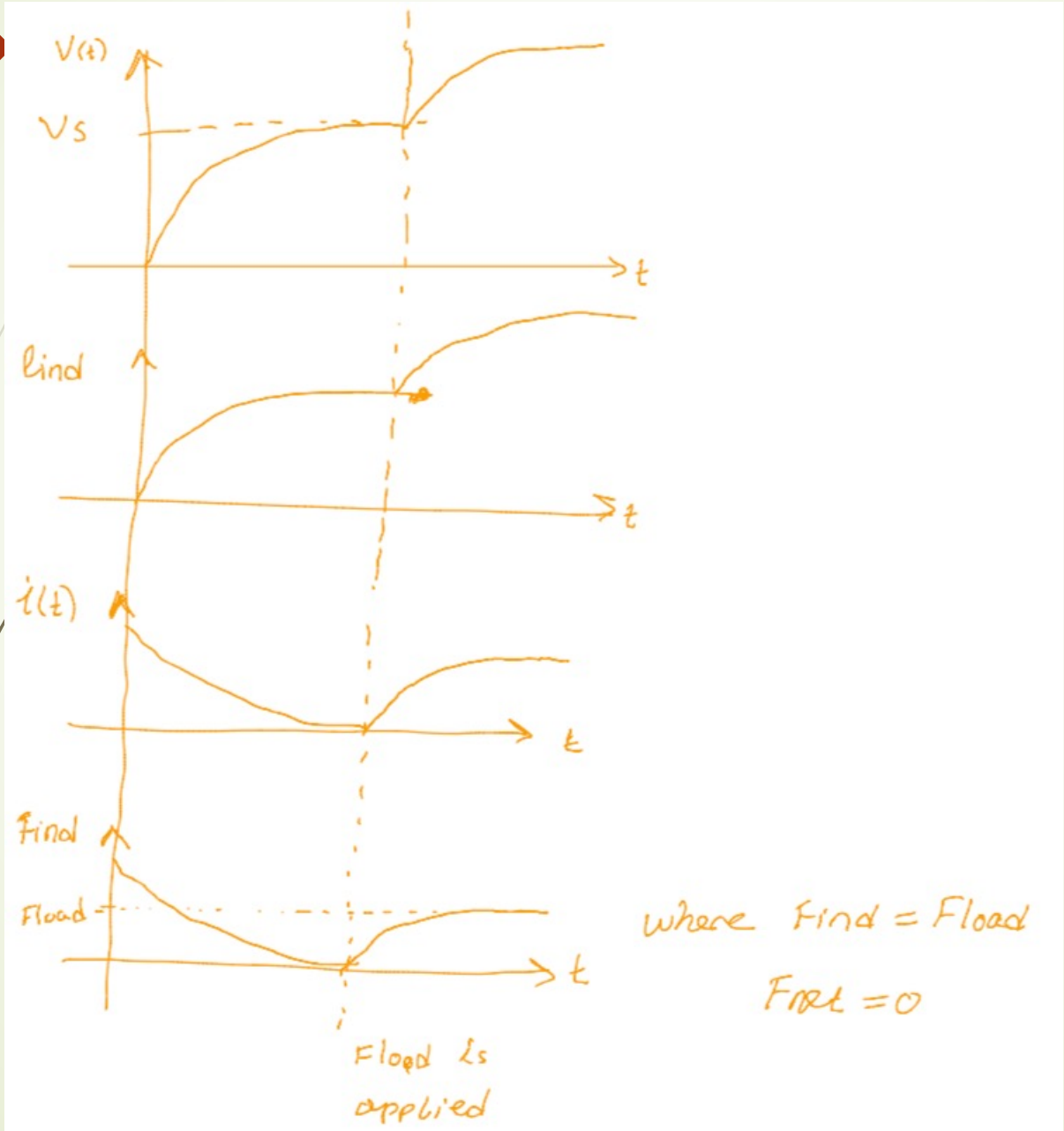
- Since this current now flows up through the bar, it induces a force in the bar given by

$$F_{\text{ind}} = ilB \qquad \text{to the left}$$

# Summarize of a dc generator operation

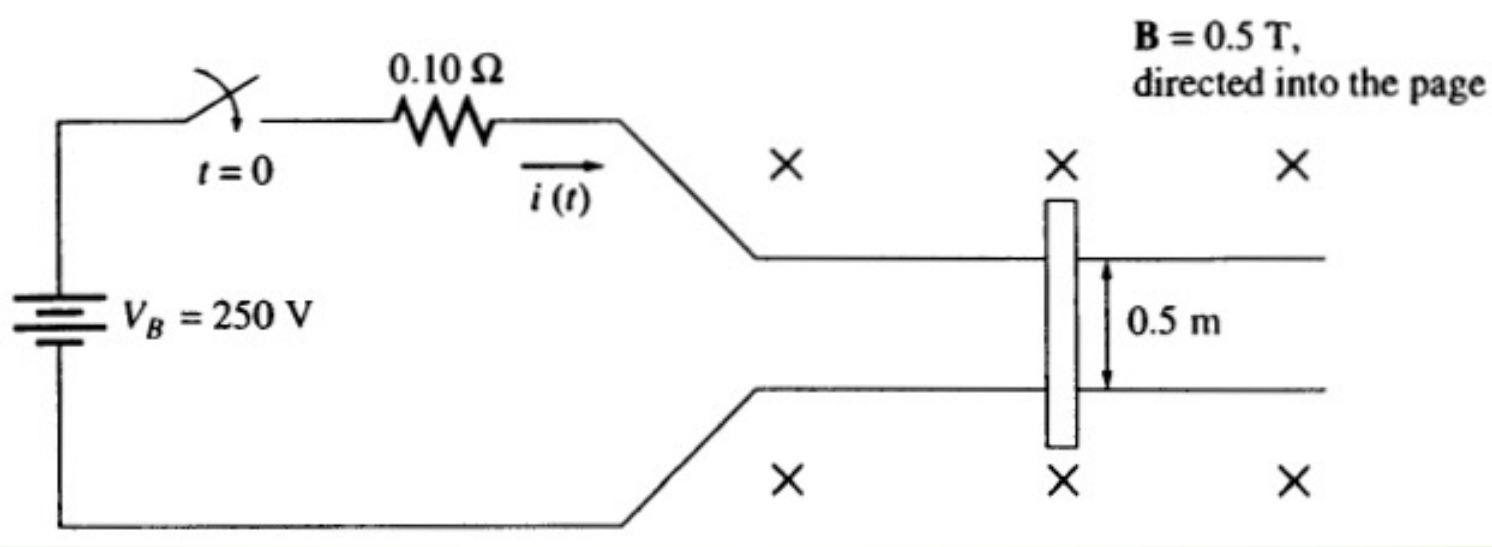
1. A force $\mathbf{F}_{app}$ is applied in the direction of motion; $\mathbf{F}_{net}$ is in the direction of motion.
2. Acceleration $a = F_{net}/m$ is positive, so the bar speeds up ($v\uparrow$).
3. The voltage $e_{ind} = v\uparrow Bl$ increases, and so $i = (e_{ind}\uparrow - V_B)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a higher speed $v$.
5. An amount of mechanical power equal to $F_{ind}v$ is now being converted to electric power $e_{ind}i$, and the machine is acting as a generator.

# Graphical plots for generator action



# Starting problems with dc linear machine

At starting conditions, the speed of the bar is zero, so $e_{ind} = 0$.




Thus, the current flow at starting is

$$i_{start} = \frac{V_B}{R} = \frac{250\ V}{0.1\ \Omega} = 2500\ A$$

Such currents can cause severe damage to a motor.

# Starting problems with dc linear machine

Inserting an extra resistance into the circuit during starting to limit the current flow until $e_{ind}$ builds up enough to limit it.

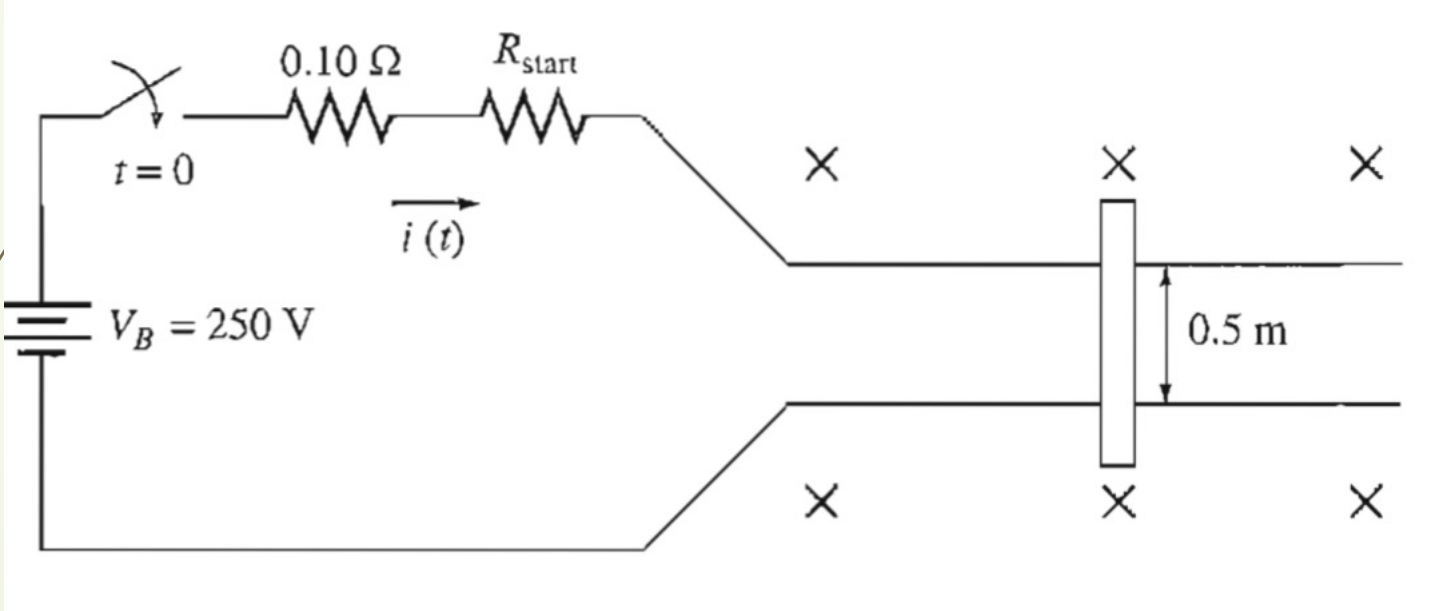



Fig 1-26 A linear dc machine with an extra series resistor inserted to control the starting current.

# Example 1-10

**Example 1–10.** The linear dc machine shown in Figure 1–27a has a battery voltage of 120 V, an internal resistance of 0.3 Ω, and a magnetic flux density of 0.1 T.

*(a)* What is this machine's maximum starting current? What is its steady-state velocity at no load?

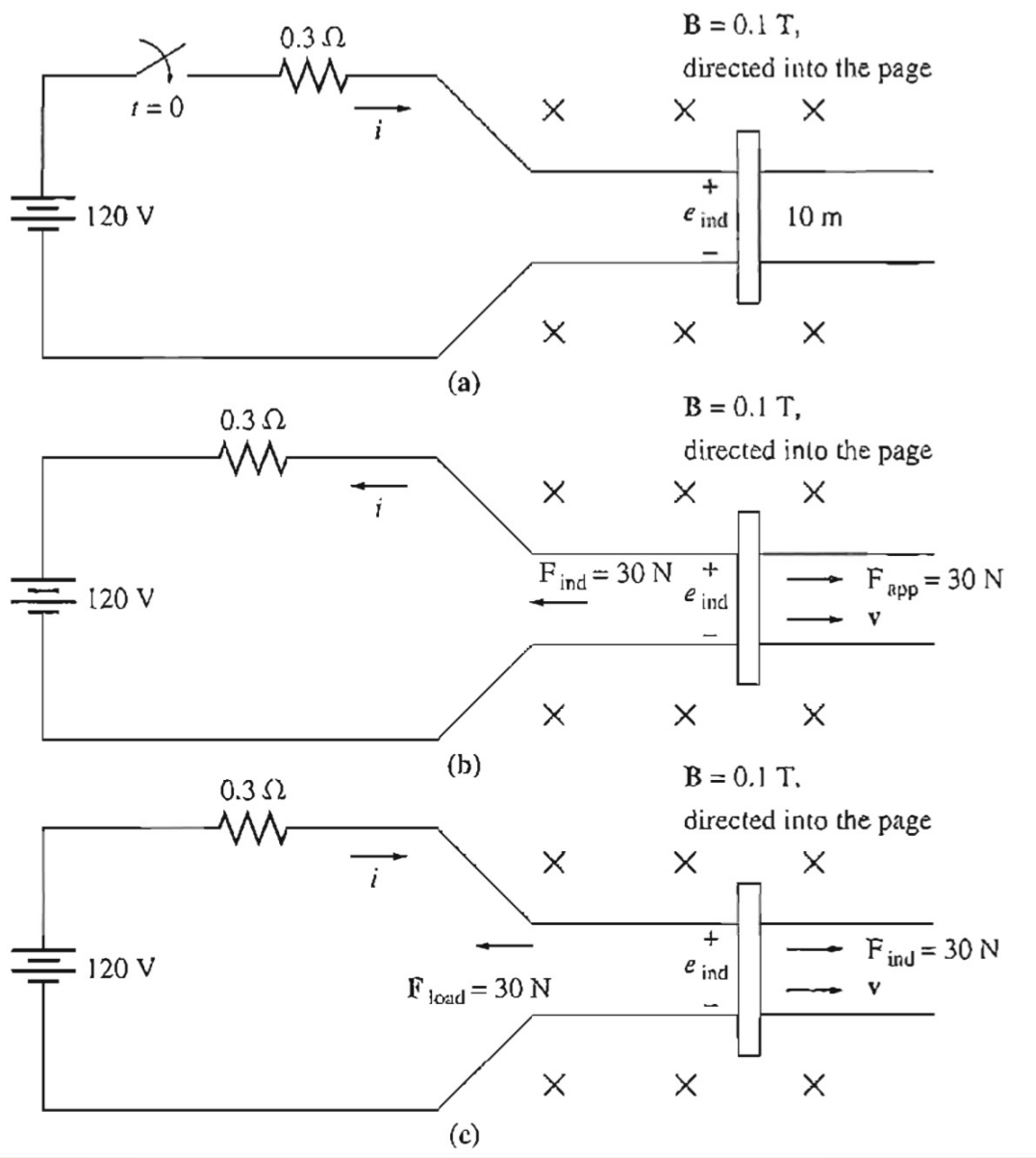
*(b)* Suppose that a 30-N force pointing to the right were applied to the bar. What would the steady-state speed be? How much power would the bar be producing or consuming? How much power would the battery be producing or consuming? Explain the difference between these two figures. Is this machine acting as a motor or as a generator?

*(c)* Now suppose a 30-N force pointing to the left were applied to the bar. What would the new steady-state speed be? Is this machine a motor or a generator now?

*(d)* Assume that a force pointing to the left is applied to the bar. Calculate speed of the bar as a function of the force for values from 0 N to 50 N in 10-N steps. Plot the velocity of the bar versus the applied force.

*(e)* Assume that the bar is unloaded and that it suddenly runs into a region where the magnetic field is weakened to 0.08 T. How fast will the bar go now?

# Example 1-10




$i' = \frac{250}{0.1} = 2500\ A$

10 times the rated current of the m/c

Fig. 1-27 The linear dc machnine of Example 1-10, (a) Starting conditions; (b) operating as a generator; (c) operating as a motor.

# Example 1-10

***Solution***

*(a)* At starting conditions, the velocity of the bar is 0, so $e_{ind} = 0$. Therefore,

$$i = \frac{V_B - e_{ind}}{R} = \frac{120\ \text{V} - 0\ \text{V}}{0.3\ \Omega} = 400\ \text{A}$$

When the machine reaches steady state, $\mathbf{F}_{ind} = 0$ and $i = 0$. Therefore,

$$VB = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl}$$

$$= \frac{120\ \text{V}}{(0.1\ \text{T})(10\ \text{m})} = 120\ \text{m/s}$$

# Example 1-10

*(b)* Refer to Figure 1–27b. If a 30-N force to the right is applied to the bar, the final steady state will occur when the induced force $\mathbf{F}_{ind}$ is equal and opposite to the applied force $\mathbf{F}_{app}$, so that the net force on the bar is zero:

$$F_{app} = F_{ind} = ilB$$

Therefore,

$$i = \frac{F_{ind}}{lB} = \frac{30\ \text{N}}{(10\text{m})(0.1\ \text{T})}$$

$$= 30\ \text{A} \quad \text{flowing up through the bar}$$

The induced voltage $e_{ind}$ on the bar must be

$$e_{ind} = V_B + iR$$

$$= 120\ \text{V} + (30\text{A})(0.3\ \Omega) = 129\ \text{V}$$

and the final steady-state speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{129\ \text{V}}{(0.1\ \text{T})(10\ \text{m})} = 129\ \text{m/s}$$

The bar is *producing* $P = (129\ \text{V})(30\ \text{A}) = 3870\ \text{W}$ of power, and the battery is *consuming* $P = (120\ \text{V})(30\ \text{A}) = 3600\ \text{W}$. The difference between these two numbers is the 270 W of losses in the resistor. This machine is acting as a *generator.*

# Example 1-10

*(c)* Refer to Figure 1–25c. This time, the force is applied to the left, and the induced force is to the right. At steady state,

$$F_{app} = F_{ind} = ilB$$

$$i = \frac{F_{ind}}{lB} = \frac{30\ \text{N}}{(10\ \text{m})(0.1\ \text{T})}$$

$$= 30\ \text{A} \quad \text{flowing down through the bar}$$

The induced voltage $e_{ind}$ on the bar must be

$$e_{ind} = V_B - iR$$

$$= 120\ \text{V} - (30\ \text{A})(0.3\ \Omega) = 111\ \text{V}$$

and the final speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{111\ \text{V}}{(0.1\ \text{T})(10\ \text{m})} = 111\ \text{m/s}$$

This machine is now acting as a *motor*, converting electric energy from the battery into mechanical energy of motion on the bar.

# Example 1-10

**(e) If the bar is initially unloaded, then $e_{ind} = V_B$. If the bar suddenly hits a region of weaker magnetic field, a transient will occur. Once the transient is over, though, $e_{ind}$ will again equal $V_B$.**

This fact can be used to determine the final speed of the bar. The *initial speed* was 120 m/s. The *final speed* is

$$VB = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl}$$

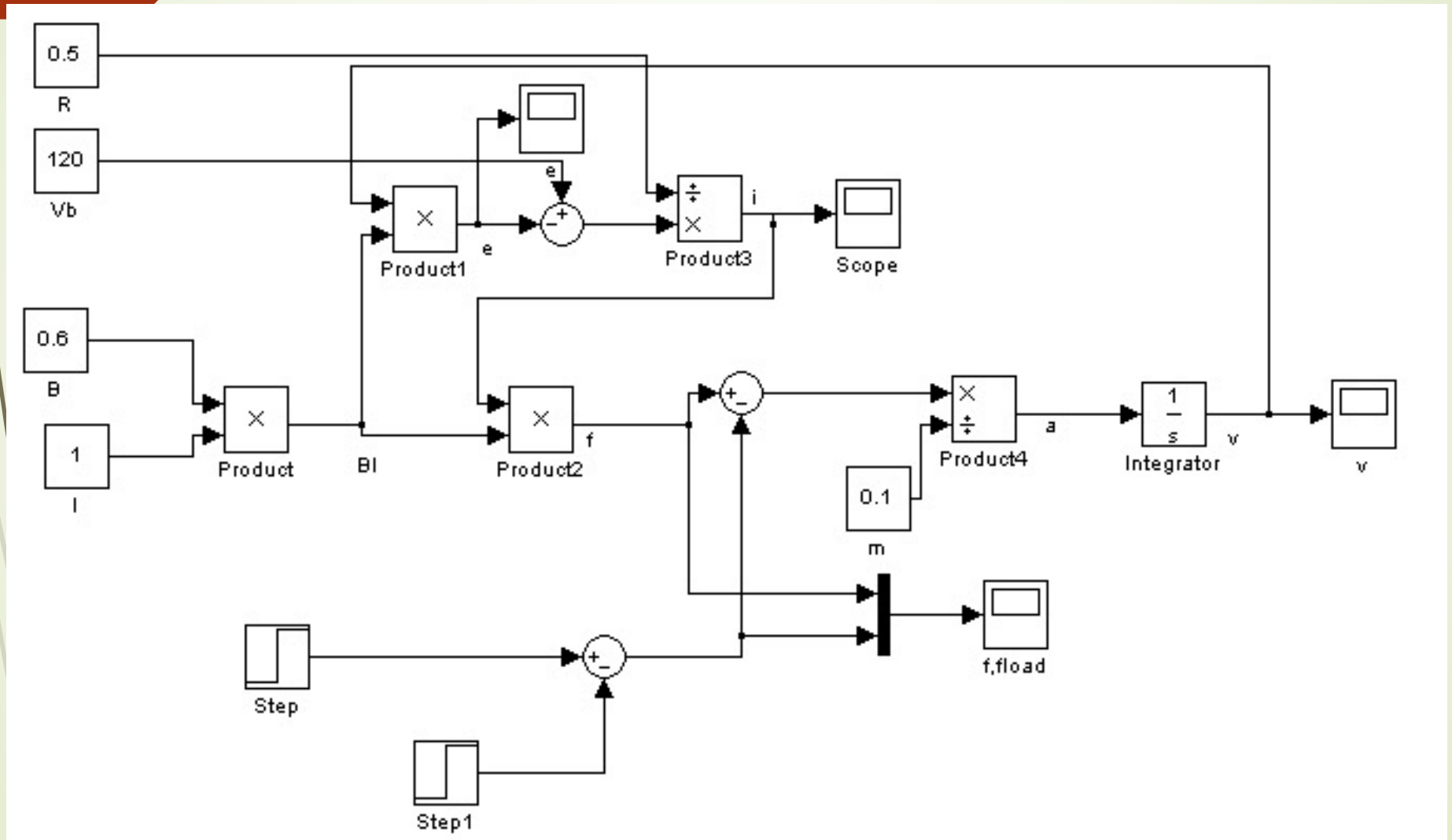
$$= \frac{120\ \text{V}}{(0.08\ \text{T})(10\ \text{m})} = 150\ \text{m/s}$$

Thus, when the flux in the linear motor weakens, the bar speeds up. The same behavior occurs in real dc motors: When the field flux of a dc motor weakens, it turns faster. Here, again, the linear machine behaves in much the same way as a real dc motor.

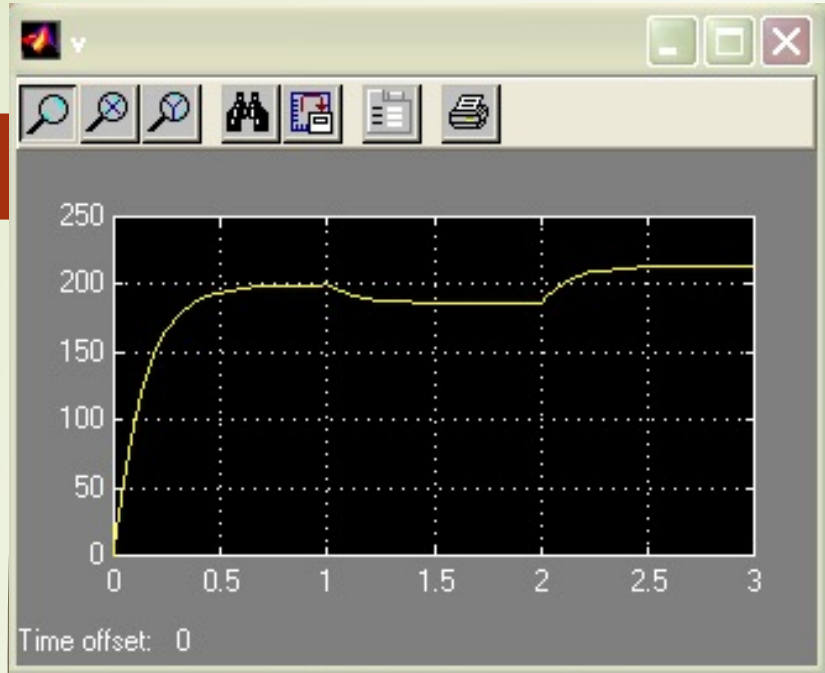
# Linear dc motor simulation

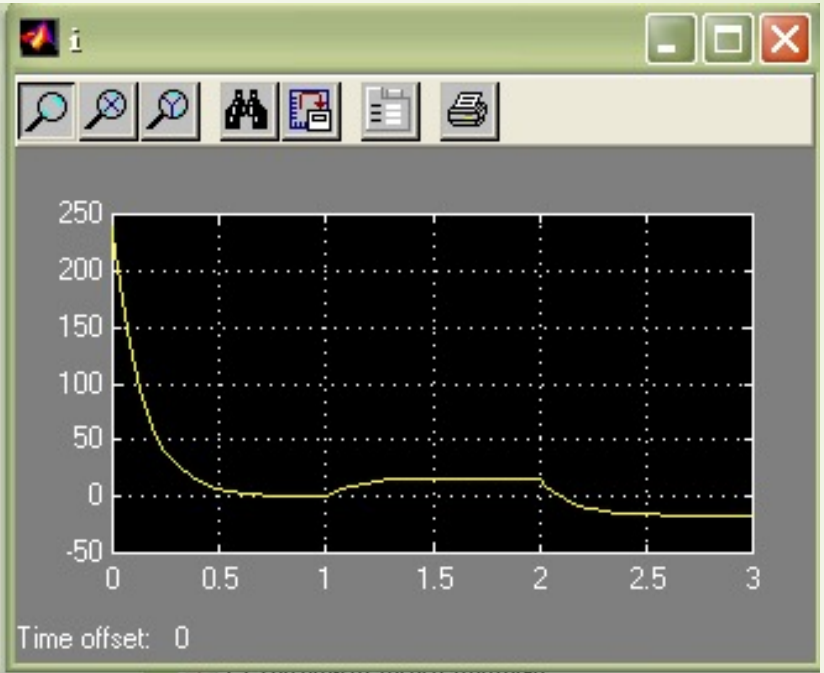
- **Equations:**
  - F = ilB, l: bar length
  - e = vBl
  - i = (Vb-e) / R
  - dv/dt = (F-Fload)/m
- **Simulation parameters:**
  - Vb=120V, R=0.3Ω, l = 1m
  - B=0.6T, m=0.1kg
  - Fload=10(u-1)-20(u-2) N

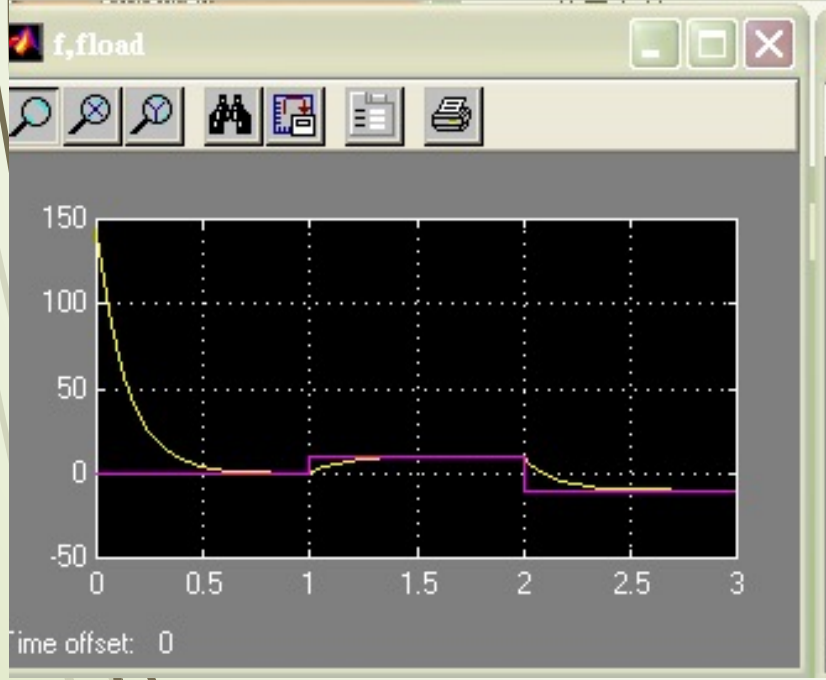
# Linear dc motor simulation

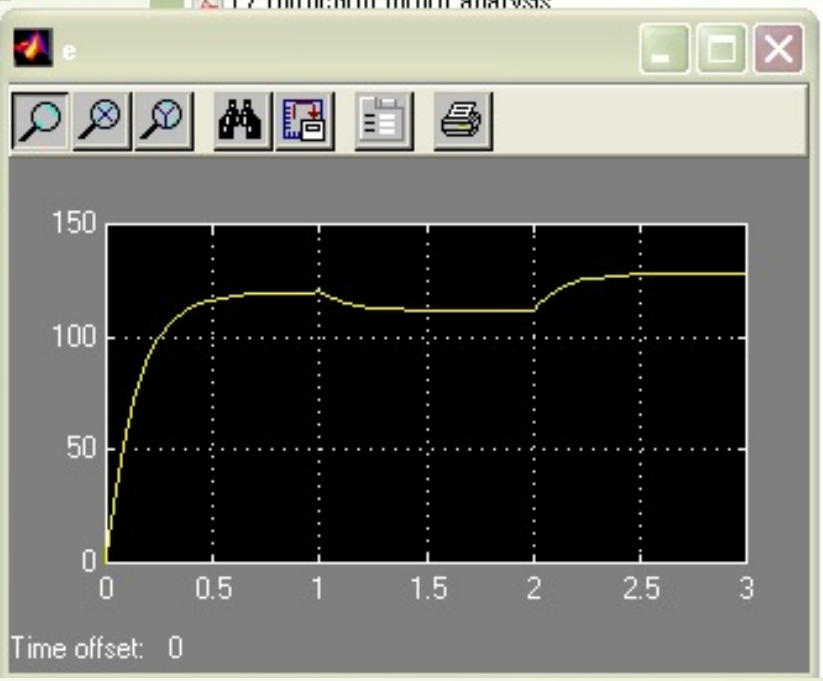



e=Blv, (Vb-e)/R=I, Bli=f, f-(fstep_t1- fstep_t2)/m=a, a/s=v

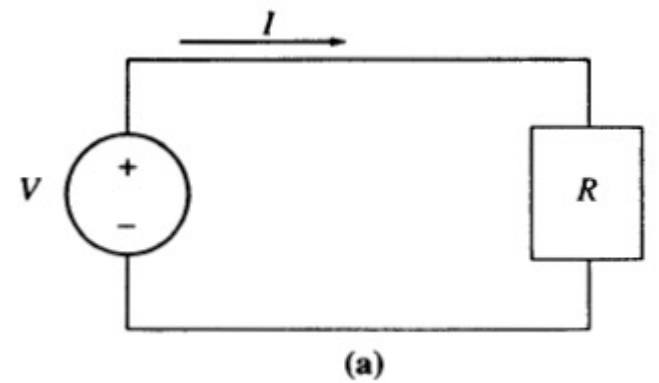

v
250
200
150
100
50
0
0
0.5
1
1.5
2
2.5
3
Time offset: 0


i
250
200
150
100
50
0
-50
0
0.5
1
1.5
2
2.5
3
Time offset: 0


f,fload
150
100
50
0
-50
0
0.5
1
1.5
2
2.5
3
ime offset: 0


e
150
100
50
0
0
0.5
1
1.5
2
2.5
3
Time offset: 0

# Real, reactive and apparent power in AC circuits

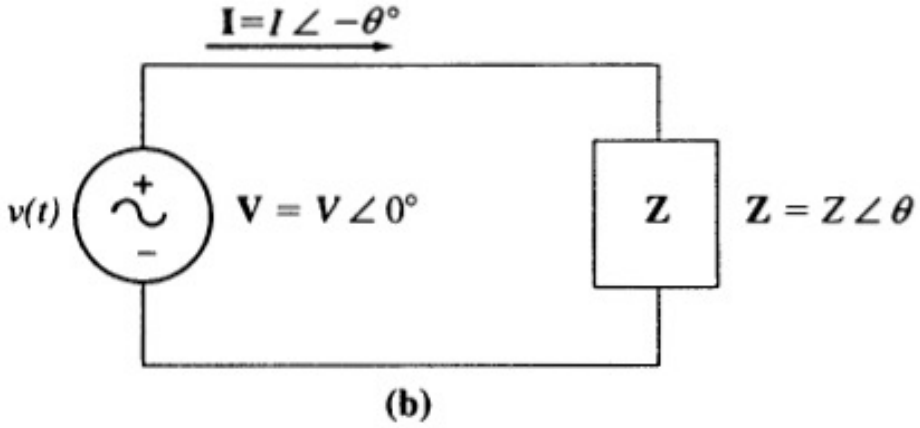


(a)

$$P = VI$$

# Real, reactive and apparent power in AC circuits

- AC source applies power to an impedance Z




$$v(t) = \sqrt{2}V\cos\omega t$$

$$i(t) = \sqrt{2}I\cos(\omega t - \theta)$$

# Instantaneous power

The instantaneous power supplied to this load at any time $t$ is

$$p(t) = v(t)i(t) = 2VI \cos \omega t \cos(\omega t - \theta) \qquad (1\text{–}58)$$

The angle $\theta$ in this equation is the *impedance angle* of the load. For inductive loads, the impedance angle is positive, and the current waveform lags the voltage waveform by $\theta$ degrees.

If we apply trigonometric identities to Equation (1–58), it can be manipulated into an expression of the form

$$p(t) = VI \cos \theta (1 + \cos 2\omega t) + VI \sin \theta \sin 2\omega t \qquad (1\text{–}59)$$

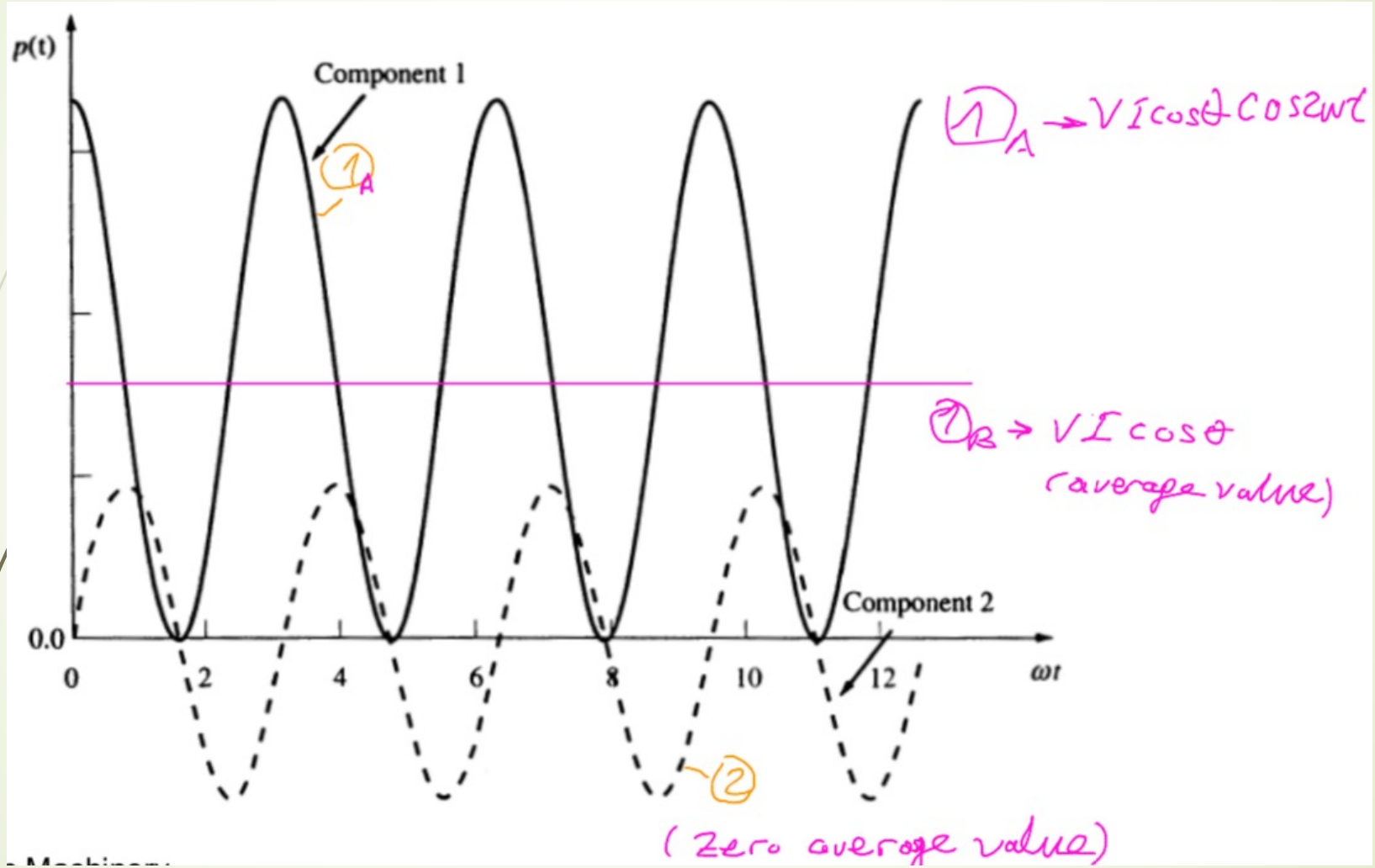
(1) (2)

where $\cos(\omega t - \theta) = \cos \omega t \cos \theta + \sin \omega t \sin \theta$

$p(t) = 2VI(\cos^2 \omega t \cos \theta + \cos \omega t \sin \omega t \sin \theta)$

where $\cos^2 \omega t = \frac{1}{2}[1 + \cos(2\omega t)]$ and $\cos \omega t \sin \omega t = \frac{1}{2} \sin 2\omega t$

# Instantaneous power



# Average power and reactive power

Note that the *first* term of the instantaneous power expression is always positive, but it produces pulses of power instead of a constant value. The average value of this term is

$$P = VI \cos \theta \qquad (1\text{–}60)$$

which is the *average* or *real* power ($P$) supplied to the load by term 1 of the Equation (1–59). The units of real power are watts (W), where 1 W = 1 V × 1 A.

Note that the *second* term of the instantaneous power expression is positive half of the time and negative half of the time, so that *the average power supplied by this term is zero.* This term represents power that is first transferred from the source to the load, and then returned from the load to the source. The power that continually bounces back and forth between the source and the load is known as *reactive power* ($Q$). Reactive power represents the energy that is first stored and then released in the magnetic field of an inductor, or in the electric field of a capacitor.

# Reactive power Q and apparent power S

1. Reactive power Q (var) is defined from instantaneous power

$$Q = VI \sin \theta$$

2. Apparent power S (VA) is defined to represent the product of voltage and current magnitudes

$$S = VI$$

# Alternative Forms of the Power Equations

If a load has a constant impedance, then Ohm's law can be used to derive alternative expressions for the real, reactive, and apparent powers supplied to the load. Since the magnitude of the voltage across the load is given by

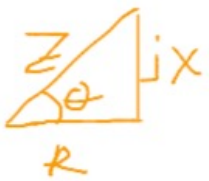
$$V = IZ \tag{1–63}$$

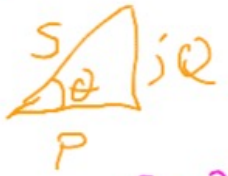
substituting Equation (1–63) into Equations (1–60) to (1–62) produces equations for real, reactive, and apparent power expressed in terms of current and impedance:

$$P = I^2 Z \cos\theta \tag{1–64}$$

$$Q = I^2 Z \sin\theta \tag{1–65}$$

$$S = I^2 Z \tag{1–66}$$




where $|Z|$ is the magnitude of the load impedance $Z$.

Since the impedance of the load $Z$ can be expressed as

$$Z = R + jX = |Z|\cos\theta + j|Z|\sin\theta$$

we see from this equation that $R = |Z|\cos\theta$ and $X = |Z|\sin\theta$, so the real and reactive powers of a load can also be expressed as

$$P = I^2 R \tag{1–67}$$

$$Q = I^2 X \tag{1–68}$$

where $R$ is the resistance and $X$ is the reactance of load $Z$.

# Power in AC circuit

$i(t)=I(1/\mathrm{sqr}(2))\cos(wt-20)$

The chosen *v* and *i* include all possibilities because

- If the load is purely resistive: $\theta = 0°$
- If the load is purely inductive: $\theta = 90°$
- If the load is purely capacitive: $\theta = -90°$

For a network that is primarily inductive, $\theta$ is positive (*v* leads *i* or *i* lags *v*)

For a network that is primarily capacitive, $\theta$ is negative (*i* leads *v*)

# Power in AC circuit

Positive power means that power has been distributed from supply into circuit.

Negative power means that the power has been distributed from circuit into supply.

There are three type of power in AC circuit:

i) Average power, P
ii) Apparent Power, S
iii) Reactive Power, Q

# Complex power representation

For simplicity in computer calculations, real and reactive power are sometimes represented together as a *complex power* **S**, where

$$\mathbf{S} = P + jQ \tag{1–69}$$

The complex power **S** supplied to a load can be calculated from the equation

$$\mathbf{S} = \mathbf{VI}^* = V_{rms} I_{rms}^* \tag{1–70}$$

where the asterisk represents the complex conjugate operator.

where $|S| = V_{rms} I_{rms} = \sqrt{P^2 + Q^2}$

# Complex power representation

To understand this equation, let's suppose that the voltage applied to a load is $\mathbf{V} = V \angle \alpha$ and the current through the load is $\mathbf{I} = I \angle \beta$. Then the complex power supplied to the load is

$$\mathbf{S} = \mathbf{VI}^* = (V\angle\alpha)(I\angle-\beta) = VI\angle(\alpha-\beta) = VI\cos(\alpha-\beta) + jVI\sin(\alpha-\beta)$$

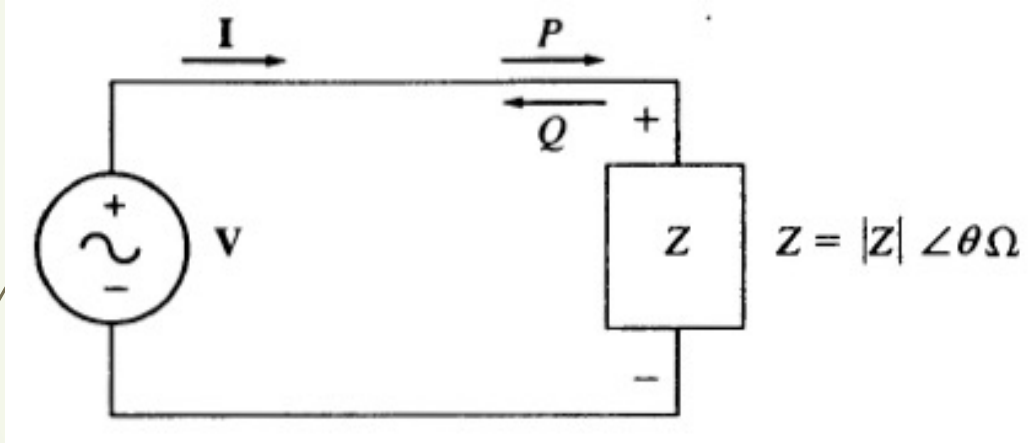
$= VI\, e^{j(\alpha-\beta)}$, with $\theta = \alpha - \beta$

The impedance angle $\theta$ is the difference between the angle of the voltage and the angle of the current ($\theta = \alpha - \beta$), so this equation reduces to

$$\mathbf{S} = VI\cos\theta + jVI\sin\theta = P + jQ$$

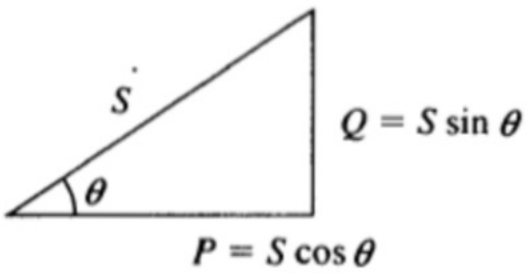
# Power direction

The direction of Q
+ve for the first cycle
-ve for the second cycle.




$$\mathbf{I} = \frac{\mathbf{V}}{Z} = \frac{V\angle 0^\circ}{|Z|\angle\theta} = \frac{V}{|Z|}\angle -\theta$$

# Power factor




$$\cos\theta = \frac{P}{S}$$

$$\sin\theta = \frac{Q}{S}$$

$$\tan\theta = \frac{Q}{P}$$

The quantity cos $\theta$ is usually known as the *power factor* of a load. The power factor is defined as the fraction of the apparent power $S$ that is actually supplying real power to a load. Thus,

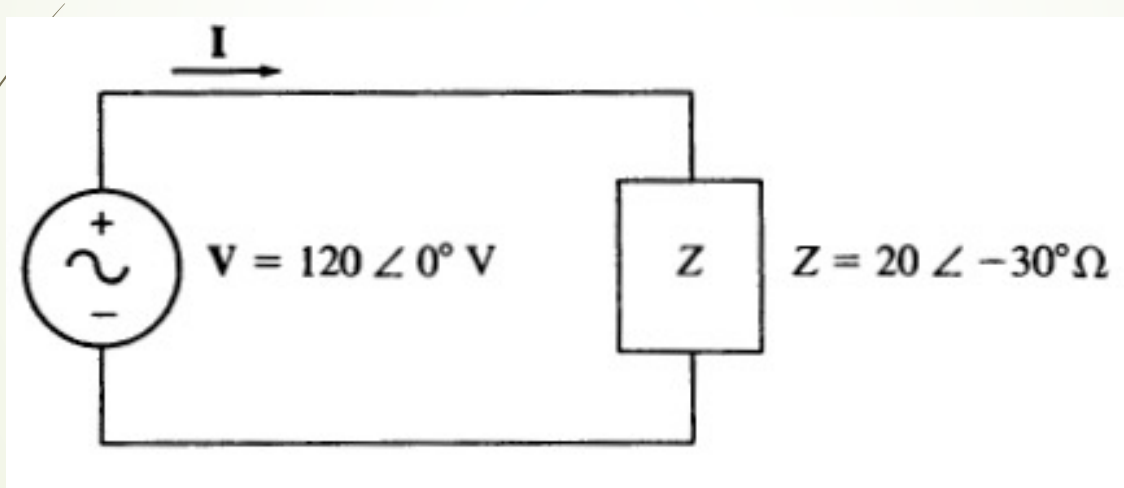
$$\text{PF} = \cos\theta = \frac{P}{|S|} \qquad (1\text{–}71)$$

where $\theta$ is the impedance angle of the load.

where $|S| = VI$

# Example 1-11

**Example 1–11.** Figure 1–34 shows an ac voltage source supplying power to a load with impedance $Z = 20\angle -30° \ \Omega$. Calculate the current **I** supplied to the load, the power factor of the load, and the real, reactive, apparent, and complex power supplied to the load.



***Solution***

The current supplied to this load is

$$\mathbf{I} = \frac{\mathbf{V}}{Z} = \frac{120\angle 0^\circ \text{ V}}{20\angle -30^\circ\ \Omega} = 6\angle 30^\circ \text{ A}$$

The power factor of the load is

$$\text{PF} = \cos\theta = \cos(-30^\circ) = 0.866 \text{ leading} \qquad (1\text{–}71)$$

(Note that this is a capacitive load, so the impedance angle $\theta$ is negative, and the current *leads* the voltage.)

The real power supplied to the load is

$$P = VI\cos\theta \qquad (1\text{–}60)$$

$$P = (120 \text{ V})(6 \text{ A})\cos(-30^\circ) = 623.5 \text{ W}$$

The reactive power supplied to the load is

$$Q = VI\sin\theta \qquad (1\text{–}61)$$

$$Q = (120 \text{ V})(6 \text{ A})\sin(-30^\circ) = -360 \text{ VAR}$$

The apparent power supplied to the load is

$$S = VI \tag{1–62}$$

$$Q = (120\ \text{V})(6\ \text{A}) = 720\ \text{VA}$$

The complex power supplied to the load is

$$\mathbf{S} = \mathbf{VI}^* \tag{1–70}$$

$$= (120\angle 0°\ \text{V})(6\angle -30°\ \text{A})^*$$

$$= (120\angle 0°\ \text{V})(6\angle 30°\ \text{A}) = 720\angle 30°\ \text{VA}$$

$$= 623.5 - j360\ \text{VA}$$

# Notes about total P, Q, and S

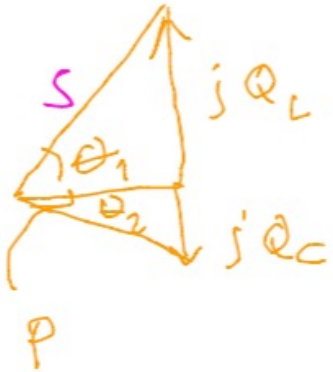
- First, the total apparent power, $S_T$ can be determined from the total average $P_T$ and total reactive powers $Q_T$ and **cannot** be determined from the apparent powers of each branch.
- Second, and more important, it is **not necessary** to consider the series-parallel arrangement of branches. In other words, the total real $P_T$ , total reactive $Q_T$ , or total apparent power $S_T$ is independent of whether the loads are in series, parallel, or series-parallel.




$\theta_1$ : positive ⇓
Inductive load




$-\theta_2$ : negative
⇓
Capacitive load




$$jQ = jQ_L - jQ_C$$

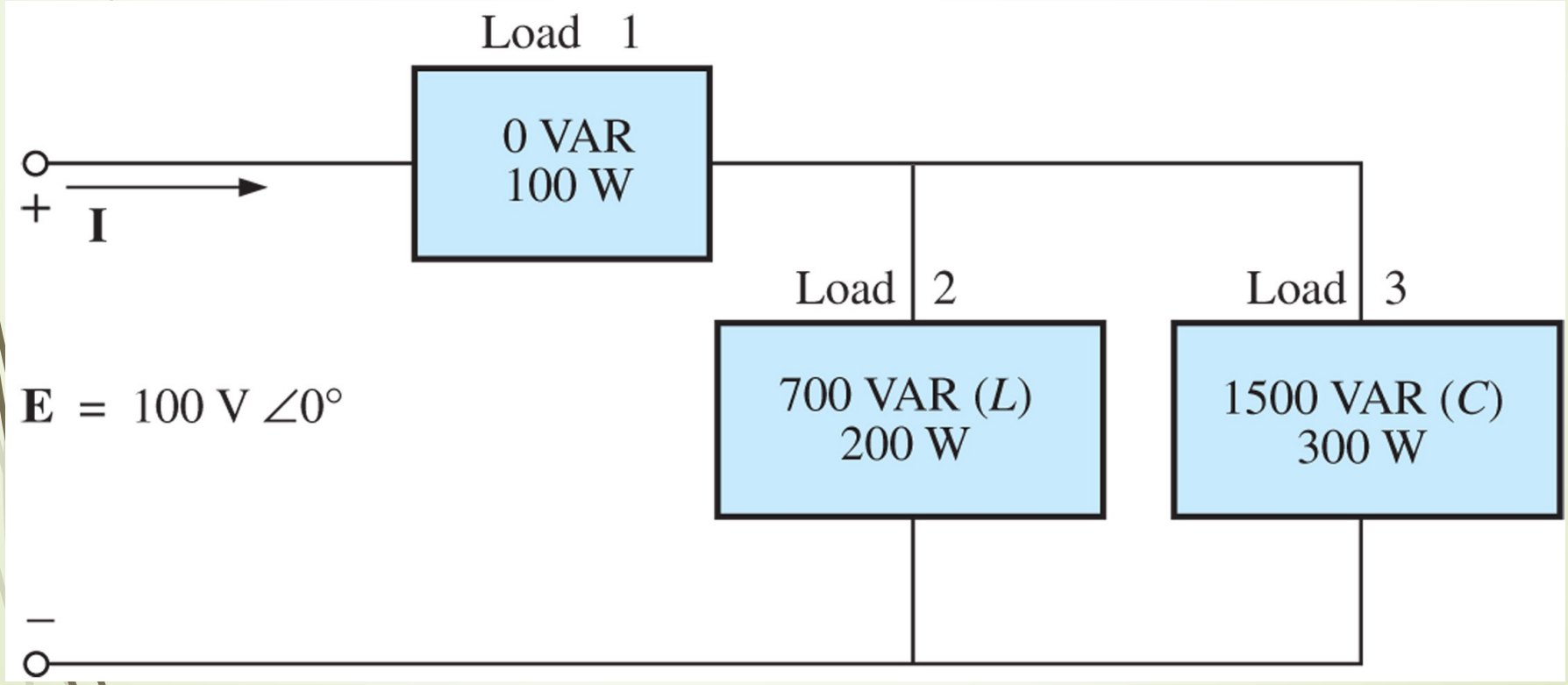
$$S = P + jQ$$

$$\theta_1 - \theta_2 = \theta$$

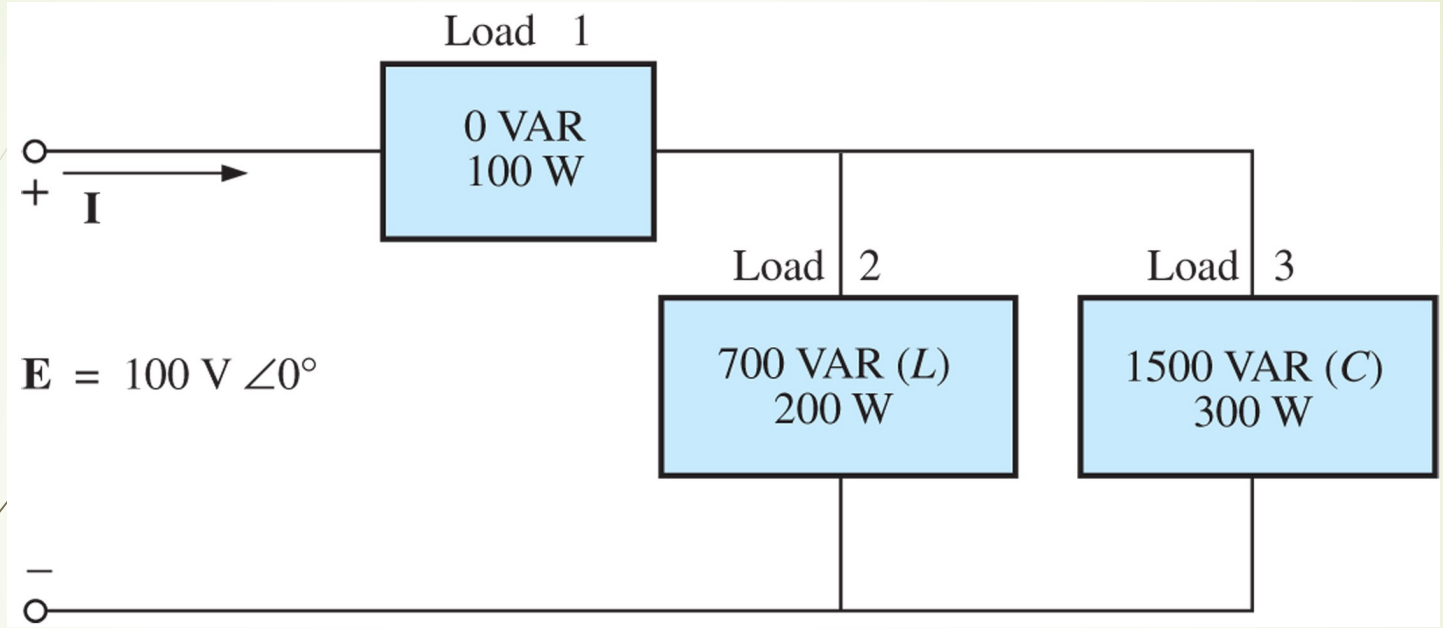


# Example

1. Find the total real power $P_T$, total reactive power $Q_T$, and volt-amperes $S_T$ and draw the power triangle.
2. Find the power factor (PF)
3. Find the current in phasor form.



# Solution




**1. Find total power**

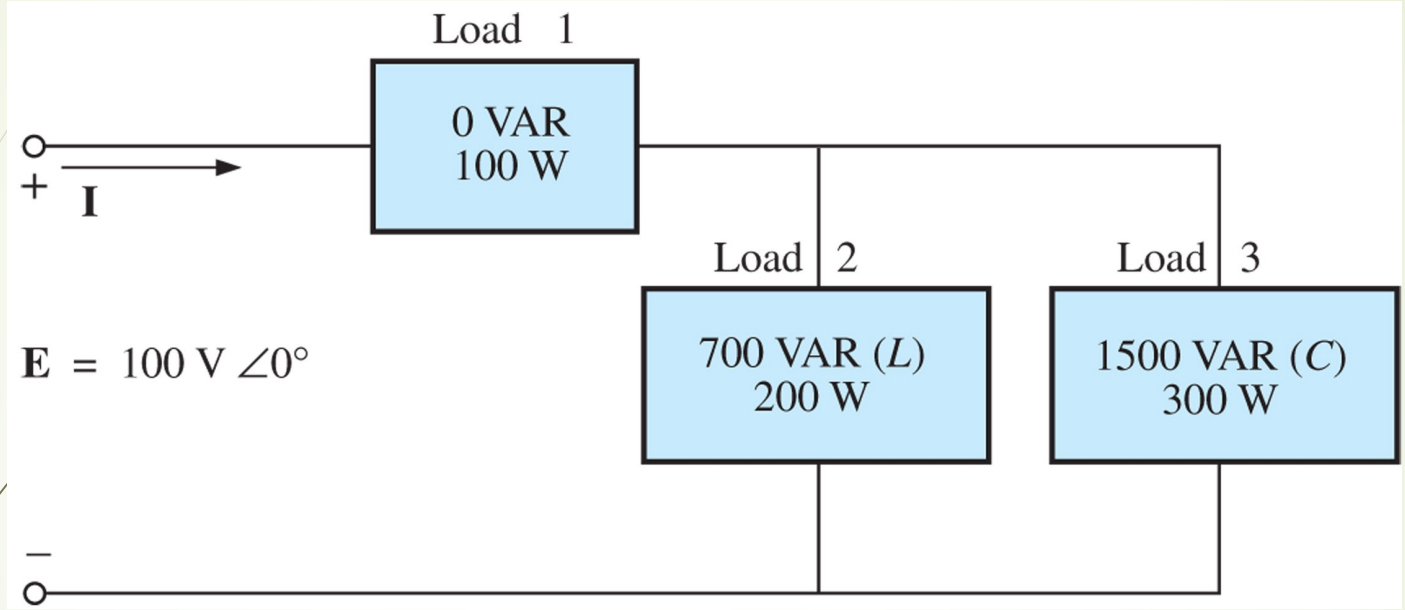
**Load 1**

$$S = \sqrt{P^2 + Q^2}$$

$$P_1 = 100 \text{ W};$$
$$Q_1 = 0 \text{ VAR}$$

$$S_1 = 100 + j0$$
$$= 100 \text{ VA}$$

# Solution (cont'd)




**Load 2**

$$S = \sqrt{P^2 + Q^2}$$

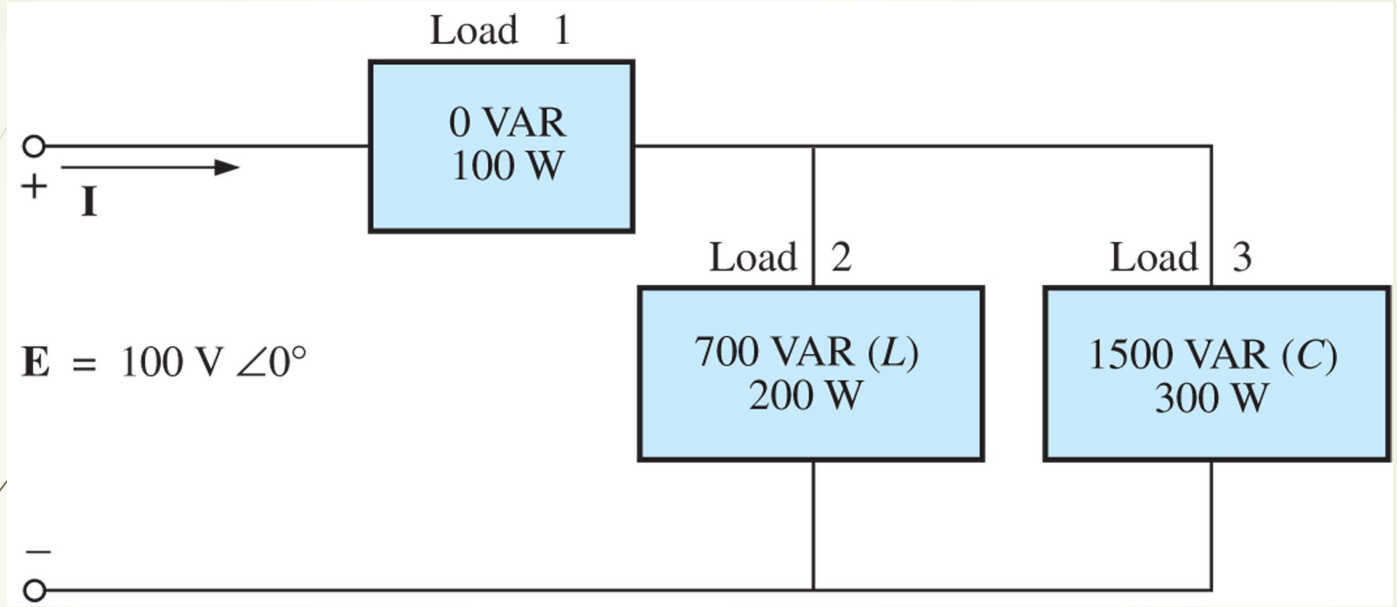
$$P_2 = 200 \text{ W};$$

$$Q_2 = 700 \text{ VAR}$$

$$S_2 = 200 + j700$$

$$= 728 \text{ VA}$$

# Solution (cont'd)




**Load 3**

$$S = \sqrt{P^2 + Q^2}$$

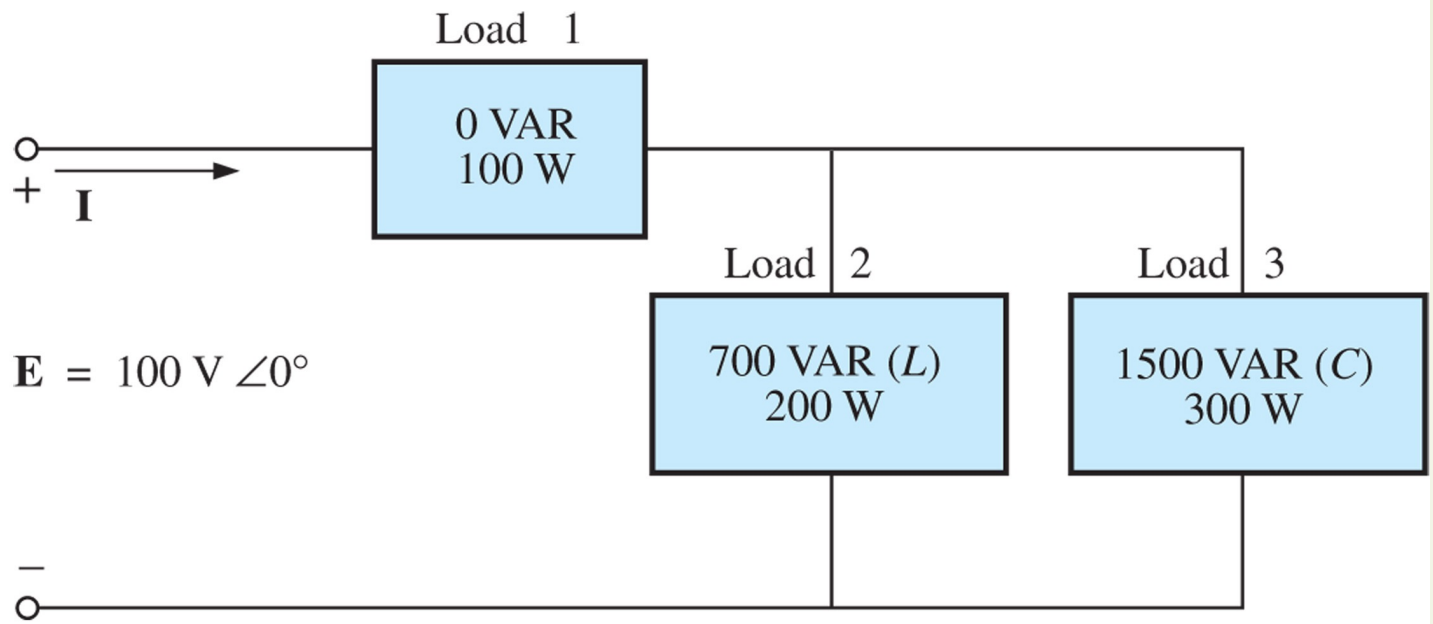
$$P_3 = 300 \text{ W};$$

$$Q_3 = 1500 \text{ VAR}$$

$$S_3 = 300 - j1500$$

$$= 1529.71 \text{ VA}$$

# Solution (cont'd)

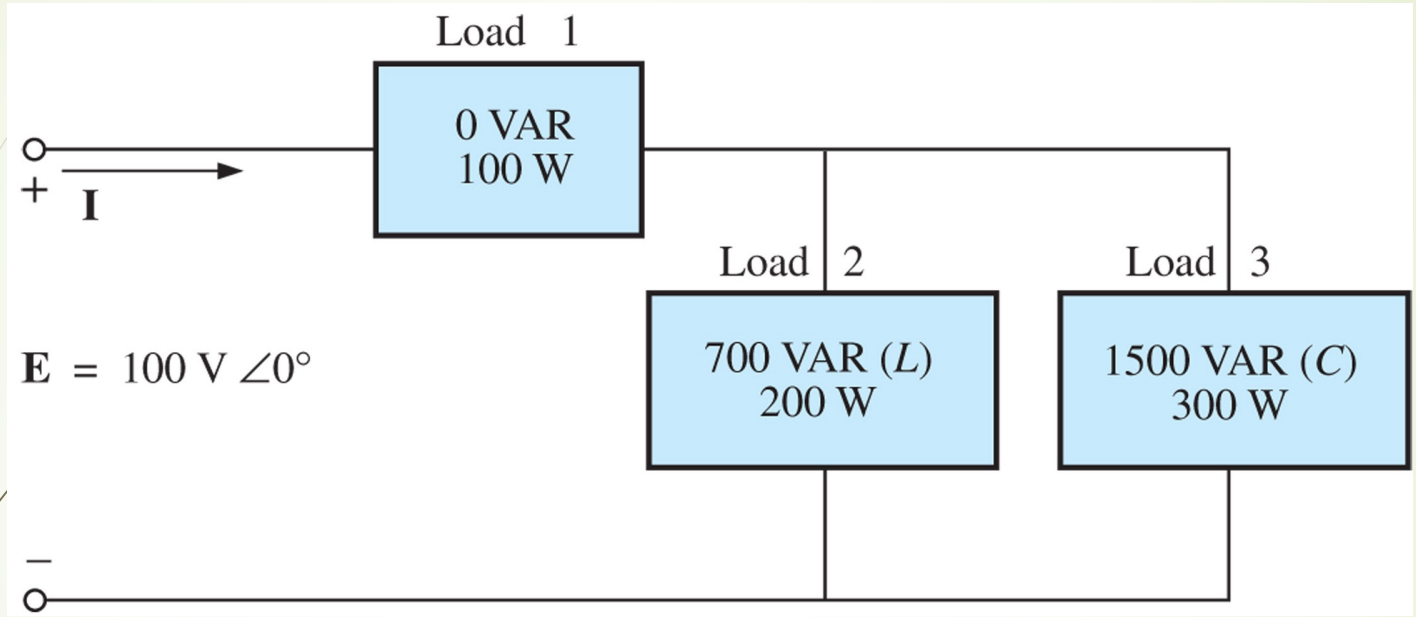



**Total real power =>**

$$P_T = P_1 + P_2 + P_3$$
$$= 100 + 200 + 300$$

$$P_T = 600 \text{ W}$$

# Solution (cont'd)




**Total reactive power =>**

$$Q_T = Q_1 + Q_2 + Q_3$$
$$= 0 + 700 - 1500$$
$$= -800 \text{ VAR}$$

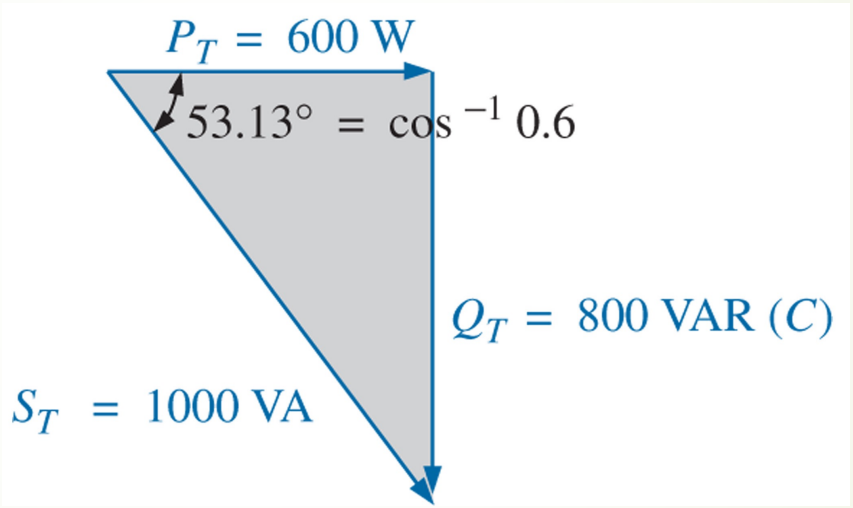
$$Q_T = 800 \text{ VAR(C)}$$

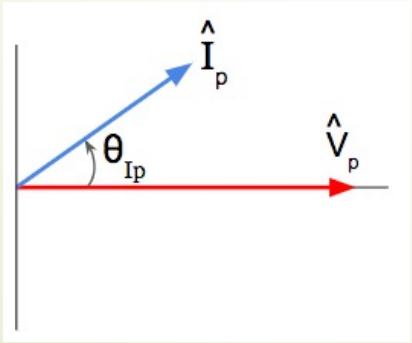
# Solution (cont'd)

## 2. The power factor (PF)

$$PF = \frac{P_T}{S_T} = \frac{600}{1000}$$

$PF = 0.6$ *leading*




$\hat{S} = |S|\angle\theta \quad \hat{I}_p = |I_p|\angle\theta_{Ip} \quad \hat{V}_p = |V_p|\angle 0°$

$\theta = -\theta_{Ip}$

$\theta_{Ip} = -\theta$

*Only when:* $\theta_{Vp} = 0°$

# Solution (cont'd)

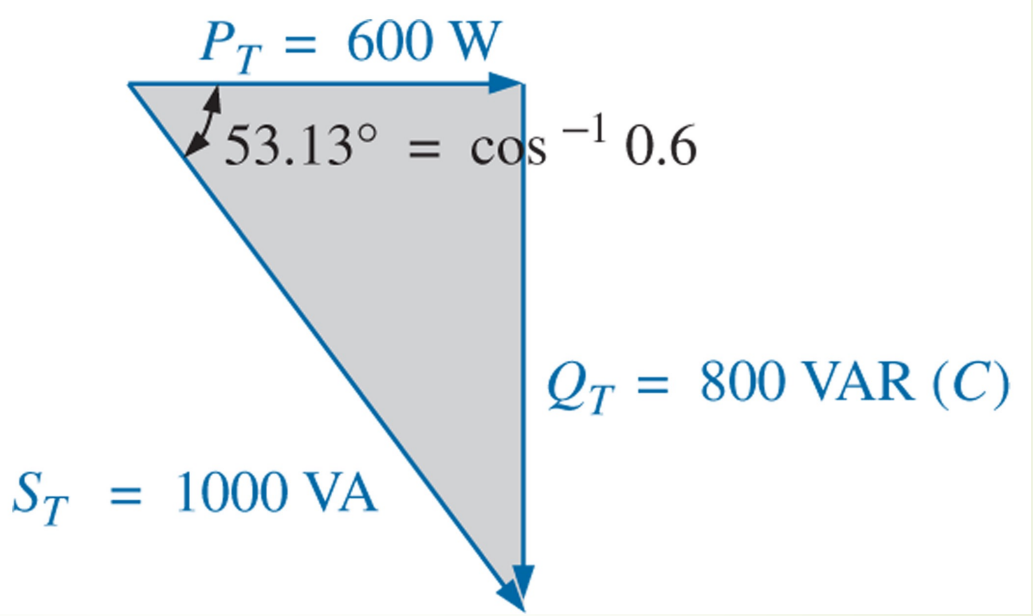
**3. The current**

$$S_T = 1000\ \text{VA}$$

$$VI = 1000\ \text{VA}$$

$$I = \frac{1000\ \text{VA}}{100\ \text{V}}$$

$$I = 10\ \text{A}$$




Since *PF* is **leading,** **I** leads **E**, predominantly **capacitive** circuit.

$$\mathbf{I} = 10\ \text{A} \angle +53.13°$$

| Type of Power | Formulas | Units |
|---|---|---|
| Average Power | $P = V_{rms} I_{rms} \cos\theta$<br>$= I_{rms}^2 R$<br>$= \frac{V_{rms}^2}{R}$ | Watt |
| Reactance Power | $Q = V_{rms} I_{rms} \sin\theta$<br>$= I_{rms}^2 X$<br>$= \frac{V_{rms}^2}{X}$ | VAR |
| Apparent Power | $S = V_{rms} I_{rms}$<br>$= I_{rms}^2 Z$<br>$= \sqrt{P^2 + Q^2}$ | VA |
| Complex Power | $S_C = V_{rms} I_{rms}^*$<br>$= S\angle \pm\theta$<br>$= P \pm Q$ | VA |